

**Genetic Optimization for Unified Pairs Trading via
Monte Carlo Simulations, Extreme Gradient Boosting and
Graph Optimization**

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I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa

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Resumo

Pairs Trading é uma estratégia de negócio, neutra, que explora desvios temporários entre ativos historicamente correlacionados. Esta tese introduz um sistema que unifica as duas fases típicas da estratégia, utilizando *machine learning*. A metodologia integra a formação de pares de ativos e otimização dos parâmetros da estratégia de *trading* num único Algoritmo Genético, permitindo a otimização simultânea de correlação e rentabilidade. A rentabilidade esperada é avaliada via simulações de Monte Carlo, modelando a dinâmica de mercado com processos *Ornstein-Uhlenbeck Variância-Gama*. No Algoritmo Genético, modelos de *Extreme Gradient Boosting* evoluem para prever movimentos dos pares, filtrando sinais e melhorando a precisão da estratégia. Estratégias candidatas são agregadas num portfólio com otimização baseada em grafos, minimizando risco sistémico e impondo diversificação. Testes no *S&P500* mostram retornos ajustados ao risco superiores em diversos regimes de mercado. Resultados destacam a vantagem da estimativa prospetiva e execução com *machine learning*, revelando que o desempenho passado gera rendimentos decrescentes. O sistema atingiu 10% de retorno médio anual com drawdowns baixos. Ao unificar o processo num sistema adaptável, esta investigação oferece uma alternativa resiliente e estabelece bases para avanços em arbitragem estatística automatizada.

Palavras-chave: *Pairs Trading*; Formação de Pares; Estratégia de *Trading*; Algoritmos Genéticos; Simulações de Monte Carlo; *Extreme Gradient Boosting*; Processo *Ornstein-Uhlenbeck* com *Variance Gamma*; Otimização de Grafos; Arbitragem Estatística; Retorno Ajustado ao Risco

Abstract

Pairs trading is a market-neutral strategy exploiting temporary deviations between two historically correlated assets. It involves pair formation and defining trading signals. This thesis introduces a unified, machine learning-enhanced framework addressing traditional limitations like static assumptions and overfitting. The methodology features a novel Genetic Algorithm chromosome encoding both asset pairs and their trading parameters, enabling simultaneous combinatorial pair and strategy optimization guided by correlation and profitability metrics. To mitigate backward-looking metric risks, expected profitability is assessed via Monte Carlo simulations modeling mean-reverting spread dynamics using Ornstein–Uhlenbeck Variance–Gamma processes, ensuring robust parameter tuning under diverse scenarios. Within the Genetic Algorithm, Extreme Gradient Boosting models evolve to forecast short-term spread movements, acting as a trade-signal filter and enhancing execution precision. Candidate strategies from the evolutionary search are assembled into a portfolio using graph-based optimization, minimizing systemic risk and enforcing diversification constraints. Empirical tests on historical *S&P500* data demonstrate the framework consistently achieves superior risk-adjusted returns across market regimes, significantly outperforming benchmark pairs trading heuristics. Results highlight the advantages of forward-looking profitability estimation and ML-guided execution, revealing that reliance on past performance yields diminishing returns. The full system achieved an average yearly return of 10% with relatively low drawdowns. By unifying the entire pipeline into an adaptive, scalable system, this research offers a resilient alternative to heuristic approaches and lays groundwork for advances in automated statistical arbitrage.

Keywords: Pairs Trading; Pair Formation; Trading Strategy; Genetic Algorithms; Monte Carlo Simulations; Ornstein–Uhlenbeck Variance Gamma Process; Extreme Gradient Boosting; Graph-Based Portfolio Optimization; Statistical Arbitrage; Risk-Adjusted Return

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Nomenclature

General Terms

PF Pair Formation

TS Trading Strategy

Performance Metrics

MDD Maximum Drawdown

ROI Return on Investment

SR Sharpe Ratio

Genetic Algorithm Components

BO Bi-Objective

G Chromosome configuration

GA Genetic Algorithm

MO Multi-Objective

NSGA-II Non-dominated Sorting Genetic Algorithm II

NSGA-III Non-dominated Sorting Genetic Algorithm III

POF Pareto-optimal front

Q Magnitude penalization factor

SO Single-Objective

System Variants

GXMU Unified system via Monte Carlo simulations, Extreme Gradient Boosting and graph optimization

MU Unified system via Monte Carlo simulations

SU Simple unified system

XMU Unified system via Monte Carlo simulations and Extreme Gradient Boosting

Portfolio Management

GP Graph-based Portfolio management

L Capital allocation ratio

N_{days} Re-optimization frequency (days)

N_{runs} Number of runs

Spread Modeling

β Hedge ratio or cointegration coefficient

$Coint$ Cointegration

HL Half-life

Mag Magnitude of the spread

S_t Spread at time t

SDD Sum of squared distances (distance metric)

Vol Volatility of the spread

w Rolling window size for spread calculation

Monte Carlo Simulations

Δt Time step in simulations

η Jump scaling factor

γ Risk aversion parameter in value function

λ Mean reversion speed in Ornstein-Uhlenbeck process

μ Long-term equilibrium level of spread

σ Volatility parameter

b Jump intensity in Variance-Gamma process

m Number of Monte Carlo simulations

MC Monte Carlo

r Discount rate for future profits

T Forecast horizon (days)

V Value function

X_0 Initial spread value for simulations

Trading Parameters

Δc	Exit step size for threshold adjustment
c	Close thresholds for long/short positions
d	Open thresholds for long/short positions
k	Labeling scheme thresholds
l	Stop-loss thresholds for long/short positions

Machine Learning Models

ANN	Artificial Neural Network
$ARIMA$	Autoregressive Integrated Moving Average
$DDQN$	Double Deep Q-Network
DQN	Deep Q-Network
$LSTM$	Long Short-Term Memory network
ML	Machine learning
RL	Reinforcement Learning
XGB	Extreme Gradient Boosting

Technical Indicators

EMA	Exponential Moving Average
MA	Moving Average
$MACD$	Moving Average Convergence Divergence
RSI	Relative Strength Index
SMA	Simple Moving Average

Chapter 1

Introduction

Pairs trading is a market-neutral investment strategy first introduced in the mid-1980s by quantitative researchers at Morgan Stanley. By capitalizing on relative price movements between correlated assets, this strategy generates profits irrespective of broader market direction, making it a cornerstone of statistical arbitrage. Despite its conceptual simplicity, successful pairs trading hinges on two critical challenges: (1) the identification of robust asset pairs with stable long-term relationships, and (2) the precise timing of trades to exploit temporary divergences while mitigating risk. Traditional approaches, often reliant on cointegration or distance metrics, suffer from computational inefficiencies, overfitting to historical data, and an inability to adapt to dynamic market regimes.

This thesis addresses these limitations by proposing a novel, machine learning-driven framework that unifies pair selection, trade execution, and portfolio optimization into a cohesive, adaptive system, by leveraging multi-objective Genetic Algorithms (*GA*), Monte Carlo simulations, Extreme Gradient Boosting (*XGBoost*) machine learning models, and a Graph-based portfolio optimization.

1.1 Topic Overview

Pairs trading exploits the equilibrium relationship between two correlated assets, such as Coca-Cola (*KO*) and Pepsi (*PEP*), whose stock prices historically move in unison (Figure 1.1). This is often the case in securities from the same industry, that are affected by the same market forces, although not necessary.

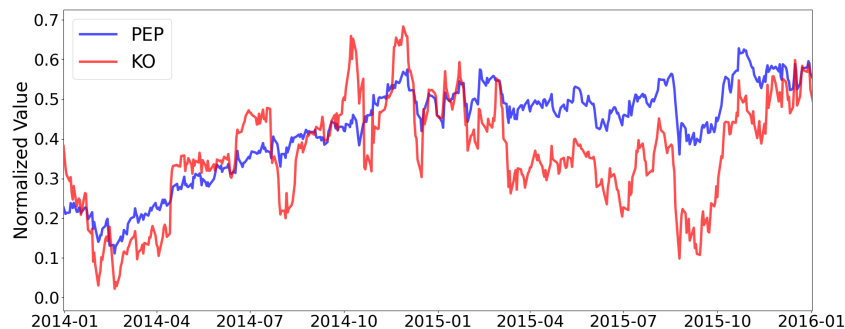


Figure 1.1: *KO* and *PEP* stock price (2014 – 2016)

When a gap is found between the pair, and assuming the persistence of their equilibrium relationship, a trader can exploit the correlation discrepancy for a profit. A pairs trading strategy is comprised of two stages: the pair formation, and the trading phase. The pair formation is accomplished by grouping pairs of assets whose prices have historically moved together, by means of various methods. In the trading phase, the price difference for the assets of each pair is monitored. If it is detected that this spread widens according to the implemented strategy, the investor will open a short position on the overvalued stock, and a long position on the undervalued one. Next, the trader waits for the value of the assets to converge, having both positions immediately closed - representing a profit.

1.2 Motivation

This simple two-step procedure has shown promising results since its inception. While conceptually elegant, its practical implementation continues to face significant challenges. Despite decades of research, traditional pairs trading methods often fall short in several critical areas. Pair selection based on conventional metrics—such as Euclidean distance or cointegration—frequently identifies spurious relationships, leading to unreliable trade signals. Execution timing is another limitation, as static thresholds typically fail to adapt in volatile or trending markets. Risk management also suffers, with strategies often overexposed to sector-specific shocks that undermine portfolio diversification. Furthermore, the decoupled nature of pair selection and strategy optimization can lead to misaligned objectives, resulting in statistically valid pairs that nonetheless underperform in live trading. Historical bias remains a persistent issue, as reliance on backtested returns and fixed thresholds reduces adaptability in non-stationary environments, often leading to overfitting and unstable real-time performance. Scalability is also a concern; brute-force screening across large asset universes—such as the S&P 500—places considerable computational demands on the system, hindering timely reoptimization.

Finally, conventional risk management tools like simple stop-loss mechanisms and naive allocation rules are typically inadequate in addressing systematic exposures, especially during regime shifts or structural market breaks. In contrast, modern machine learning and evolutionary computation offer promising alternatives by enabling a unified, adaptive, and forward-looking approach to optimizing both pair formation and trading strategies.

1.3 Objectives

Our framework addresses these issues through modular, machine learning-enhanced components, each validated in independent case studies (Section 4).

This thesis aims to design, implement, and evaluate a novel, unified framework for pairs trading that integrates advanced computational techniques across all key stages: pair identification, trade execution, and portfolio construction. The specific objectives are:

1. **Unified Multi-Objective Optimization:** Develop a co-optimized genetic algorithm (*GA*) that si-

multaneously selects asset pairs and calibrates trading thresholds, aligning statistical and profit objectives in a single evolutionary process. An adaptable framework is designed, allowing for adaptable search space constraints, objective and gene configurations, trading rules and portfolio management.

2. **Simple Implementation:** A simple system is designed from the adaptable framework. This serves as a baseline for further improvements, as it employs a simple version of each component of the framework. The profitability objective relies on past performance. For the trading rules, the typical threshold-based trading strategy is employed. Finally, a simple portfolio is constructed from the resulting evolutionary population.
3. **Forward-Looking Profitability:** Replace historical performance metric with expected profit (EP) derived from Monte Carlo simulations of mean-reverting spread dynamics, enhancing robustness to future market scenarios.
4. **Adaptive Execution via Machine Learning:** Incorporate XGBoost models to predict short-term spread trends, refining entry and exit signals beyond static or purely threshold-based rules.
5. **Graph-Constrained Portfolio Management:** Introduce a graph-based allocation scheme that optimally balances capital among concurrent pairs, minimizing systemic risk and improving diversification.

1.4 Contributions

This work makes the following key contributions to the pairs trading literature and practice:

- **Integrated Pairs Trading:** A novel chromosome design that encodes both asset selection and dynamic trading parameters, enabling unified optimization via NSGA-II across dual objectives: statistical co-integration and profitability. The integrated system allows for solutions to be derived, that represent a pair of securities, and it's optimal trading triggers.
- **Monte Carlo-Driven Fitness:** A forward-looking objective function based on expected profit from Ornstein–Uhlenbeck Variance Gamma simulations, which mitigates path-dependency and improves out-of-sample stability. It is integrated into the genetic optimization, allowing trading triggers to be fine-tuned according to possible future paths, and pairs to be selected according to their expected profitability, instead of past performance.
- **Evolutionary optimization of XGB models:** A XGBoost-based predictive layer that forecasts threshold dynamics is introduced in the system, acting as a gatekeeper for trade decisions and integrating seamlessly with the genetic framework. This integration allows the genetic optimization to select the model, out of an ensemble, that maximizes the trading strategy profitability.

- **Graph-Optimized Portfolio Management:** A portfolio extraction method leveraging graph theory to solve an edge-maximization problem, balancing trade profitability and diversification constraints in real time.

1.5 Thesis Outline

The remainder of the document is structured as follows: Chapter 2 (Background and Related Work) reviews pairs trading fundamentals and machine learning applications; Chapter 3 (Implementation) details the framework's architecture, separated in four Case Studies; Chapter 4 (Results) presents empirical results across four case studies, validating each module and their synergies; and Chapter 5 (Conclusions) summarizes contributions and outlines future research directions.

Chapter 2

Background

2.1 Pairs Trading

Pairs Trading (*PT*) is a statistical arbitrage trading strategy that capitalizes on temporary historical deviations from an equilibrium relationship between correlated assets. The strategy involves identifying such assets and establishing a long position in an under-performing security and a corresponding short position in an over-performing counterpart, given the prices diverge beyond a statistically significant threshold, and profit is realized upon subsequent historical mean convergence (thereby closing all positions) [1]. Gatev et al. [1] introduced the seminal pairs trading framework, formalizing the strategy through a structured two-phase process:

- **Pair Formation (*PF*):** Statistical measures are employed to identify potential pairs, by screening securities for high correlation over a predefined pair formation period. This step ensures the selection of pairs exhibiting a stable long-term equilibrium relationship, a prerequisite for reliable mean reversion.
- **Trading Strategy (*TS*):** The strategy monitors the pair's spread (difference between two securities' quoted prices). A divergence exceeding a predetermined threshold triggers a contrarian trade: shorting the outperforming asset and longing the underperforming one. Positions are closed when the spread reverts to its mean, thereby capturing the convergence. The profitability of this mechanism hinges on the statistical likelihood of mean reversion, which is implicitly assumed to outweigh transaction costs and execution risks, for truly correlated assets.

The appeal of pairs trading lies in its dual capacity to generate risk-adjusted returns uncorrelated with broader market movements, thus maintaining a market-neutral profile. By simultaneously holding long and short positions for correlated assets, the strategy hedges against systematic risk factors, rendering it theoretically immune to directional market shifts [1]. Empirical studies corroborate this neutrality, demonstrating low correlation between pairs trading returns and market indexes (such as the *S&P500* index) [2].

2.1.1 Challenges

Empirical evidence over decades reveals a systematic decline in pairs trading profitability, challenging its past reputation as a robust market-neutral strategy. While the approach yielded consistent excess returns in 1980–1990, performance sharply deteriorated post-2000 due to changes in market microstructure and increased arbitrage risks. Do and Faff [3] identify two main drivers: (1) heightened fundamental risk from macroeconomic shocks disrupting historical price relationships, and (2) noise-trader risk where irrational behaviors exaggerate transient divergences beyond statistical norms. These factors undermine simplistic heuristics like correlation-based pair selection [4]. A core vulnerability lies in unstable cointegration even among historically correlated securities. Do and Faff [4] show that equity pairs strategies returned 60% less from 1989–2002 compared to 1962–1988, not due to crowding but the breakdown of the *Law of One Price*, which states identical assets should converge in price after adjusting for costs [5]. Structural shifts—regulatory changes, asymmetric information, and liquidity fragmentation—have weakened this mechanism, reducing mean reversion reliability and increasing unrealized losses as fewer pairs revert within fixed windows.

Expanding financial data complicates pair selection further. Though larger datasets offer more opportunities, Sarmento [5] warn of a dimensionality paradox: wider universes require stricter filtering to avoid noise, potentially excluding viable non-linear or regime-dependent pairs. Figueria and Horta [6] highlight risks of persistent divergence caused by fundamental repricing, necessitating dynamic stop-loss rules to control drawdowns.

Naicker [7] synthesizes key operational variables to address these challenges: training period length must balance depth and relevance; security subsets often align with industry but can cross sectors; correlation and stability thresholds need careful tuning to avoid excluding or weakening pairs. Additionally, trading duration, entry/exit triggers, and risk controls like stop-losses and diversification critically affect resilience.

These insights mark a paradigm shift where modern pairs trading requires integrating machine learning and stochastic modeling to counter the erosion of linear mean-reversion assumptions. Ongoing innovation in detecting non-equilibrium relationships and managing asymmetric risks remains vital for sustaining strategy effectiveness amid evolving markets.

2.1.2 Search Space

The construction of a pairs trading strategy necessitates rigorous delineation of its search space—the universe of candidate pairs and methodologies governing their selection—operationalizing the trade-off between computational tractability and statistical robustness.

Regarding the pairing of securities, there are two predominant frameworks in the literature. While broader searches expand opportunity sets, they demand robust safeguards against overfitting—a challenge increasingly addressed through machine learning techniques. Conversely, narrow sector-based approaches benefit from dimensionality reduction but require adaptive reclassification systems to account for evolving industry boundaries, particularly in technology-driven sectors.

Security coupling often is done for two asset pairs, denoted as univariate pairing. However, the strategy can be employed by finding various constituents for both legs of the trade, denoted as multivariate pairs. The distinction between univariate and multivariate pairs hinges on the dimensionality of the cointegration relationship.

- **Exhaustive Search:** This unconstrained approach evaluates all possible pairwise combinations within a predefined asset universe, irrespective of sectoral affiliations. Advocates including [8] and [9] emphasize its capacity to uncover non-obvious relationships across industries, particularly in markets with latent cross-sectoral linkages. However, empirical evidence highlights a critical drawback: the increase of dimensionality amplifies false discovery rates, as spurious correlations arising from multiple hypothesis testing often yield pairs that exhibit transient cointegration during formation periods but fail to persist in trading phases [3].
- **Sector-based Search:** By restricting pair formation to securities within the same industry or sector, this methodology leverages shared exposure to sector-specific risk factors (e.g., commodity prices, regulatory shifts) to enhance cointegration stability. Seminal works by [1] and [3] demonstrate that intra-sector pairs exhibit stronger mean-reversion tendencies due to inherent economic linkages, translating to higher risk-adjusted returns. Nevertheless, excessive constraints risk overlooking inter-sector pairs with valid equilibrium relationships, particularly in vertically integrated industries or during macroeconomic regime shifts.
- **Univariate Pairs:** Traditional strategies focus on bivariate relationships, where two assets form a stationary spread via linear combination. This parsimonious framework dominates academic research due to its computational simplicity and well-established theoretical underpinnings in Engle-Granger cointegration theory [1]. However, univariate models face limitations in capturing complex interdependencies among multiple assets, particularly in markets dominated by sector ETFs or multinational conglomerates with diversified revenue streams.
- **Multivariate Pairs:** Extending the cointegration framework to three or more assets, multivariate pairs model spreads as $S_t = \sum_{i=1}^n \beta_i P_{i,t}$, where β_i are coefficients ensuring stationarity. [6] and others demonstrate that multivariate configurations can capture higher-order equilibrium relationships (e.g., triangular arbitrage in related commodities) and diversify idiosyncratic risk through basket-based hedging. Despite these advantages, multivariate strategies face steep barriers: (1) computational complexity escalates combinatorially with asset count; (2) estimation errors in β_i coefficients amplify transaction costs; (3) sparse liquidity in multi-legged trades increases execution slippage. Quasi-multivariate hybrids, such as portfolios of bivariate pairs or machine learning-driven asset clustering offer pragmatic middle grounds by simulating multivariate exposure through aggregated univariate positions [6]. The academic paucity of multivariate studies—notably [6] and [5]—reflects both methodological inertia and practical constraints. Yet, the rise of high-dimensional cointegration tests and deep learning architectures capable of latent factor extraction suggests untapped potential. As markets grow increasingly interconnected, the integration of multivariate frameworks may prove critical in sustaining pairs trading efficacy.

2.1.3 Machine Learning

Pairs trading has evolved substantially to address the increasing complexities and uncertainties of modern financial markets. While traditional methods remain foundational, they often struggle to adapt to rapidly shifting market dynamics. Consequently, recent frameworks have integrated machine learning techniques to improve pair selection accuracy, optimize trading strategies, and enhance overall performance under contemporary conditions.

Advances in computational power have facilitated the development of sophisticated machine learning models capable of analyzing extensive historical data to reveal complex patterns and relationships between securities—patterns that traditional methods might miss. Utilizing advanced statistical models, neural networks, and ensemble algorithms, these models generate dynamic, data-driven forecasts of potential price divergences in paired securities. They continuously adapt to changing market environments, refining predictions to enable more precise timing of entries and exits alongside improved risk management. Table 2.1 summarizes key recent contributions, showcasing the variety of machine learning approaches and their promising outcomes. Although numerous other studies propose novel machine learning applications to pairs trading, the works presented here offer the most relevant insights for the solution architecture developed in this thesis.

Table 2.1: Table containing ML strategies in the pairs trading literature. PF represents the pair formation phase, while TS represents the trading strategy.

Authors		Implementation		Data	
2025	<i>Present Work</i>	<i>PF</i>	<i>NSGA II, XGB</i>	2000 – 2024	<i>S&P500</i>
2025	<i>Present Work</i>	<i>TS</i>	<i>NSGA II, XGB</i>	2000 – 2024	<i>S&P500</i>
2022	<i>Figueira et al. [6]</i>	<i>PF</i>	<i>NSGA II, NSGA III</i>	2015 – 2022	<i>S&P500</i>
2022	<i>Figueira et al. [6]</i>	<i>TS</i>	<i>ARIMA, XGB</i>	2014 – 2020	<i>S&P500</i>
2021	<i>Sermpinis et al. [10]</i>	<i>TS</i>	<i>GA, RL</i>	1980 – 2018	Bloomberg
2021	<i>Chen et al. [11]</i>	<i>TS</i>	<i>GA</i>	2009 – 2020	Taiwanese market
2021	<i>Flori et al. [2]</i>	<i>TS</i>	<i>LSTM</i>	2000 – 2019	<i>S&P500</i>
2020	<i>Sarmento [5]</i>	<i>PF</i>	<i>OPTICS</i>	2009 – 2019	Commodity ETFs
2020	<i>Sarmento [5]</i>	<i>TS</i>	<i>ARMA, LSTM</i>	2012 – 2018	Commodity ETFs
2020	<i>Brim [8]</i>	<i>TS</i>	<i>DDQN, DQN</i>	2014 – 2017	<i>S&P500</i>
2020	<i>Jirapongpan et al. [12]</i>	<i>TS</i>	<i>ANN, XGB</i>	2015 – 2018	<i>FX market</i>
2019	<i>Chen et al. [13]</i>	<i>TS</i>	<i>GA</i>	—	—
2019	<i>Goldkamp et al. [14]</i>	<i>PF</i>	<i>NSGA II</i>	2012 – 2016	<i>S&P500</i>
2015	<i>Huang et al. [15]</i>	<i>TS</i>	<i>GA</i>	2003 – 2012	Taiwanese market
2015	<i>Dunis et al. [16]</i>	<i>PF</i>	<i>MLP, HONN, GPA</i>	2005 – 2010	CBOT Corn, Ethanol
2009	<i>Dunis et al. [17]</i>	<i>PF</i>	<i>RNN</i>	—	CBOT Soybean/oil
2009	<i>Huck [18]</i>	<i>PF</i>	<i>ANN, ELECTRE III</i>	1999 – 2007	<i>S&P100</i>
2006	<i>Dunis et al. [19]</i>	<i>TS</i>	<i>NN</i>	1995 – 2003	NYMEX WTI, GAS

The integration of machine learning into pairs trading frameworks thus represents a significant step forward, combining predictive power with adaptability to maximize returns and exploit market inefficiencies. This potential has been substantiated by a growing body of research demonstrating successful applications of various machine learning algorithms to pairs trading.

2.2 Pair Formation

In order to find profitable pairs, first, we find all the possible candidates. Having selected the search space, we can then apply one of many techniques for pair selection, relying on different criteria.

Before delving into the pair selection methods, we point out the recent work of Figueira & Horta [6], which posed the question whether multivariate pair (comprised of more than two securities) could outperform univariate pairs (two securities). In their research, they find conclusive evidence that univariate pairs outperform the multivariable ones. In their study, Goldkamp & Dehghanimohammadabadi [14] discover that multivariate pairs can offer more trading opportunities, although of lesser quality. There is still insufficient research on multivariate pairs to disregard the viability of multivariate pairs.

Krauss [9] provides a thorough overview of pairs trading literature which clusters the most common pair selection techniques, such as distance, cointegration, and other approaches. Similarly, Rad et al. [20] provide a robust study of the performance of three different pairs trading strategies – the distance, cointegration, and copula method.

2.2.1 Distance Approach

The distance approach has been introduced by Gatev et al. [1] in 2006. It proposes pair selection based on the minimization of the distance criterion. Krauss [9] synthesized the methodology, in two key steps: first, the calculation of a cumulative total return index for each stock, normalized to the first day of a 12-month formation period; second, the computation of the sum of Euclidean squared distances (SDD) for the price time series of all possible pairs among n stocks. Afterwards, we select the top 20 exhibiting the lowest historical distance metric during a six-month trading period. Then, the prices are again normalized to the first day of this period. Trades are initiated when the spread diverges by more than two historical standard deviations, and are closed either upon mean-reversion, at the end of the trading period, or when the stock is delisted. This defines a threshold based strategy, which will be explained further in detail in this work.

This simple method proved to be able to generate statistically significant risk-adjusted excess returns over a 40-year period, solidifying pairs trading as a legitimate anomaly in the capital market. However, Krauss [9] implies that there are identified areas for improvement, particularly in the choice of Euclidean squared distance for identifying pairs, which he deems suboptimal. Krauss [9] defines the spread variance of two securities, $(P_{it}$ and $P_{jt})$, which denote the normalized price time series of the securities i and j of a pair. He expresses the empirical spread variance as follows:

$$V(P_{it} - P_{jt}) = \frac{1}{T} \sum_{t=1}^T (P_{it} - P_{jt})^2 - \left(\frac{1}{T} \sum_{t=1}^T P_{it} - P_{jt} \right)^2 \quad (2.1)$$

And solving for the average sum of squared distances for the formation period:

$$\overline{SSD}_{ijt} = \frac{1}{T} \sum_{t=1}^T (P_{it} - P_{jt})^2 = V(P_{it} - P_{jt}) + \left(\frac{1}{T} \sum_{t=1}^T P_{it} - P_{jt} \right)^2 \quad (2.2)$$

Krauss [9] claims, from 2.2, that constraining for low SSD is the same as minimizing the sum of the spread variance or the sum of the squared spread mean. This means that this method is likely to form pairs with low spread variance - and consequently, limited profit potential. [9] also points out that, given this method, a spread of zero would result in no profits, even though it would represent the perfect pair.

2.2.2 Cointegration Approach

The cointegration method, developed by Nobel laureates Engle and Granger [21], is a mathematical model in economics which states that a set of variables is considered cointegrated if a linear combination of those variables have a lower order of integration. In simpler terms, it implies that non-stationary variables can be used to model a stationary variable. A stochastic process is said to be stationary if its mean and variance are constant over time [22]. The cointegration equation for signals x_t and y_t can be represented as:

$$y_t - \beta x_t = \mu + \epsilon_t, \quad (2.3)$$

In 2.3, β represents the cointegration factor, μ the mean, and ϵ_t is a stationary, mean-reverting process. Cointegration is valuable for creating artificial stationary time series, especially for trading purposes. Testing for cointegration helps identify stable, long-run equilibrium between a pair of assets. Common methods for testing include the Engle-Granger two-step method and the Johansen's method, where the former involves setting up a cointegration regression and testing residuals for stationarity.

In short, this approach allows for cointegrated pair. If two securities, X_t and Y_t , are found to be cointegrated, then, by definition, the resulting series from the linear combination described in 2.4.

$$S_t = Y_t - \beta X_t, \quad (2.4)$$

This formulation of the spread series is very useful, as the single signal is sufficient to represent a pair's component deviations. In pairs trading, the cointegration factor is commonly called hedge ratio, and it describes the amount of stock Y to purchase or sell for every unit of X .

Vidyamurthy's work [23] is very prominent in this approach, offering a widely referenced work that presents a univariate cointegration method to pairs trading. While his theoretical framework lacks empirical applications and is somewhat spontaneous for practitioners, it provides valuable insights. The approach involves three main steps: (i) Preselecting potentially cointegrated pairs based on statistical or fundamental information; (ii) Testing for tradability using a proprietary approach; and (iii) Designing trading rules with nonparametric methods.

Notably, Vidyamurthy [23] doesn't emphasize rigorous cointegration testing in his process, opting for a more practical approach. Nevertheless, the underlying principle guiding his framework is the concept of cointegrated pairs.

Apart from the base cointegration method explained above, there is more literature concerning other cointegration approaches. Carvalho [24] summarizes in his work different cointegration techniques, as well as their drawbacks.

Rad et al. [20] also performed a study comparing pair matching approaches. Their findings indicate that the distance method yields a slightly higher monthly return compared to the cointegration approach, albeit at a cost of having a slightly lower sharpe ratio. However, on a risk-adjusted basis, the two strategies perform equally well over the full sample period of 1962 – 2014.

Krauss [9], concludes in his work that the cointegration approach is more rigorous than the distance approach. The main reason is that cointegration identifies econometrically more sound equilibrium relationships.

2.2.3 Other Methods

Before exploring machine learning techniques, it is useful to acknowledge several alternative methodologies in pairs trading research. One prominent category is fundamental-based pairing, where securities are selected based on similarity in fundamental metrics—such as price-to-earnings ratios or return-on-equity—under the assumption that firms with comparable fundamentals will exhibit price convergence. Another class of techniques leverages copula-based dependency modeling, which allows for flexible, non-linear dependence structures and captures asymmetries or tail dependencies that linear methods overlook. Some works highlight how copula methods can generate more stable, albeit often smaller, returns compared to distance or cointegration models [20]. Additionally, clustering-based approaches—including k-means, hierarchical, and graph-based clustering such as spectral clustering on co-trading networks—can structure the search space by grouping assets with similar return behavior, after which pairs are selected within clusters [5]. Finally, hybrid models combine these methods—for example, clustering to delimit the universe followed by copula or cointegration testing to select specific pairs—balancing the interpretability and robustness advantages of traditional approaches with the flexibility of advanced statistical tools. While these alternatives are conceptually compelling, their adoption has been limited by increased complexity, parameter sensitivity, and data requirements, which likely explains why distance and cointegration remain the most prevalent strategies in practice.

2.2.4 Machine Learning

Apart from the classic approaches, the introduction of machine learning in the area has brought positive results. Huck [18] first integrated *ML* into pair formation by proposing a three-step framework of forecasting, outranking, and trading. In his method, an artificial neural network forecasts individual stock returns, and the *ELECTRE-III* outranking algorithm ranks stocks by potential growth, forming “top-bottom” rather than co-moving pairs, yielding promising returns in the *S&P100* index.

Regarding genetic optimization in Pairs Trading, Goldkamp and Dehghanimohammadabadi [14] recast pair formation as a multi-objective optimization problem, employing a mixed-integer programming formulation and *NSGA-II* to maximize expected profit while minimizing risk. Their system permits truly multivariate pairs across sectors and demonstrated superior Pareto fronts and outperformance of single-objective benchmarks on *S&P500* data [14]. Building on this, Figueira and Horta [6] compared *NSGA-II* and *NSGA-III* formulations with different objective combinations—minimizing cointegration distance, maximizing spread volatility, and minimizing spread half-life—and found that a carefully balanced bi-objective model often outperformed both single and higher dimensional multi-objective approaches, without compromising robustness to market crashes [6].

2.3 Trading Strategies

After exploring the most relevant techniques for selecting pairs, we will examine the existing strategies for trading execution. In most literature, the trading strategy is usually based on the threshold trading model.

2.3.1 Threshold-Based Trading

The simple strategy proposed by Gatev et al. [1] is the traditional framework for trading execution in a pairs trading strategy. The spread series between the two components of the pair is determined, and acts as a trading trigger. The criterion for opening a position is based on the deviation of the trading signal. A position is typically initiated when the trading signal deviates by more than two historical standard deviations. The position is closed when it converges to the mean. Moreover, if a position has been opened but the trading signal does not revert to the mean within the trading window, the portfolio position is forcefully closed, potentially resulting in losses, as frequently observed in literature. For securities X and Y , which have already been identified as forming a potential pair, the threshold-based strategy, illustrated in Figure 2.1, can be described more formally as follows:

- Define the thresholds that open/close long and short positions. The typical entry triggers used are the upper and lower Bollinger bands, represented by $S_t = \mu \pm 2\sigma$, based on the spread volatility, σ . Positions are usually closed when the spread reverts back to its mean, $S_t = \mu$. The typical stop loss threshold is set at $S_t = \mu \pm 3\sigma$;
- Calculate the spread, $S_t = Y_t - X_t$, as well as its mean, μ , and standard deviation, σ . In literature, these variables can either be calculated for the whole pair formation period, as proposed by [1], or as a rolling window for past data, with a defined period, as proposed by subsequent works;
- For the trading period, monitor the spread evolution in order to detect whenever a threshold is crossed. If the long threshold is crossed, buy Y and sell X . Conversely, if the short threshold is crossed, sell Y and buy X . When holding a position, in case the exit threshold is triggered, it is closed.

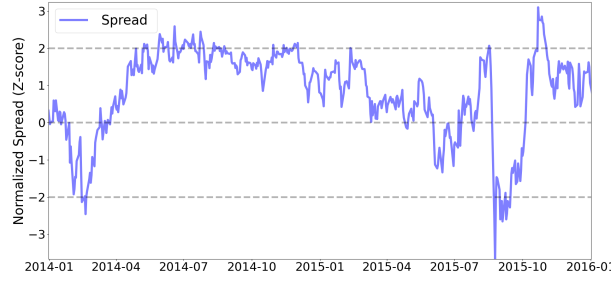


Figure 2.1: Threshold Based Trading Strategy with constant thresholds for Pair $PEP - KO$.

In the example, when the spread crosses the upper threshold, the trade is initiated by opening a short position in the underperforming security and a long position in the overperforming one. The spread later converges to the mean, at which point the positions are closed, securing a profit.

It is important to note that the spread's definition is intricately linked to the method employed for pair identification. Following the distance approach, the spread would be defined as $S_t = Y_t - X_t$. As for the cointegration approach, the spread would be defined by $S_t = Y_t - \beta X_t$. Consequently, this distinction also influences the trading strategy. In the case of the spread defined as $S_t = Y_t - X_t$, opening a long position entails buying Y and selling X in an equivalent number of shares. Gatev et al. [1] advocate for this methodology. In the case where the spread is defined as $S_t = Y_t - \beta X_t$, there appears to be a lack of consensus in the literature regarding the most appropriate approach to handle the hedge ratio. Rad et al. [20] assert that, as per definition, the long and short positions inherently possess unequal values. Consequently, the authors suggest investing \$1 in the long-positioned asset and determining the corresponding value for the short-positioned one based on the hedge ratio. Conversely, other researchers like Rudy et al. [25], opt to disregard the hedge ratio entirely, opting for a money-neutral position by investing an equal amount of money in each stock. Similarly, Chan [26] also claims the investor should invest in β shares of X for each share of Y .

Most pairs trading strategies rely on the threshold-based triggers, as it still proves to be reliable. However, the formerly described strategy exhibits certain limitations. One downside of the threshold-based trading model, is that the entry points are not defined with much precision. The only criterion to enter a position is crossing a predefined threshold, independently of the spread's current direction [5]. Furthermore, it has no concern with optimizing entry points, as it doesn't guarantee the threshold crossing is aligned with the ideal time for entering a position [5]. The spread may continue to diverge, or take too long to revert, resulting in a capital loss.

2.3.2 Other methods

The Time Series method optimizes trading signals by applying different methods of time series analysis. The Stochastic Control approach studies the optimal portfolio holdings in each component of the pair. Naturally, it draws from stochastic control theory. A notable example in the literature include Mudchanatongsuk et al. [27], which optimised a pairs trading system as a stochastic control problem. Holý & Tomanová [28] suggested that high-frequency data are contaminated by the market microstructure noise, which causes significant bias in parameter estimation when not taken into account.

It's relevant to mention Leung & Li [29], where the spread is modeled as an Ornstein–Uhlenbeck process (via Monte Carlo simulations), and the authors formulate a double stopping problem to decide when to enter and exit a position. By modeling the spread trajectory, the threshold triggers can be optimized for the next trade. Importantly, a stop-loss constraint is imposed as a state constraint. Their analysis shows that the “entry region” must lie strictly above the stop-loss level and that a higher stop-loss (i.e. allowing for a larger adverse move before cutting losses) usually implies a lower take-profit level. Though the fine-tuning depends on parameter values (such as mean reversion speed and volatility), many optimal strategies calibrated in similar models tend to use entry signals around 2σ and stop-loss limits that might roughly correspond to 3σ levels—or, alternatively, are set as a fixed percentage of the capital allocated (for instance, a 20% loss limit).

2.3.3 Machine Learning

Similarly to the pair formation phase, machine learning algorithms have seen recent success in the trading strategy. *ML*'s strength in forecasting is well suited to anticipating spread reversion and generating timely entry/exit signals [5]. Huck [18] trading strategy diverges from mean-reversion thresholds by simply going long on top-ranked stocks and short on bottom-ranked, once divergence thresholds are exceeded, effectively implementing a “Top–Bottom” threshold rule that achieved significant directional accuracy improvements. Regarding Genetic algorithms, Huang et al. [15] developed a genetic algorithm tailored for defining optimal trading triggers, encoding moving averages, Bollinger Bands, and stock weighting coefficients. The *GA* optimized for defining optimal trading triggers improves annual returns on Taiwanese semiconductor stock market [15]. Similarly, Chen et al. [13] integrates various trigger decisions into the genetic encoding of the algorithm. Sermpinis et al. [10] compared simple thresholds optimized by *GA* against *DRL* agents trained via advantage actor–critic for commodity pairs trading; the *DRL*-based *TS* dominated both *GA*-optimized and classic benchmarks (distance, cointegration), demonstrating the power of *RL* in action-selection under uncertainty [10]. Finally, regarding *XGB* predictive models, Figueira [6] develops an *ARIMA* and *XGB* model that is able to predict spread dynamics (regression). Jirapongpan et al. [12] also develops a *XGBoost*-based trading strategy (classification).

2.4 Genetic Algorithms

Genetic Algorithms, a subset of Evolutionary Algorithms, are optimization techniques inspired by natural selection processes, and have been effectively applied across various domains, including financial modeling, due to their adaptability and robustness in handling complex, high-dimensional problems.

GA's operate by representing candidate solutions as chromosomes, where each decision variable is encoded as a gene. The optimization process is driven by genetic operators—most notably crossover and mutation. Crossover exchanges genetic material between parent chromosomes to generate novel offspring, promoting diversity within the population.

Mutation introduces stochastic alterations in genes, which helps to escape local optima and enhances global search capabilities. The iterative process continues until a termination criterion is met, either through convergence to a satisfactory fitness level or the completion of a predetermined number of generations.

In financial applications, *GA*'s have been utilized to optimize trading strategies by fine-tuning parameters of technical indicators such as *MACD* and *EMA*, aiming to maximize returns while managing risk. Their population-based approach allows for efficient exploration of vast solution spaces, making them suitable for problems such as pairs trading, where identifying optimal asset pairs or trading criteria is crucial. The flexibility of *GA*'s in incorporating problem-specific constraints and their compatibility with parallel computing architectures further enhance their applicability in financial optimization tasks. Their ability to adapt to dynamic market conditions and integrate multiple objectives makes them a valuable tool in developing robust financial models, where solution spaces may be non-convex, discontinuous, and high-dimensional [6]. Their adaptability is further enhanced by the ease of integrating problem-specific constraints and the potential for large-scale parallelization. For pairs trading specifically, [6] advocates for the use of genetic optimization for navigating large search spaces with conflicting objectives.

2.4.1 Chromosome

The chromosome structure is a crucial component of the genetic algorithm's design. For *PF*, the existing chromosome implementations consist in encoding the pair connections, while in *TS*, the genome guides the thresholds used as trading triggers. Applications in literature for pairs trading include:

- (*PF*) **MO multivariate pairing** [14] Each chromosome consists of three coordinated gene-blocks per side (long vs. short): α (cardinality) is a real value in $[0, 1]$ that decodes to the integer number of stocks in that component; θ (selection) is a list of real values that map to specific stock indices, choosing which assets enter the basket; and β (allocation) is a list of raw weights that are normalized to sum to one, specifying each stock's capital allocation. This variable-length encoding lets the GA pick *how many* stocks (α), *which* ones (θ) and *how much* to allocate to each (β) in one unified structure. Whenever α changes, a cheap reconstruction step automatically drops the lowest-weight entries or samples new stocks (assigning them the average weight) to keep θ and β aligned—thus always satisfying dollar-neutrality and cardinality constraints. By cleanly separating discrete selection from continuous weighting and rebuilding on-the-fly, we avoid unnecessary gene mutations, enforce feasibility under tight constraints, and give the GA the flexibility to explore both uni- and multivariate pair spaces efficiently. Figure 2.6 illustrates this encoding: the *genotype* (α, θ, β) is shown above, and its decoded *phenotype*—integer component sizes, concrete stock indices, and normalized weights—is shown below, ready for fitness evaluation. We estimate the search space to be 10^{30} for their size 7 pairs, and 10^5 for their univariate configuration (size 1).

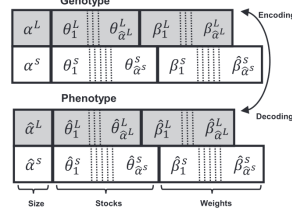


Figure 2.2: Solution representation. [14]

- **(PF) MO multivariate pairing** [6] Each candidate pair is encoded as a binary chromosome of length $2N$ (first N bits for the long leg, next N bits for the short), with hard constraints that each side must contain between l and u stocks and no stock may appear in both legs. Figure 2.3 represents an example for a universe of three stocks. We estimate the search space to be 10^5 for a univariate implementation, and 10^{200} for their unconstrained multi-variate implementation.



Figure 2.3: Example chromosome for a universe of three stocks. [6]

- **(TS) SO Threshold-based Trading** [10] The proposed method uses a GA for threshold fine-tuning in pairs trading, optimizing entry (open) and exit (close) thresholds to enhance trading signals. The GA refines dynamic thresholds to adapt to market conditions, avoiding overfitting and maintaining computational feasibility. The GA employs a binary-coded chromosome (Figure X) where: (i) Open threshold (b_1^o to b_k^o) is optimized within $[1, 3]$ (multiples of historical standard deviation); (ii) Close threshold (b_1^c to b_k^c) is optimized within $[0, 2]$. Trades trigger when the spread exceeds the open threshold (entry) or falls below the close threshold (exit). The precision of optimization is controlled by $k = 10$ bits per threshold, balancing accuracy and computational cost. We estimate the search space to be 10^6 .

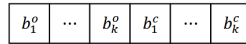


Figure 2.4: Chromosome encoding. [10]

- **(TS) SO Threshold-based Trading** [11, 13] The chromosome representation encodes six parameters critical to optimizing a pairs trading strategy across. Given variables are $cDay$, that specifies the historical period (in days) to calculate correlation coefficients between stocks, while $cLimit$ sets a threshold to qualify pairs. For signal generation using Bollinger Bands, $mDay$ defines the moving average window, $BBentryWidth$ and $BBoutWidth$ determine entry and exit channel widths (as multiples of standard deviations), and $oDay$ establishes a dynamic range $[E_{T-oDay}, E_T]$ to validate signal triggers. Together, these parameters enable the genetic algorithm to evolve strategies that balance correlation-based pair identification with Bollinger Band-driven market timing. The fitness function evaluates how well these tuned parameters perform in generating profitable trades, ensuring adaptability to market dynamics. We estimate the search space to be 10^{15} .

$cDay$	$cLimit$	$BBentryWidth$	$BBoutWidth$	$mDay$	$oDay$
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Figure 2.5: Chromosome representation. [11]

- **(TS) SO Threshold-based Trading** [15] The chromosome structure encodes the trading system's parameters into a unified binary string, partitioned into distinct blocks for simultaneous optimization. Each block represents a critical component: the integer period n for the moving average window, the real-valued thresholds x and y defining Bollinger Band entry/exit multiples (as standard deviations), and a series of raw weights β_i for capital allocation across m stocks. These blocks are concatenated into a single binary sequence allowing the genetic algorithm to holistically evolve timing rules and portfolio weights. The binary segments are converted to actionable values via scaling: each block's decimal equivalent maps to predefined parameter ranges (e.g., n as an integer window size, β_i normalized to sum to one). This encoding enables the GA to dynamically balance market timing (adjusting n , x , y) with asset allocation (tuning β_i), generalizing beyond traditional two-stock pairs to handle groups of m stocks. By integrating these parameters into a single genotype, the system efficiently explores high-dimensional search spaces, ensuring robust adaptation to market dynamics while maintaining feasibility. The design emphasizes precision and flexibility, allowing the GA to uncover strategies that exploit mean-reverting spreads across diverse stock combinations, validated through temporal testing to ensure stability in dynamic financial environments. We estimate the search space to be 10^{39} .

b_n^1	$\dots$	$b_n^{n_n}$	b_x^1	$\dots$	$b_x^{n_x}$	b_y^1	$\dots$	$b_y^{n_y}$	$b_{\beta_1}^1$	$\dots$	$b_{\beta_1}^{n_{\beta_1}}$	$\dots$	$b_{\beta_m}^1$	$\dots$	$b_{\beta_m}^{n_{\beta_m}}$
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Figure 2.6: Chromosome encoding. [15]

2.4.2 Objectives

The design of objective functions in pairs trading strategies presents a critical challenge: reconciling competing aims of profit maximization with risk mitigation, particularly against non-reverting spreads. Optimal performance necessitates a careful balance between these objectives, where oversimplification or redundancy can degrade both model robustness and profitability [14]. This section delineates the methodological considerations and empirical insights guiding the selection and integration of objectives.

In multi-objective (MO) optimization, the interplay between objectives profoundly influences search space topology and Pareto front characteristics. Correlated objectives, such as minimizing spread stationarity deviation (SSD) while maximizing the number of zero crossings (NZC), introduce redundancy that undermines evolutionary efficiency [14]. Conversely, objectives that encapsulate distinct facets of trading performance enable the discovery of solutions that balance profitability, stability, and adaptability. For instance, while [14] and [6] establish that multi-objective models outperform single-objective (SO) counterparts, their findings reveal diminishing returns with excessive objective counts: bi-objective (BO) frameworks frequently surpass models incorporating three or more objectives (MO).

Common objectives for the pairs trading problem that can be found in literature are:

- *(PF) Coint* [14, 6] Minimizing cointegration residuals ensures stable mean-reverting spreads while accommodating dynamic β optimization. Pairs with low cointegration represent robust long-term correlation. This metric, widely adopted in non-GA frameworks [9, 3, 4], enforces statistical rigor in pair selection.
- *(PF) Vol, HL* [14, 6] Maximizing volatility amplifies profit potential through frequent divergence-convergence cycles, while minimizing half-life accelerates mean reversion. Combining these metrics ensures frequent, valuable trades.
- *(PF) SSD, NZC* [14, 6] Though SSD minimization and NZC maximization both target spread stationarity, their correlation risks redundancy. NZC prioritizes zero-crossing frequency as a proxy for mean reversion, whereas SSD emphasizes short-term spread similarity. Similarly to the cointegration metric, the SSD is also widely used (outside of evolutionary implementation) for finding profitable correlated pairs [1, 9].
- *(PF) Mag* [14] Minimizing the terminal training spread divergence precludes unsustainable deviations but may delay trade initiation during convergence phases.
- *(TS) ROI* [10, 11, 13, 15] ROI maximization ensures profitability alignment with real-world execution constraints, transcending statistical validity to reflect economic viability.

2.4.3 NSGA II, NSGA III

Among the multi-objective population selection algorithms, two prominent ones are the *NSGA-II* (Non-dominated Sorting Genetic Algorithm II) and *NSGA-III* (Non-dominated Sorting Genetic Algorithm III). Since its inception in 2002 by Deb et al. [30], *NSGA-II* has stood as the benchmark in multi-objective optimization, as noted by Pilát [31] in 2010. Its distinguished features include (i) the elitist principle, affording elite individuals in a population the opportunity to persist into the next generation; (ii) incorporation of an explicit diversity-preserving mechanism; and (iii) emphasis on non-dominated solutions.

The elitism principle contributes to enhanced and expeditious performance, ensuring that promising solutions endure throughout the algorithm's execution. The diversity-preserving mechanism guarantees a well-distributed set of solutions along the optimal Pareto curve. However, as the number of objective functions increases, the count of non-dominated solutions grows exponentially, leading to newly generated solutions having diminishing representation in the evolving population and the preservation of diversity becoming progressively intricate. *NSGA-III* (Deb & Jain, [32]) emerges as an evolution of its predecessor, specifically designed for many-objective optimization problems, distinguishing itself by modifying how the population is filtered and by refining the diversity preserving mechanism. These algorithms have proven reliable in a pairs trading setting [6].

2.4.4 Pareto Front

Pareto dominance is a relationship between two multi-objective solutions. A solution is dominating if and only if it has at least one objective better than the dominated solution, with all the other objectives no worse [14]. A candidate solution is considered pareto optimal if it is non-dominated, such that no other candidates show improvements without degrading another objective.

In multi-objective optimization, the pareto front is the set of all pareto efficient solutions of a given generation. Multi-objective optimization approaches tackle each objective independently, resulting in a frontier of non-dominated solutions. Pareto fronts, which represent the combination of objective functions, can exhibit various shapes, ranging from linear and convex to non-convex and discontinuous. This diversity allows for a comprehensive exploration of the trade-offs inherent in the complex landscape of multi-objective problems.

2.5 Extreme Gradient Boosting

Extreme Gradient Boosting (*XGB*) is an optimized implementation of gradient-boosted decision trees, offering state-of-the-art predictive performance, computational efficiency, and built-in regularization to mitigate overfitting in noisy, non-stationary financial data [33]. It employs second-order Taylor approximations of the loss to guide tree splits, supports sparse data handling via a sparsity-aware split finder, and parallelizes both histogram construction and tree learning to scale gracefully to large datasets.

2.5.1 Advantages

- **Regularization and Complexity Control:** *XGBoost*'s objective includes γ (minimum loss reduction for splits) and λ (tree weight decay), which penalize overly complex trees and help prevent overfitting on financial time series with spurious patterns [34].
- **Handling of Missing/Sparse Inputs:** Its default “sparse-aware” algorithm natively learns optimal default directions for missing values, obviating separate imputation steps and preserving speed on high-frequency tick data [33].
- **Custom Objectives and Evaluation Metrics:** Users may define loss functions tailored to financial objectives (e.g. negative Sharpe ratio, drawdown penalties) and bespoke evaluation metrics, aligning the model directly with trading performance goals [33].
- **Scalability:** Block-compressed data structures, out-of-core learning, and GPU support enable *XGBoost* to train on very large datasets with limited memory, crucial for intraday and tick-level modeling [33].

2.5.2 Applications

Pairs trading predictive models forecast the spread, and base the trading decisions on the prediction. The typical modeling pipeline includes:

1. **Target Definition:** Directly predict the one-step ahead spread (rather than each price separately) to minimize error propagation [6]. Conversely, instead of solely predicting spread price, a set of classes can be designed as to allow the model to perform classification instead [12].
2. **Feature Engineering:** Construct a feature matrix $\mathbf{X}_t$ comprising [6, 12]: (i) Lagged spreads, rolling statistics (mean, variance), and half-life of mean reversion; (ii) Technical indicators: momentum metrics (*ROC*, *RSI*), volatility proxies (*ATR*, realized volatility), and volume-based features; and (iii) Macroeconomic controls: interest rates, *FX* rates, or sector indices to capture regime shifts .
3. **Cross-Validation:** Employ walk-forward (rolling) validation—sequentially expanding the training window and testing on the next period—to simulate live trading and avoid look-ahead bias.
4. **Hyperparameter Tuning:** Optimize $\{\eta, \text{max_depth}, \text{subsample}, \text{colsample_bytree}, \lambda, \gamma\}$ via grid search or Bayesian methods to balance bias–variance trade-offs.
5. **Early Stopping:** Halt training when the validation loss does not improve for n consecutive rounds, preventing overfitting to past regimes [33].

Figueira [6] proposed a ML-based pairs trading framework. They combined *ARIMA* and *XGB* regression to predict the spread trend (more specifically, the difference between the current day’s spread and the next day’s spread), directly feeding the prediction into a rule-based trading engine. This hybrid approach realized a 23% annualized return over 2018–2021 and outperformed static threshold benchmarks. Furthermore, through dynamic thresholding, they calibrated entry/exit bands around the mean using forecast quantiles, reducing drawdown during COVID-19 shocks to under 5.4%.

Jirapongpan and Phumchusri [12] evaluated *XGB* versus ANN for predicting pairs trading profitability. Their classifier used features such as historical returns, volatilities, OLS beta, and rolling correlations to label trades as profitable or not. Although in-sample accuracy reached 80%, out-of-sample performance fluctuated around 60%, underscoring the challenges of generalizing ML models in dynamic markets. Overall, *XGBoost*’s blend of regularization, scalability, and model-agnostic interpretability makes it a powerful tool for building and understanding pairs trading systems that adapt to evolving market conditions.

2.6 Monte Carlo

Monte Carlo (*MC*) simulation has emerged as an indispensable tool for evaluating trading strategies under stochastic models where analytical solutions are intractable [35]. In pairs trading, where profitability hinges on the statistical properties of mean-reverting spreads, *MC* methods enable three critical functions: (1) simulating spread trajectories under realistic dynamics, (2) optimizing entry/exit thresholds, and (3) estimating risk-adjusted returns while accounting for jump risks and heavy-tailed distributions.

This section establishes the theoretical foundations for Lévy-driven Ornstein-Uhlenbeck (OU) processes and their simulation, providing context for modern pairs trading frameworks.

2.6.1 From Gaussian to Lévy-Driven OU Processes

Traditional pairs trading models assume spreads follow a Gaussian OU process:

$$dX(t) = \lambda(\mu - X(t))dt + \sigma dW(t), \quad (2.5)$$

where λ is the mean reversion speed, μ the equilibrium level, and $W(t)$ a Wiener process. While analytically convenient, 2.5 fails to capture the discontinuous jumps and heavy tails observed in empirical spreads [29]. This limitation motivates the use of Lévy processes, which generalize Brownian motion by permitting both continuous diffusion and discontinuous jumps.

The variance gamma (VG) process, introduced by [36], provides a particularly suitable driver for OU dynamics. A VG process $Z(t)$ is constructed by subordinating Brownian motion with a gamma process $G(t)$:

$$Z(t) = \mu G(t) + \sigma W(G(t)), \quad G(t) \sim \Gamma\left(\frac{t}{\nu}, \nu\right), \quad (2.6)$$

where μ controls drift, σ volatility, and ν the variance rate of the gamma subordinator. The VG-OU process then becomes:

$$dX(t) = -\lambda X(t)dt + dZ(\lambda t), \quad (2.7)$$

combining mean reversion with the VG's ability to model skewness and kurtosis [29].

2.6.2 Exact Simulation of OU-VG Dynamics

Euler-Maruyama discretization of 2.7 introduces biases due to the VG process's jump component. Instead, exact simulation uses the closed-form transition density derived from the characteristic function [37]:

$$\phi_{Z^*}(\mu) = \exp\left(\int_0^{\lambda\Delta t} \psi_Z(e^s \mu) ds\right), \quad (2.8)$$

where ψ_Z is the characteristic exponent of $Z(t)$. The discretized process evolves as:

$$X(t_i) = e^{-\lambda\Delta t} X(t_{i-1}) + e^{-\lambda\Delta t} Z^*(\Delta t), \quad (2.9)$$

with innovations $Z^*(\Delta t)$ simulated via gamma subordinators:

$$Z^*(\Delta t) = \eta(e^{\lambda\Delta t} - 1) + G_+^*(\Delta t) - G_-^*(\Delta t), \quad (2.10)$$

$$G_+^*(\Delta t) \sim \Gamma(b\Delta t, b_+), \quad G_-^*(\Delta t) \sim \Gamma(b\Delta t, b_-), \quad (2.11)$$

where $b_{\pm} = \frac{2b}{\sqrt{\mu^2 + 2\sigma^2 b_{\pm} \mu}}$. The drift term $\eta(e^{\lambda\Delta t} - 1)$ ensures alignment with the OU process's equilibrium level μ .

2.6.3 Risk-Adjusted Performance via Value Functions

A core challenge in pairs trading lies in balancing expected profitability against risk exposure. This trade-off is formalized through value functions, which synthesize stochastic profit outcomes into a scalar metric that guides strategy optimization [29]. For a trading strategy $\mathcal{TS}$ applied to simulated spread paths $\{S_i^{\text{sim}}\}_{i=1}^m$, the risk-adjusted value function V integrates three critical components:

- **Discounted Profits:** Future profits are discounted at rate r to account for time preference and capital opportunity costs:

$$P_i = e^{-r\tau_i} \cdot \mathcal{TS}(S_i^{\text{sim}}), \quad (2.12)$$

where τ_i denotes the exit time for the i -th simulation. This penalizes strategies relying on prolonged trade durations [38]. The discount factor $e^{-r\tau_i}$ introduces dynamic programming considerations, as holding a position for longer diminishes the value of the expected profitability.

- **Expected Profitability:** The expected profits for a given pair is estimated by averaging each simulated outcome:

$$\mathbb{E}[P] = \frac{1}{m} \sum_{i=1}^m P_i, \quad (2.13)$$

This metric reflects the strategy's average performance across the simulated market regimes [35].

- **Risk Penalization:** Profit variance quantifies outcome uncertainty, indicating greater risk exposure:

$$\text{Var}(P) = \frac{1}{m-1} \sum_{i=1}^m (P_i - \mathbb{E}[P])^2, \quad (2.14)$$

The value function V combines these elements through a risk-aversion parameter $\gamma \geq 0$:

$$V = \mathbb{E}[P] - \gamma \cdot \text{Var}(P). \quad (2.15)$$

- When $\gamma = 0$, V reduces to raw expected profit maximization.
- For $\gamma > 0$, V embeds mean-variance optimization principles, penalizing strategies with volatile returns [39].

Monte Carlo methods enable efficient estimation of V by simulating m spread trajectories under the assumed stochastic model (Section 2.6.2).

The parameter γ can be calibrated to reflect specific risk preferences—higher values prioritize profit stability over absolute returns, as demonstrated in [40] for energy spread trading. This framework provides a mathematically rigorous basis for comparing strategies while accounting for both temporal and stochastic uncertainties inherent in pairs trading. [41] proposed simulating the value function for different threshold configurations in order to determine the optimal threshold levels.

2.6.4 Parameter Estimation for the Ornstein-Uhlenbeck Process

To calibrate the Ornstein-Uhlenbeck (OU) process to empirical spread data, as proposed by [42], the methodology follows a three-two step process involving linear regression and parameter transformation. This aligns with the discretization approach outlined in [23].

The OU process is discretized into an autoregressive model:

$$S_{t+1} = aS_t + b + \epsilon, \quad (2.16)$$

where:

- $a = e^{-\lambda\delta}$ represents the mean reversion decay rate
- $b = \mu(1 - e^{-\lambda\delta})$ captures the equilibrium level adjustment
- $\epsilon \sim \mathcal{N}\left(0, \sigma^2 \frac{1 - e^{-2\lambda\delta}}{2\lambda}\right)$ is the noise term

Using historical spread data $\{S_t\}_{t=1}^N$, ordinary least squares (OLS) regression is applied to estimate coefficients $\hat{a}$ and $\hat{b}$. The residuals $\hat{\epsilon}$ provide the standard deviation estimate $\hat{\sigma}_\epsilon$.

The OU parameters λ , μ , and σ are derived from regression outputs:

$$\lambda = -\frac{\ln(\hat{a})}{\delta}, \quad (\text{mean reversion speed}) \quad (2.17)$$

$$\mu = \frac{\hat{b}}{1 - \hat{a}}, \quad (\text{long-term equilibrium}) \quad (2.18)$$

$$\sigma = \hat{\sigma}_\epsilon \sqrt{\frac{-2 \ln(\hat{a})}{\delta(1 - \hat{a}^2)}}, \quad (\text{volatility}) \quad (2.19)$$

where δ is the time step between observations (e.g., $\delta = 1/252$ for daily data in annualized terms).

For the estimations, the rolling window of 30 days for spread calculation ensures parameter adaptivity to recent market conditions and daily re-estimation recommended for non-stationary environments [43].

2.6.5 Parameter Sensitivities and Strategic Implications

Both analytical results for Brownian-driven OU spreads and studies for variance–gamma-driven OU spreads (such as [29, 41, 12, 43, 44]) highlight how key model inputs shape optimal entry/exit thresholds. Key findings include:

- **Jump intensity (b):** Smaller b (more frequent small jumps) raises average overshoot, pushing the open threshold upward to maximize profitability, while lowering expected returns.
- **Mean-reversion (λ):** Faster λ concentrates hitting-time distributions, reducing trade duration and variance, thus modestly widening profitable entry bands.
- **Equilibrium (μ):** Nonzero μ induces asymmetry: for negative skew one finds $d_+^* > d_-^*$ to offset heavier left-tail jumps.

- **Volatility** (σ): Higher volatility increases the spread's variability, leading to a broader distribution of potential outcomes. This necessitates wider entry and exit thresholds to maintain a desired risk-reward profile. However, increased volatility also demands a higher number of simulation paths to achieve statistical accuracy in MC simulations.
- **Jump scaling factor** (η): Adjusts the location (drift) component of the jumps in the variance gamma process. Larger $|\eta|$ induces directional drift in jump behavior, which can skew the stationary distribution and impact the symmetry of the optimal thresholds. When η is nonzero, especially when combined with nonzero μ , it further enhances asymmetry in optimal entry/exit points due to a biased jump structure.
- **Initial Spread** (X_0): The simulations can be performed by starting at $X = 0$, in order to optimize the trading strategy. However, simulations can be generated from various initial spreads $X_0 \neq 0$, achieving a more targeted simulated environment.
- **Discount-rate** (r): A higher discount rate diminishes the present value of future profits, prompting earlier exits and narrower entry thresholds to capitalize on near-term gains. Conversely, a lower r allows for wider thresholds, accommodating longer holding periods. This dynamic is crucial in environments with fluctuating interest rates.
- **Risk-aversion** (γ): Incorporating a $-\gamma Var$ penalty reduces both entry and exit thresholds as γ rises, reflecting a desire for lower volatility of returns.

In combination, these analytical and simulation-based sensitivities provide a comprehensive guide to calibrating entry/exit rules in both Gaussian and heavy-tailed mean-reverting models, yielding robust pairs-trading strategies across market regimes.

2.7 Graph Portfolio Management

Qureshi et al. [45] proposed a novel graph-based portfolio management technique. Empirical analysis performed revealed the matching-based pair selection method delivers substantially improved risk-adjusted performance compared to the traditional correlation threshold approaches. Theoretical parameter estimates show that eliminating shared-asset covariance reduces portfolio variance by roughly 50%, raising the annualized Sharpe ratio from 0.50 to 1.18. Backtests on *S&P500* data (2017 – 2023) confirm these findings: the matching strategy achieves a gross Sharpe of 1.23 (versus 0.48 for the baseline and 0.59 for the index), net cumulative returns near 66% after realistic transaction costs, and maximum drawdown under 8.3%, whereas the baseline approach suffers negative net returns and the index draws down 19.5%. Moreover, median monthly turnover under matching is only one-third that of the baseline, sharply reducing trading costs, and single-stock concentration is capped at one pair per asset.

The universe of N assets is represented as a weighted graph $G = (V, E)$, where each edge (i, j) carries weight $w_{ij} = -t_{ij}$. All N assets are modeled as nodes in a graph and every candidate pair as an edge weighted by $-t_{ij}$, which defines the cointegration strength.

By computing a maximum-weight matching (via Edmonds' blossom algorithm), we simultaneously (i) maximize aggregate cointegration signal and (ii) enforce that no two selected pairs share an asset, thereby directly minimizing portfolio variance from pairwise covariance. This graph-theoretic framework offers a clear separation of signal (edge selection) and risk (node disjointness), scales in polynomial time to large universes, and generalizes naturally to hypergraph matching for multi-asset arbitrage strategies.

2.8 Evaluation Metrics

In order to assess the reliability of the proposed models, it will be vital to take use of the following evaluation metrics.

- **Return of Investment** (ROI): The Return on Investment (ROI) is a valuable metric for investment analysis. ROI signifies the profitability of the strategy, indicating efficient capital utilization. It will be main criteria used to reflect the success of this works proposed frameworks, as well as any trading strategy. It is calculated by dividing the net profit by the initial investment.

$$ROI = \frac{Net\ Profit}{Initial\ Investment} \times 100 \quad (2.20)$$

- **Sharpe Ratio** (SR): The Sharpe ratio, formulated in 1966 by Nobel laureate William F. Sharpe [46], serves as a risk-adjusted metric for assessing the return on investment. It considers not only the returns generated but also the associated risk.

$$SR = \frac{R_p - R_f}{\sigma} \quad (2.21)$$

R_p represents the annual portfolio returns, R_f is the risk-free rate, and σ the standart deviation of the portfolio. Higher values for the Sharpe ratio indicate favorable risk-return profiles. Thus, it will be employed when testing for model reliability.

- **Maximum Drawdown** (MDD): A maximum drawdown (MDD) is the largest observed decrease from a peak to a trough in a portfolio until a new peak is reached. This metric helps quantify the maximum loss an investor could have experienced during the trading stage. A lower MDD suggests more stable and less risky performance.

$$MDD = \frac{Through\ Value}{Peak\ Value} - 1 \quad (2.22)$$

Chapter 3

Implementation

This research advances the field of pairs trading and market prediction by introducing a scalable, modular framework that enhances both the robustness and profitability of trading strategies through genetic optimization. The core innovation lies in the unified integration of pair formation and trading parameter optimization within a single evolutionary process. This approach directly addresses a major limitation of conventional pairs trading methodologies: the separation of statistical pair selection and strategy evaluation. Unlike traditional systems, which first identify statistically coherent asset pairs and only later assign the pair a trading strategy, the proposed framework aligns pair selection with profitability from the outset. This ensures that identified asset combinations are not only statistically sound but also strategically advantageous under realistic market conditions. The architecture of this unified optimization framework is depicted in Figure 3.1.

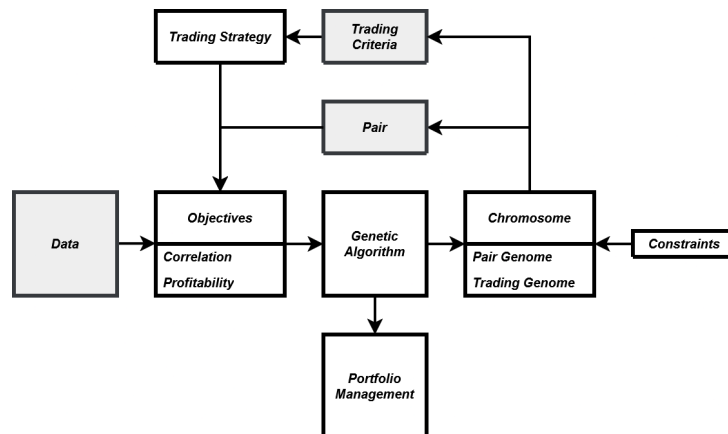


Figure 3.1: Adaptable Unified Pairs Trading Framework.

The central insight behind this design is that pair selection and trading strategy development must be co-optimized. Traditional approaches suffer from a misalignment between the two: a pair may exhibit strong statistical properties but fail to generate returns when paired with fixed or loosely adaptive strategies. Conversely, a robust trading strategy may not generalize well across different pair dynamics. By integrating both the pair and its corresponding optimal trading strategy within a single evolutionary framework, the system overcomes these limitations.

At its core, the system employs a co-optimized chromosome structure in which each candidate solution encodes both a potential trading pair and its corresponding trading configuration. This allows each pair to benefit from a specifically tailored trading strategy. The genetic algorithm evolves these chromosomes using two mandatory fitness objectives: (i) a correlation-based metric that captures statistical similarity, and (ii) a profitability metric based on dynamically testing the candidate strategy on the encoded spread. This dual-objective evaluation ensures that discovered pairs are both statistically valid and practically profitable.

Furthermore, the framework is designed to be highly modular and extensible. Researchers and practitioners can plug in improvements at any stage—whether by modifying the fitness functions, altering the chromosome structure, experimenting with new strategies, adjusting optimization constraints, or integrating advanced portfolio management techniques. This chapter presents several iterations of the system, each enhancing a specific component of the system, while keeping the various methodologies synergistic, in order to extract maximum returns. An overview of the framework’s evolution is provided in Figure 3.2.

- **3.1 Simple Unified Pairs Trading System (*SU*)** Introduces a baseline configuration of the unified framework, which allows for concurrent pair formation and strategy fine-tuning, showcasing the core functionality of the integrated optimization process. A threshold-based trading strategy is used, with dynamically adjusted open/close thresholds, and a simple yet effective reversion test supports entry and exit timing. The genome encodes both the trading criteria and pair composition, enabling formation of cointegrated spread pairs $S = A - \beta B$, without sector constraints. However, search space constraints are used disallow trades contrary to momentum. Statistical strength is evaluated via cointegration, while historical profitability (penalized by unfavorable magnitude spreads) serves as the primary performance metric. A simple tradable portfolio is generated, by applying concurrent re-optimization, extracting the pareto front, and equally assigning capital to each of the final solutions. This iteration of the system serves as a baseline for an unified pairs trading system, with the purpose of addressing key limitations—namely, (i) the suboptimal reliance on past performance for optimization; (ii) the rigidity of threshold-based trading strategies; and (iii) the simplistic aggregation of optimization solutions into trading portfolios.
- **3.2 Expected Profitability via Monte Carlo Simulations (*MU*)** Enhances the unified system by modifying the chromosome structure, strategy logic, objective functions, and portfolio handling. It introduces an Expected Profitability (*EP*) metric, replacing historical returns with forward-looking projections generated through Monte Carlo simulations. Replacing past performance with an expected profitability metric allows the optimization to select more robust pairs and allows for the trading strategy to be fine-tuned according to the possible future paths. This enhancement also enables features like stop-loss optimization and improved generalization to unseen data. This metric becomes central to the subsequent system implementations, as the proposed implementations rely on this effective ranking metric.
- **3.3 Sentiment-based Trading via Extreme Gradient Boosting (*XMU*)** Introduces a suite of

XGBoost models to evaluate the mean-reverting properties of spread behavior, aiding the genetic threshold-based trading strategy. Model selection is embedded within the chromosome and dynamically optimized by the genetic algorithm based on performance under simulated price paths, resulting in a further evolved system.

- **3.4 Graph-based Portfolio Management (*GXMU*)** Implements a novel graph-based portfolio management method, leveraging the Expected Profit metric. The approach solves an edge maximization problem to determine optimal pair allocations, considering active and possible trades, available capital, performance metrics, and diversification constraints. This leads to the final, most advanced version of the system integrating all prior improvements with adaptive, graph-optimized portfolio construction.

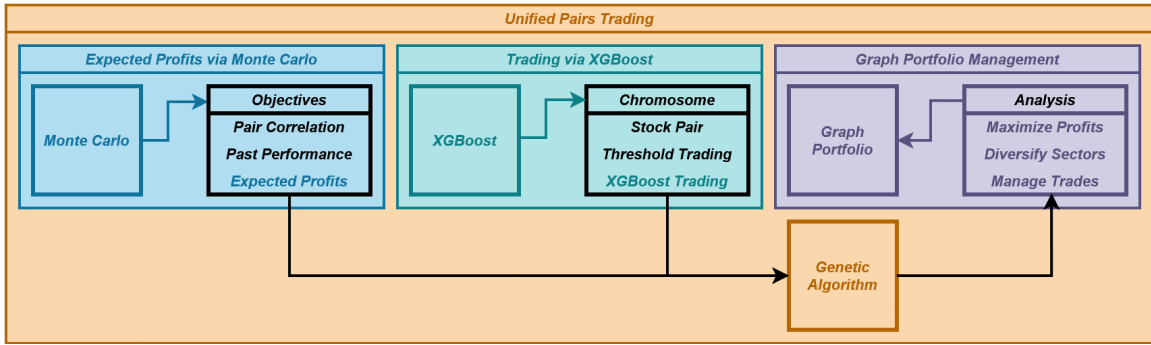


Figure 3.2: Case study structure: Simple Unified Pairs Trading System; Expected Profits via Monte Carlo, Trading via XGBoost; and Graph Portfolio Management.

3.1 Simple Unified Pairs Trading System

Traditional methods conduct optimization for PF and TS separately, leading to misalignment between statistically significant pairs and their real-world performance, in spite of enabling independent fine-tuning. In contrast, our unified pairs trading system directly connects pair selection with trading performance, ensuring that the pair's fitness evaluation is always aligned with the goal of maximizing profit. Unlike methods that rely solely on correlation metrics, our framework prioritizes profitability from the outset and allows for exploration of specific pairs that might only be profitable taking advantage of certain trading strategy parameters. However, this increases computational complexity and introduces the risk of overfitting to historical data.

In this section, we detail every component of a simple unified pairs trading framework implementation (SU), by proposing a trading strategy, objective configuration, chromosome structure, search space constraints and a simple portfolio management methodology (SP). Figure 3.4 illustrates the proposed model architecture.

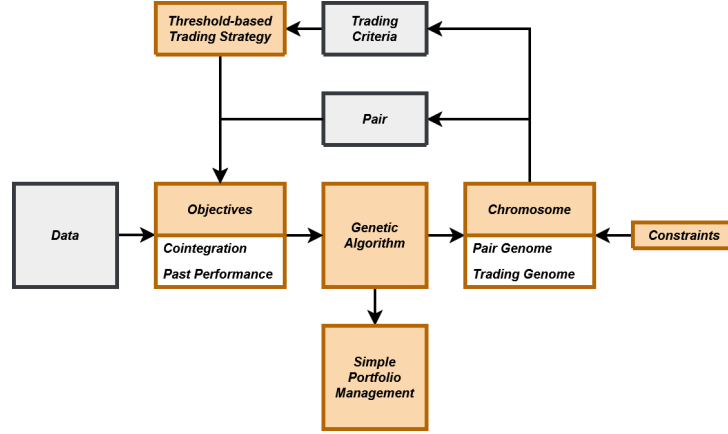


Figure 3.3: Simple Unified Pairs Trading System Architecture.

The genetic algorithm is responsible for deriving solutions that represent a pair and their respective trading strategies (Section 3.1.1). Univariate pairs are selected from the *S&P500* index, without sector restrictions, and various search space constraints are proposed (Section 3.1.2). The trading strategy employed is a simple threshold-based approach, with optimized thresholds determined via genetic optimization (Section 3.1.3). The genetic algorithm is responsible for training both the pair selection and trading criteria, integrating them into a single genome structure (Section 3.1.4). The *GA* optimizes a predefined objective set comprising functions that cater to both pair selection and trading execution (Section 3.1.5). The final population for an optimization consists of individuals with maximized fitness values based on training stock data, ensuring that the resulting solutions optimize the desired objectives. From the final population, a subset of individuals is extracted to construct a tradable portfolio (Section 3.1.6). Tests are performed from 2000 – 2024 (Section 3.1.7), and data cleaning is performed on the data set (Section 3.1.8). A series of benchmark tests are performed in order to compare a separate and unified approach (Section 3.1.9). By comparing the profitability of different portfolio configurations, we evaluate the reliability of various objective sets and the effectiveness of integrating *PF* and *TS* into a unified model and assess the broader applicability of this framework (Section 4.3).

3.1.1 Genetic Algorithm Architecture

We employ Genetic Algorithms (GAs) to address the core challenges in trading strategy and pair design: the exponential growth of the solution space and the presence of conflicting optimization objectives [6, 14]. Specifically, the *NSGA-II* [30] is adopted for its proven efficacy in exploring high-dimensional search spaces. *NSGA-II* efficiently maintains solution diversity, preserves elite individuals across generations, and facilitates convergence toward globally optimal solutions [47], overcoming the computational inefficiencies and non-elitist limitations inherent in traditional multi-objective evolutionary algorithms [6]. Crucially, a multi-objective *GA* framework, unlike many alternative optimizers, generates a diverse population of optimal solutions, representing distinct trade-offs between objectives, rather than a single optimum. Post-optimization, we rigorously analyze this set to identify robust and practically viable trading strategies, moving beyond purely theoretical optimality.

The framework (Figure 3.4) systematically identifies optimal bi-objective pareto pairs that satisfy competing criteria [6].

- **Randomized Population:** Generate an initial population of 200 randomized individuals;
- **Constrains:** Discard invalid individuals, according to the search space constrains set;
- **Genetic Operators:** Apply uniform crossover and randomized mutation to create new individuals, also complying with the constrains;
- **Fitness:** Evaluate each individual's fitness using predefined objective functions (cointegration and return on investment);
- **Selection:** Select the best-performing individuals using the *NSGA-II* selection algorithm;
- **Stop Criteria:** Iterate until the stopping criterion is met (maximum generations or convergence).

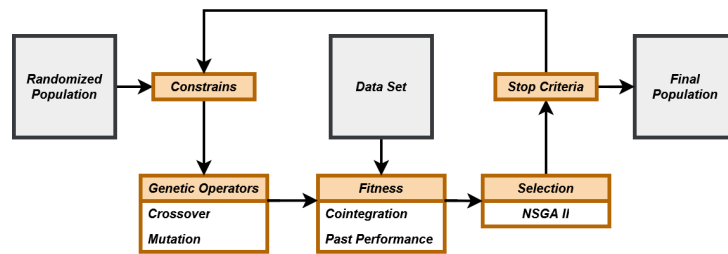


Figure 3.4: Genetic Algorithm Architecture.

3.1.2 Search Space

The search space comprises all constituents of the *S&P500* index, with no imposed sector constraints. This unrestricted approach is motivated by evidence in the literature suggesting that limiting pair selection to narrow sectoral groups can significantly reduce the availability of profitable opportunities [5]. By allowing cross-sector pairings, the model captures more diverse mean-reversion relationships, potentially uncovering structural inefficiencies overlooked by traditional, sector-constrained approaches.

However, this broader universe also introduces a considerable number of candidate pairs, many of which may exhibit temporary or coincidental relationships. Genetic algorithms are well-suited to navigating such large and noisy search spaces due to their inherent ability to balance exploration and exploitation under conflicting objectives [6]. Finally, given the vast number of pair combinations, additional restrictions are applied to the trading strategy configuration to maintain computational feasibility. As prior research has shown, the majority of profitability in pairs trading arises from selecting strong pairs rather than fine-tuning strategy parameters [6]. Therefore, the framework allocates more computational resources to optimizing the pair formation component, while the search space for trading thresholds remains deliberately compact. This approach preserves the unified nature of the optimization, where both pair formation (*PF*) and trading strategy (*TS*) are co-evolved, but it acknowledges the diminishing returns of over-optimization in *TS* when *PF* is poorly chosen. However, slight variations in thresholds can still influence the fitness rankings, reinforcing the importance of integrated optimization.

Pair Formation

In light of recent studies that do not provide consistent results for multivariate pairs, as well as to reduce the search space, we will only test univariate pair formation. This optimization will focus on finding strong univariate pairs. Combining multivariate pairs with the conflicting objectives and already extensive search space would likely lead to an exponentially larger solution space. Furthermore, solutions for a univariate pairs trading optimization can be later integrated into a multivariate portfolio if necessary, in the Portfolio Management step. However, adapting the framework to be able to generate multivariate pairs could be explored in a separate work, although it requires to implement measures that effectively reduce the system complexity. Furthermore, to enhance the robustness of the pair-selection process, we compute the spread as

$$S = A - \beta B,$$

where β is estimated via a cointegration test rather than treated as a fixed distance metric. Although estimating β in this way increases computational complexity and restricts our optimization to pairs that pass cointegration screening, it yields two key benefits. First, cointegration-based spreads are far more interpretable and have been shown empirically to identify reliable, economically viable trading pairs ([1, 9, 23, 18, 20]). For instance, Huck [18] reports a higher rate of convergent trades, and Rad et al. [20] demonstrate greater robustness in volatile markets. Second, by deriving β directly from the cointegration test—selecting the value that minimizes the test's p -value—we avoid enlarging the genetic algorithm's search space. As there will be more correlated spreads, due to every spread being scaled by the optimal β factor, constraining the search prevents needless evaluation of predominantly unprofitable solutions without sacrificing the quality of generated pairs.

Constraints

To make the search process both effective and computationally tractable, an additional constraint is introduced to refine pair selection.

The proposed constraint is guided by empirical trading principles, and validated experimentally. To reduce the risk of entering trades against strong market trends, short positions are restricted from assets with bullish momentum, and long positions from those exhibiting bearish momentum. This prevents counterproductive trades that oppose prevailing trends. Momentum is identified using a simple price momentum indicator, where the current spread S_t exceeds its 30-day simple moving average $MA_{30,t}$. The conditions are formalized as:

$$Bull_t \rightarrow S_t > MA_{30,t} \quad (3.1)$$

$$Bear_t \rightarrow S_t < MA_{30,t} \quad (3.2)$$

While the current approach employs a basic momentum indicator, the constraint streamlines the search space, prioritizes statistically and economically viable pairs. Future enhancements could integrate advanced dynamic thresholds, machine learning for predictive trend analysis, or adaptive momentum metrics.

3.1.3 Trading Strategy

A conventional threshold-based trading strategy is implemented where predetermined upper and lower thresholds trigger long or short positions while two closing thresholds and two stop-loss thresholds govern position exits. For a given spread, $S = A - \beta B$, representing the pair $A \rightarrow B$ (long position on security A and short position on security B), the entry and exit triggers are subsequently refined using genetic optimization to maximize the pair's profitability.

Typical Thresholds

Its important to note that because these values are usually determined by backtesting on historical data, many researchers report that "optimal" thresholds are model and market specific. Based on the literature such as [1, 9, 41], and assuming a normalized spread:

$$S_{norm} = (S - \mu_w) / \sigma_w \quad (3.3)$$

with rolling window w that derive the rolling mean, μ_w , and the rolling standard deviation, σ_w . The standard thresholds for the threshold-based trading strategy are commonly set as follows [9, 14, 1, 29]:

$$\left\{ \begin{array}{ll} \text{Open Long} & \rightarrow S_t > 2 \\ \text{Close Long} & \rightarrow S_t < 0 \quad \vee \quad S_t > 3 \\ \text{Open Short} & \rightarrow S_t < -2 \\ \text{Close Short} & \rightarrow S_t > 0 \quad \vee \quad S_t < -3 \end{array} \right. \quad (3.4)$$

Simple Reversion Test

To enhance the threshold-based strategy, a simple reversion confirmation test is introduced for both trade entries and exits. Rather than executing a trade immediately upon the first threshold breach, the system waits for a subsequent crossing.

- **Open Trigger:** As illustrated in Figure 3.5, this delay allows the spread to further diverge from its mean before entry—potentially increasing profit margins. Conversely, if the spread continues to diverge without reverting after the initial breach, no position is opened, thereby preventing unnecessary losses as depicted in Figure 3.6.

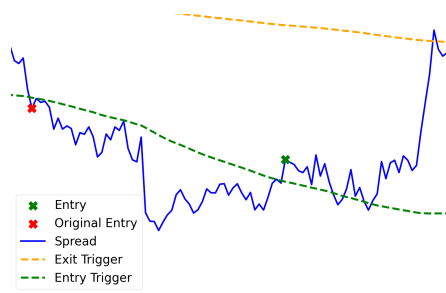


Figure 3.5: Example of delayed entry, showing a position opened after the second threshold crossing to maximize profit potential.

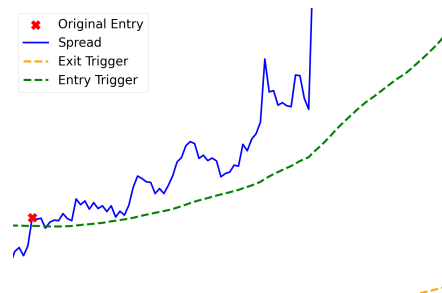


Figure 3.6: Example of avoiding a trade entry when the spread diverges without reverting, preventing unnecessary losses.

Waiting for the second crossing confirms the spread is reverting before entry, reducing false signals. When the spread exceeds the initial open trigger, the threshold shifts upward (due to high standard deviation). This delayed entry allows buying the spread at a wider value than at the first cross, improving potential profit and avoiding unsustainable momentum.

- Close Trigger:** A similar rationale can be applied to trade exits. However, when the spread reverts and surpasses the sell threshold (mean), the trigger threshold would normally remain near its initial level (due low standard deviation), contrary to the open delay cases. Thus, a delay would almost always result in a worse trade. To address this, the sell threshold is gradually lowered in increments of Δc standard deviations (σ) each time it is crossed, until it ultimately reaches the original open trigger on the opposite side of the spread. This adaptive adjustment is illustrated in Figure 3.7 and Figure 3.8. Such a mechanism also increases possible trade profitability, and allows for full reversals of the spread (up to 4σ), thereby enabling a more optimal exit than the traditional pairs trading triggers.

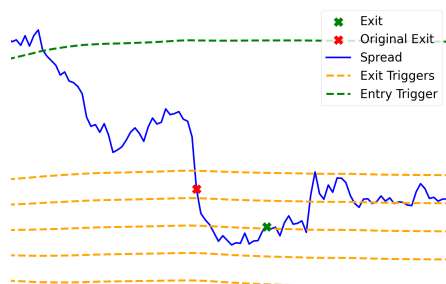


Figure 3.7: Example of delayed exit, showing a position closed after a first threshold adjustment.

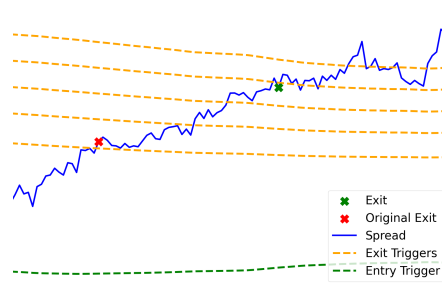


Figure 3.8: Example of delayed exit, showing a position closed after a the final threshold adjustment.

- Stop Loss Trigger:** The stop-loss trigger operates independently of the above rules. Since a stop-loss signal indicates a trade has already become significantly unprofitable, delaying its execution would only exacerbate losses.
- Signal Smoothing:** In order to reduce false triggers caused by market noise, the spread values are smoothed using an 8-day exponential moving average. Figures 3.9 and 3.10 demonstrate that such smoothing helps prevent premature triggering of the confirmation (second crossing) signal.

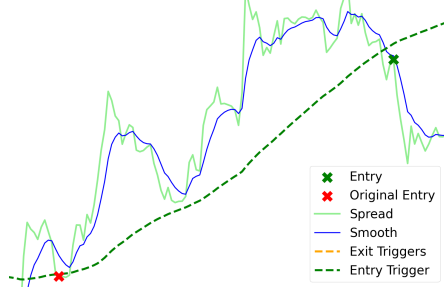


Figure 3.9: Trading strategy entry signals with smoothed thresholds, showing the second crossing for confirmed trade entry.

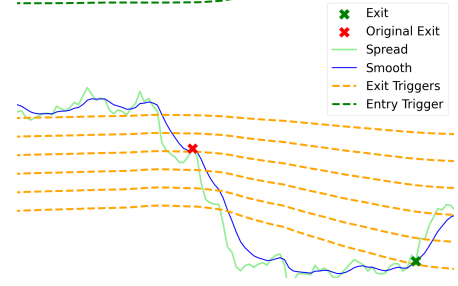


Figure 3.10: Trading strategy exit signals with smoothed thresholds, showing the second crossing for confirmed trade exit.

Following the rationale of Brock et al. [48], this filtering component is critical in ensuring that only significant trends generate trade signals. The filtering methods they employ are designed to ensure that only meaningful trends (rather than transient fluctuations) trigger signals—this is conceptually similar to preventing thresholds from connecting in an overly sensitive manner. However, it's important to note that signal filtering won't fix every case, especially those close to the threshold.

Summary

Following [1, 9], positions are sized using current prices such that the dollar allocation to security A is β times that to B , consistent with the spread definition $S = A - \beta B$. The proposed trading strategy can be summarized by the following set of rules:

$$\left\{ \begin{array}{ll} \text{Open Long} & \rightarrow \quad EMA_8(S_t) > -d_+ > EMA_8(S_{t-1}) \\ \text{Close Long} & \rightarrow \quad EMA_8(S_t) < -c_+ + n_C < EMA_8(S_{t-1}) \quad \vee \quad S_t < -l_+ \\ \text{Open Short} & \rightarrow \quad EMA_8(S_t) < c_- < EMA_8(S_{t-1}) \\ \text{Close Short} & \rightarrow \quad EMA_8(S_t) < c_- - n_C < EMA_8(S_{t-1}) \quad \vee \quad S_t > l_- \end{array} \right. \quad (3.5)$$

where $EMA_8(S_t)$ denotes the 8-day exponential moving average of the spread, d_- , d_+ , c_- , c_+ , l_- , l_+ the respective trading triggers, and n_C is the number of times the $EMA_N(S_t)$ has exceeded the exit threshold (c_- for short or c_+ for long) in intervals of Δc , meaning:

$$n_c(T) = \left\lfloor \frac{|EMA_N(S_t) - c_{+,-}|}{\Delta c} \right\rfloor \quad (3.6)$$

In summary, the threshold-based trading strategy with the simple reversion confirmation test demonstrates robust performance and serves as a baseline for subsequent trading strategy enhancements. However, the trading strategy remains rigid, as it relies on a single trading trigger for opening positions (entry), and two for closing (exit and stop loss), highlighting potential for further methodological refinement. Section 3.3 extends this methodology by incorporating sentiment predictions from extreme gradient boosting models, further refining the trading rules.

3.1.4 Chromosome

Each chromosome encodes a candidate solution as a tuple of nine genes ($G = [G^1, G^2, \dots, G^8]$) representing both pair constituents of the spread, and the respective trading parameters. The structure is designed to balance expressiveness with computational tractability, while enabling a unified optimization of pair formation (PF) and the trading strategy (TS). We propose, similarly to existing literature, to find the optimal thresholds for buying or selling, by having a multiplier to the open and close thresholds. However, the proposed unified optimization will ensure each pair has a specific threshold levels to perform trades. The genetic configuration for the proposed strategy is presented in Table 3.1 and has a theoretical solution space of $\prod_{i=1}^9 |G_i| \approx 3.2 \times 10^{12}$ (2.5×10^6 pairs and 1.2×10^6 trading characteristics).

Table 3.1: Gene configuration for the Simple Unified Pairs Trading System

	Pair	Open	Close	Exit Step	Window
<i>Genes</i>	G_1, G_2	G_3, G_4	G_5, G_6	G_7	G_8
<i>Ranges</i>	$[1, N]$	$[1.5, 2.5]$	$[-0.2, 0.2]$	$[0, 0.3]$	$[200, 400]$
<i>Step</i>	1	0.05	0.05	0.05	25
<i>Space</i>	$\approx 500^2$	20^2	8^2	6	8

- **Stock Pair (G_1, G_2):** Selects the two securities comprising the pair, drawn from the whole universe of stocks. Selects the optimal β factor that minimizes the cointegration residuals, and enforces the constraint $G_1 \neq G_2$, and excludes trades contrary to momentum.
- **Open Multipliers (G_3, G_4):** Scale the standard entry thresholds for long and short trades, allowing for adaptive risk sensitivity, with either conservative or aggressive triggers.
- **Close Multipliers (G_5, G_6):** Scale the exit thresholds for long and short trades, enabling earlier or later closings.
- **Exit Step (G_7):** Scale the Δc factor for delaying trade exits, (same factor for long and short trades), given by equation (3.6).
- **Rolling Window (G_8):** Determines the number of days for computing the moving average and standard deviation of the spread. A shorter rolling window may result in frequent but lower profit trades, while a longer window promotes larger thresholds and potentially higher profit trades, albeit at a reduced frequency. Balances between more reactive (shorter) and more stable (longer) market adaptation strategies.

The trading strategy governed by these genes is defined by the following rules of equation 3.5, replacing $d_+ = G_3$, $d_- = G_4$, $c_+ = G_5$, $c_- = G_6$, $\Delta c = G_7$, and normalizing the spread according to equation 3.3, with rolling window $w = G_8$. It is important to note that the stop-loss thresholds, (l_+ and l_-), are not directly optimized by this initial iteration of the model, as populations tend to converge on highly correlated pairs that rarely trigger the stop-loss condition, rendering the genetic values quasi-random.

It is also possible that over the course of the training period, a pair's open/close triggers may not lead to any executed trades (d_+ , d_- , c_+ and c_-), further rendering the triggers random-unless a particular threshold configuration for that pair happens to result in a backtested trade. In Section 3.2, the system is adjusted to allow for full threshold optimization, via Monte Carlo simulations, allowing the system to have fully dynamic trading triggers.

3.1.5 Objectives

As previously discussed, the core of this unified methodology is the co-maximization of a correlation and a profitability metric. While pair formation primarily drives profitability, trading parameters play a complementary role in refining performance. Based on literature findings and experimentation, the following objective functions are proposed, either for the final implementation, or for benchmarking purposes:

- *ROI*: Maximizing returns is fundamental to any trading strategy, but in the context of pairs trading, this must be balanced against statistical robustness. Incorporating ROI ensures that trading decisions are not just statistically sound but also profitable under realistic execution. This fitness function computes the hypothetical profit over the training period based on the individual's encoded trading parameters. Simultaneous optimization of *ROI* and *Coint* prevents overfitting to spurious patterns and instead favors pairs with both statistical validity and economic value. However, using this metric in the proposed system locks the trigger optimization to that specific pair's training period, which might not be realistic of the profitability of the trade. The randomness inherent in the analysis of a single pair for optimization results in a suboptimal ranking system for individuals, underscoring the need for a more robust profitability metric.
- *Coint*: A correlation metric is imperative for a pairs trading strategy, allowing for reliable pairs to be formed. Cointegration enforces the selection of statistically significant pairs exhibiting strong mean-reverting behavior. Minimizing this metric ensures robust long-term correlation—a necessary condition for effective pairs trading, unlike the Distance metric, which focus on short-term similarity. Naturally accommodates β optimization, enhancing pair relation flexibility. However, by using the optimal β factor, the cointegration minimization becomes mandatory.
- *Mag*: Maximization of the initial magnitude of the spread influences the feasibility of a trade in the early stages of evaluation, as high spread magnitude represents a pair is probably ready to trade, while minimizing it will generate solutions that are less likely to diverge. However, maximization of this objective will have the optimization return the highest magnitude pairs, which isn't always the optimal, although the pair presents a large profit potential (large distance from the mean), while minimizing it will return spreads that aren't ready to trade. Rather than directly optimizing *Mag*, it is proposed that the strategy penalizes solutions where the initial spread $|S| < 1$ or $|S| > 3$ by reducing the *ROI* fitness score by Q . This maintains an open search space and avoids prematurely discarding potentially viable evolution paths, while biasing the population towards ready to trade solutions.

- *MDD*: Minimizing maximum drawdown introduces a risk-aware component to the optimization. Measures the largest peak-to-trough decline of capital for a given individual during the training period. Encourages the evolution of strategies with controlled downside risk, complementing the profit-focused *ROI* objective. Since the *ROI* is maximized in the context of the framework, this metric becomes less useful, as the optimal solutions didn't even get a chance to have any losing trade in the training window, thus rendering this metric useless in some cases. Not selected as a primary objective but used for benchmarking and qualitative evaluation.
- *Vol*: Maximizing spread variance increases the likelihood of encountering trading opportunities. Higher spread volatility, when paired with cointegration, allows the strategy to exploit wider deviations from the mean. The *Vol* of a spread can be calculated as such:

$$Vol = \sqrt{\frac{1}{N-1} \sum_{t=1}^N (S_t - \bar{S})^2} \quad (3.7)$$

Supported by prior research [6, 14], this metric serves as a profitability enhancer when used appropriately. Looking for pairs with high spread variance and strong mean-reverting characteristics can lead to frequent and significant trading opportunities, by allowing for a wider gap to occur whilst maintaining their equilibrium relationship. Like *MDD*, this metric is used for comparative benchmarking purposes.

- *HL*: A lower half-life implies faster mean reversion, translating to more frequent trades and potential profit. Studies such as [6] and [14] indicate that minimizing the half-life improves profitability. However, since half-life is calculated based on historical data, there is a risk of non-persistence during out-of-sample testing. The cointegration is ultimately more useful as a correlation metric, although the half-life could be a candidate for a *MO* optimization, enabling better optimization of reliable pairs. Like *MDD* and *Vol*, this metric is used for comparative benchmarking purposes.

Given the mixed evidence regarding multi-objective (MO) genetic algorithms in this domain, this work adopts a bi-objective (BO) configuration as a practical and effective compromise. This avoids excessive complexity while still capturing the dual goals of valid pair formation and profitability. Summarizing, we have the following objective configuration:

$$O = (ROI [1 - Q \cdot \mathbf{1}_{\{Mag > 3 \vee Mag < 1\}}], coint) \quad (3.8)$$

where $\mathbf{1}_{\{Mag > 3 \vee Mag < 1\}}$ represents 0 if the *Mag* is in the normalized range $[1, 3]$ (no penalization), and 1 otherwise (penalization).

A known limitation of the chosen objectives is their sensitivity to overfitting on the training period. To mitigate this, as elaborated in Section 3.2, we refine the *ROI* metric by incorporating expected profit values over multiple simulated future paths. This Monte Carlo-based adjustment yields a more reliable estimate of long-term profitability, reducing reliance on historical performance alone.

3.1.6 Portfolio Extraction

The construction and management of a portfolio are crucial to the success of any trading strategy. A portfolio is not merely a collection of individual trades, but an integral part of a well-defined, reliable strategy. Through careful management, these optimized solutions can be further refined and combined to create a portfolio that outperforms the broader population.

For pairs trading, this process involves selecting the most promising pairs, balancing risk, and leveraging the strengths of each solution to create a portfolio that not only delivers consistent returns but also mitigates risk, ensuring that the strategy remains robust under various market conditions.

For the Simple Unified Pairs Trading system (*SU*), we propose a simple portfolio management methodology. Figure 3.11 and Table 3.10 summarize the key elements of this approach.

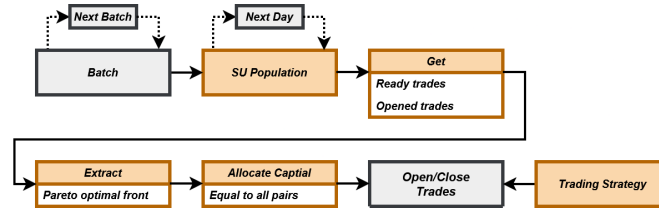


Figure 3.11: Simple Portfolio Management Architecture.

Table 3.2: Simple Portfolio Management Summary.

Batches	Training Windows	Portfolio Management
N_{day} day frequency	$GA : 900$ days	<i>get</i> : ready (current batch) and opened trades
$SU (N_{runs}$ runs)		<i>extract</i> : <i>POF</i>
		<i>allocate</i> : $Total\ Capital / L$ per pair
		<i>monitor</i> : exit triggers for opened trades

- Dynamic Optimization and Monitoring:** The optimization runs every N_{days} trading days, performing N_{runs} iterations to thoroughly explore the search space. While prior studies [6, 5] emphasize long testing windows to accommodate mean-reversion, our shorter re-optimization horizon reduces exposure to temporal de-correlation and stale signals and allows for re-calibration of optimal thresholds. During each cycle, candidate solutions are evaluated at market open. Trades are triggered only if entry conditions are met for a constituent of the final portfolio, and they remain active until either stop-loss, spread reversion, or the end of the full testing period. Furthermore, at a later cycle the optimal threshold configuration might have changed. Also, the pair might become unviable and therefore not show up as a solution of the genetic optimization. Pairs that do not trigger trades within a given cycle are discarded, and expected to be reselected in subsequent optimizations, as long as they maintain strong cointegration and high profitability. It's relevant to note that all indicated values are obtained through backtesting. If the optimal value doesn't vary on backtested data, it is kept fixed. Conversely, if it varies, it is chosen as value that maximizes a defined past period.

- **Portfolio Extraction:** The portfolio comprises both new trading pairs (those that meet the entry criteria during the cycle) and active trades carried over from the current and previous batches. The ROI and correlation values don't need to be re-calculated in the monitoring days, given that the N_{days} period is small enough that it barely varies. The final portfolio (*POF*) is extracted from the filtered set by selecting non-dominated (Pareto-optimal) solutions. Performing this step at after the filtering methods avoids prematurely discarding viable trading candidates. After extracting the pareto front, no duplicate pairs will be present in the population, as same pairs have the same cointegration value, therefore the pareto front selects the highest *ROI* candidate.
- **Capital Allocation:** Capital is then allocated evenly among the selected pairs. Following [1], the allocation per pair is computed as:

$$\text{Allocation per Pair} = \frac{\text{Total Capital}}{L} \quad (3.9)$$

In summary, the portfolio management process first identifies the most promising pairs based on both trading performance and statistical robustness. It then extracts a set of non-dominated solutions, ensuring diversity and reliability, and finally allocates capital evenly across the chosen pairs. This simple methodology for continuously re-calibrating the portfolio enhances the overall returns of the strategy, thereby underpinning the portfolio decisions made in the portfolio management proposed in Section 3.4.

3.1.7 Time Frame

To mitigate the risk of performance distortion due to temporal path dependency, the overall 2000–2024 period is segmented into multiple non-overlapping subperiods for validation. This partitioning addresses a critical shortcoming of single-timeframe analysis: an algorithm's early performance may disproportionately influence its cumulative results. For example, a strategy that performs exceptionally well during a favorable initial phase might inflate subsequent returns, while a robust strategy could appear underperforming if adverse market conditions occur at the outset.

Accordingly, the 24-year span is divided into three distinct periods: 2000–2008, 2008–2016, and 2016–2024. This segmentation facilitates evaluation across varying starting points and economic regimes, thereby reducing the influence of survivorship bias (only seeing data from companies that "survived") and mitigating the impact of fortuitous initial conditions. Additionally, testing the strategy over a long time period permits evaluating whether a pairs trading strategy has maintained its profitability potential over the years. Furthermore, the extensive time frame allows the present work to be directly compared to other literature results, on the same time period. Figure 3.12 illustrates the proposed temporal segmentation. In addition, iterative re-optimization is performed every five days to ensure continuous adaptation to evolving market dynamics. Furthermore, a three-year training period is employed for each segment under the assumption that selecting pairs over an extended time horizon reveals long-lasting associations [6]. This testing setup is consistently applied across all implementations of the unified pairs trading framework.

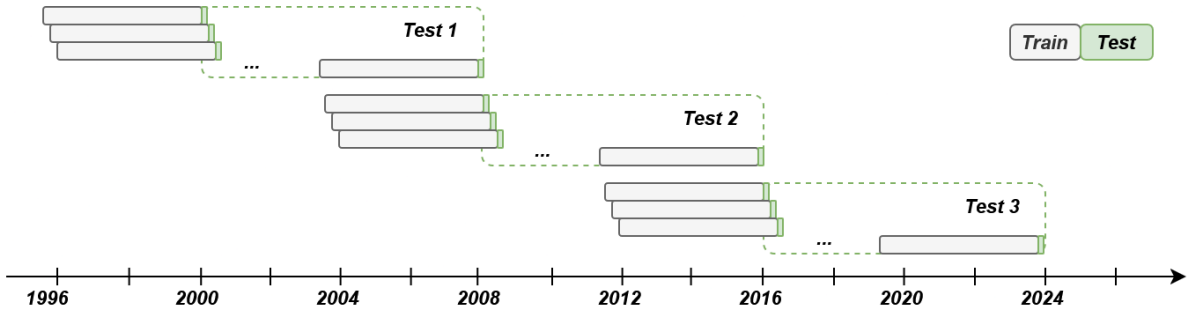


Figure 3.12: Proposed Timeframe.

3.1.8 Data Cleaning

Robust data cleaning is a fundamental pillar for generating reliable signals and ensuring valid trade execution over the entire lifespan of the trading system. Given that the analysis period spans from 2000 to 2024 and focuses on *S&P500* constituents, this framework places particular emphasis on managing missing data, and addressing structural breaks.

It is important to note that, with few exceptions in later sections (e.g., using the stock's percentage in the *S&P500* for Section 3.3 and sector classification for Section 3.4), the proposed methodologies primarily rely on spread prices. Consequently, obtaining the market open price for every stock from the *S&P500* between 2000 and 2024 is essential. Gaps in the time series are addressed using interpolation methods specifically tailored for financial data—such as forward or linear interpolation—to maintain continuity. Additionally, adjustments for stock splits, dividends, and other corporate actions are applied to ensure that the computed spreads and any derived variables accurately reflect the true underlying price dynamics. Even when a stock is delisted or removed from the *S&P500*, its complete historical price series is preserved. Similarly, for stocks that join the *S&P500* during the analysis period, historical data prior to their inclusion is also collected to provide a comprehensive view of their performance. Complementary data such as each stock's sector and its corresponding percentage within the *S&P500* are also extracted to support later analysis.

By rigorously implementing these data cleaning techniques, the framework maintains high data integrity and ensures that the evolutionary optimization process and subsequent trade evaluations are based on accurate, comprehensive, and contextually relevant datasets. This meticulous approach is particularly crucial in a high-frequency trading environment, where even minor data inconsistencies can lead to significant performance degradation.

3.1.9 Benchmark Tests

To validate the effectiveness of the unified system (*SU*), we conduct a series of benchmark tests comparing it against modular alternatives where pair formation (*PF*) and the trading strategy (*TS*) are optimized separately. The primary goal is to assess whether a joint optimization approach yields measurable advantages over a sequential, modular implementation.

The *SU* model combines genetic pair formation (*GPF*) and a genetic thresholds-based trading strategy (*GTS*) into the same optimization. In the benchmark tests, the unified system is compared against two separate implementations. First, the pair formation component is isolated by running two distinct genetic pair formations— P_1 and P_2 —which optimize only the pair-related genes (G_1 , G_2 and G_3) under the same constraints as *SU*. In these tests, the profitability objective is replaced with minimizing half-life and maximizing volatility, rather than directly targeting return on investment. After constructing these pair portfolios, a simple threshold-based trading strategy-*STS*-with the simple fixed thresholds is applied, as described in equation 3.4. Then, the trading strategy is optimized independently on the resulting candidate pairs through separate genetic trading strategy optimizations- T_1 and T_2 . In these runs, only the trading-related genes (G_4 , G_5 , G_6 , G_7 , G_8 and G_9) are optimized, where the objective is to maximize returns while minimizing the maximum drawdown. Finally, the simple portfolio management (*SP*) process is applied to integrate all final portfolios. Figure 3.13 illustrates the sequential implementation used for benchmarking, while Table 3.3 summarizes the configurations and objectives for each optimization. All tests are executed under identical market conditions from 2000 to 2024 with re-optimization every five days, ensuring a consistent comparison framework.

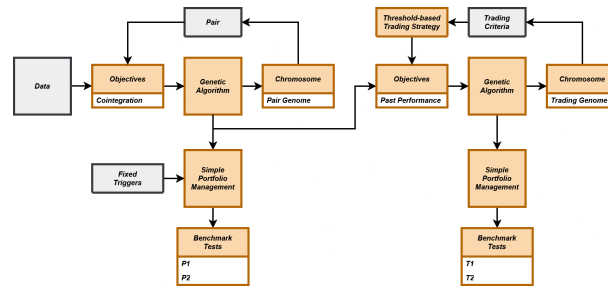


Figure 3.13: Simple Unified Pairs Trading System configuration applied in series, where the first optimization focuses on *PF* and the second on the *TS*.

Table 3.3: Benchmark implementation structure.

Name	Strategy	Objective 1	Objective 2
<i>SU</i>	<i>GPF</i> + <i>GTS</i> + <i>SP</i>	<i>min. Coint</i>	<i>max. ROI</i>
P_1	<i>GPF</i> + <i>STS</i> + <i>SP</i>	<i>min. Coint</i>	<i>min. HL</i>
P_2	<i>GPF</i> + <i>STS</i> + <i>SP</i>	<i>min. Coint</i>	<i>max. Vol</i>
T_1	P_1 + <i>GTS</i> + <i>SP</i>	<i>max. ROI</i>	<i>min. MDD</i>
T_2	P_2 + <i>GTS</i> + <i>SP</i>	<i>max. ROI</i>	<i>min. MDD</i>

This benchmark evaluates the unified system (*SU*), which simultaneously optimizes portfolio formation (*PF*) and trading strategy (*TS*) using a single bi-objective fitness function, against a modular sequential pipeline in which *PF* and *TS* are optimized separately—each with its own bi-objective function (four objectives in total). Both frameworks are tested under identical market scenarios. If *SU*, with only two objectives, consistently outperforms the sequential approach with four total objectives—delivering higher returns, lower volatility, and smaller drawdowns—this provides compelling evidence of the efficiency and efficacy of joint optimization.

3.1.10 Discussion

Traditional threshold-based pairs trading suffers from significant limitations: (i) the suboptimal reliance on historical data for parameter tuning; (ii) rigid decision-making post-entry and mandatory trading on correlation-selected pairs; and (iii) the simplistic aggregation of optimization solutions into trading portfolios. The proposed *SU* system introduces foundational improvements through continuous spread monitoring and constant re-optimization. This enables more dynamic decisions (premature closure, delayed exit, conditional entry), automatically excludes decorrelated pairs, and incorporates co-optimized elements like search space rules and magnitude-based *ROI* penalization. While *SU* partially addresses the core problems by demonstrating the value of adaptability and multi-factorial rules, its major reliance on past performance for fine-tuning and pair selection remains a suboptimal constraint. The following sections present major advancements over *SU*, each directly addressing a core limitation:

- (i) Section 3.2: Replaces backtest-dependent tuning with a real-time expected profitability metric derived from Monte Carlo spread simulations. This enables dynamic trade management (premature closure of nonviable trades, selective opening of high-probability trades) and evolves stable, profitable pairs concurrently.
- (ii) Section 3.3: Overcomes simplistic reversal triggers by integrating an XGBoost-based short-term sentiment analysis model. This multi-step spread analysis provides superior predictive power for wiser trading decisions and guides genetic optimization towards pairs scoring well across an ensemble of predictive models.
- (iii) Section 3.4: Moves beyond simple portfolio management by implementing continuous updates of open and potential trades. It selects an optimal portfolio under constraints designed to minimize overall variance while maximizing potential profitability, ensuring holistic risk-return management.

3.2 Expected Profits via Monte Carlo Simulations

Traditional optimization methods that rely solely on historical returns are vulnerable to path dependency and overfitting, potentially overlooking pairs with genuine long-term potential. To address these issues, this case study incorporates an Expected Profitability (*EP*) metric, derived from Monte Carlo simulations, thereby estimating forward-looking profit expectations. By simulating plausible future trajectories of mean-reverting spread models, the trading strategy can be evaluated on these paths, thus deriving the potential of a certain pair given certain strategy criteria. The *EP* metric inherently identifies pairs and strategies with high prospective profitability, even when historical performance is misleading. Additionally, this methodology allows for stress-testing key trading parameters—entry, exit, and stop-loss levels—across diverse market scenarios, thereby enhancing overall strategy robustness. The simulation methodology for extracting and calibrating the *EP* metric is illustrated in Figure 3.14.

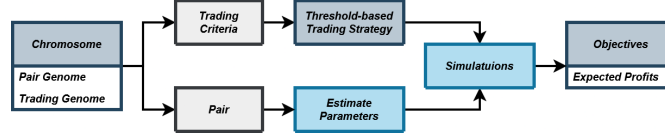


Figure 3.14: Monte Carlo Architecture.

The proposed methodology employs Monte Carlo simulations of the Ornstein-Uhlenbeck Variance Gamma ($OU-VG$) model to generate stochastic spread trajectories, upon which predefined trading strategies are executed. The modeling parameters are estimated, and the Expected Profit (EP) metric, computed by averaging trading outcomes across all simulated paths, quantifies the strategy's forward-looking profitability (Section 3.2.1). This EP metric is directly integrated into the genetic algorithm's fitness function, while individual genomes are augmented with stop-loss optimization parameters, and the portfolio management rules are updated, enabling dynamic risk management (Section 3.2.2). Comparative analysis against prior strategy iterations demonstrates the broader applicability of the EP metric across diverse market regimes (Section 4.4), validating its robustness in non-stationary environments.

3.2.1 Methodology

The methodology consist in generating multiple mean-reverting simulations for a pair, and evaluate the performance of a trading strategy guided by genes G . By synthesizing regime-switching scenarios through Monte Carlo simulations, the methodology mitigates overfitting inherent in historical backtests, guiding the optimization towards optimal pair-strategy combinations, based on expected profitability, also taking into account discounting of future profits, holding periods and uncertainty.

Spread Simulations

The proposed methodology employs Monte Carlo simulations based on an Ornstein-Uhlenbeck Variance Gamma ($OU-VG$) process [29, 37], a mean-reverting stochastic model adept at capturing heavy-tailed spread dynamics. The simulation function, denoted as $LDOUP, X \sim OU-VG(m, \Delta t, T, \lambda, b, \mu, \sigma^2, \eta, X_0)$, generates m stochastic paths over a forecast horizon T (in days) using a time step Δt .

The process initiates at X_0 , the observed spread value at the simulation's inception. Contrary to traditional approaches that initialize simulations at a neutral spread value $X_0 = 0$ for threshold optimization [41], our framework begins simulations at the current observed spread value, enabling dynamic parameter calibration tailored to prevailing market dynamics [29]. Dynamic initialization at X_0 ensures parameter relevance to current market states, addressing non-stationarity challenges [40]. The simulation evolves according to the discretized dynamics described in the Background Chapter (parameters $\lambda, b, \mu, \sigma^2, \eta$), by equations (2.9) and (2.11). Key simulation parameters and their roles are summarized in Table 3.4. The prediction horizon is kept at $T = 60$, as after 60 days most trades have concluded. It's relevant to note that the parameters $m, \Delta t, \mu, \lambda, \sigma^2$ and b are analyzed and estimated through back-testing.

Table 3.4: Description of Simulation Parameters.

Variable	Description	Role
m	Number of simulations	Determines statistical precision
Δt	Time step (days)	Temporal resolution of simulations
T	Forecast horizon (days)	Defines the simulation period
λ	Reversion speed	Controls mean-reversion rate
b	Jump intensity	Scales jump frequency in the VG component
μ	Equilibrium level	Long-term mean of the spread
σ^2	Variance	Volatility of the diffusion component
η	Jump scaling factor	Adjusts magnitude of discontinuous jumps
X_0	Initial spread value	Starting point for simulations

Gene-Driven Spread Generation

For a gene configuration G , encoding a trading pair $G_1 \rightarrow G_2$, the spread $S(G)$ is constructed as:

$$S(G) = S^{G_1} - \beta S^{G_2}, \quad (3.10)$$

where S^{G_1} and S^{G_2} denote the price series of the constituent assets, and β is the optimal hedge ratio for the pair.

The simulated T -length trajectories of the spread, denoted by $S_i^{\text{sim}}(G)$ for $i = 1, 2, \dots, m$, are generated from a discretized version of the underlying mean-reverting dynamics described by (2.9). These parameters (λ , μ and σ^2) are estimated by fitting the model to historical data (using linear regression on Equations (2.17), (2.19) and (2.18)), with the initial condition X_0 set equal to the last observed value of the actual spread $S(G)$. It's important to note that these model dynamics, as well as the optimal b value to use in the simulation, require constant recalibration, as to maintain the simulations reliable.

Risk adjusted Expected Profit Calculation

The genome guided threshold-based trading strategy, denoted as $\mathcal{TS}(S, G)$, is applied to each simulated spread trajectory $S_i^{\text{sim}}(G)$, as to obtain its profitability on the generated path. For a given gene configuration G , the discounted profit P is computed over m simulations, with each individual trade profit P_i defined as:

$$P_i = e^{-r\tau_i} \times \mathcal{TS}\left(S_i^{\text{sim}}(G), G\right), \quad (3.11)$$

- $e^{-r\tau_i}$ is the discount factor based on the exit time τ_i for the i -th simulation and the discount rate r . This term captures both the time value of money and the risk premium demanded for longer holding periods. Consequently, profits realized after an extended duration are discounted more heavily, thereby penalizing strategies that rely on extended waiting times for trade completion.

- $TS(S_i^{\text{sim}}(G), G)$ computes the returns of the pre-defined strategy on the simulated spread.

The expected discounted profit $\mathbb{E}[P]$ and the variance $Var[P]$, which quantifies the uncertainty across simulated paths, are estimated as:

$$\mathbb{E}[P] = \frac{1}{m} \sum_{i=1}^m P_i, \quad (3.12)$$

$$Var[P] = \frac{1}{m-1} \sum_{i=1}^m (P_i - \mathbb{E}[P])^2. \quad (3.13)$$

Here, $E[P]$ represents the average profitability of the gene configuration G on the simulated spread paths, integrating the mean-reverting dynamics and the inherent stochastic market noise, while $Var[P]$ captures the dispersion of profits, serving as a measure of risk.

To incorporate risk into the optimization process, we define the value function V (based on the risk-aversion parameter $\gamma > 0$) as a risk-adjusted performance metric:

$$V = E[P] - \gamma Var[P], \quad (3.14)$$

For convenience, the value function is referred to as the expected profitability (EP) of the pair. This risk-adjusted metric, which incorporates the discounting of future profits and strategy variance, provides a comprehensive basis for comparing different gene configurations G . The methodology implicitly penalizes trading strategies with longer holding periods—due to increased discounting—and those with higher profit uncertainty, thereby naturally guiding the genetic algorithm toward optimizing the entry and exit thresholds to maximize the present-value performance. Besides specific pair fine-tuning, this metric serves as a better ranking system than backtesting a strategy on a specific pair.

[41] identifies optimal threshold values that maximize the value function for the conventional threshold-based trading strategy through grid search. However, this becomes computationally intractable for the proposed adaptive dynamic trading strategy. Specifically, grid search scales exponentially with the number of parameters, requiring exhaustive evaluation of approximately $\prod_{i=3}^9 |G_i| \approx 3.2 \times 10^6$ potential strategies across m simulations for each candidate pair, $\prod_{i=1}^2 |G_i| \approx 3.2 \times 10^6$. This combinatorial explosion renders grid search impractical for real-world applications of the proposed framework, where computational efficiency is critical. Thus, it's beneficial as well that the genetic optimization is applied directly to the trading strategy, alongside the pair formation.

3.2.2 Adapting the Unified System

In this case study, the baseline unified pairs trading system is enhanced by incorporating the Expected Profitability (EP) metric into the genetic optimization process. Derived from Monte Carlo simulations of $OU-VG$ processes, EP provides a forward-looking, probabilistic measure of profitability, moving beyond reliance on historical returns. Integrated into the genetic algorithm's fitness function, EP guides the optimization toward strategies that are not only historically robust but also resilient to future market fluctuations. This metric becomes central to the implementation as is used efficiently in the unified system, as alongside cointegration minimization, concurrently promotes pairs that are stable, and offer

high profitability, thus being used for both pair selection, and informing the optimal trading strategy criteria. Figure 3.15 provides an overview of the updated unified pairs trading system (*MU*).

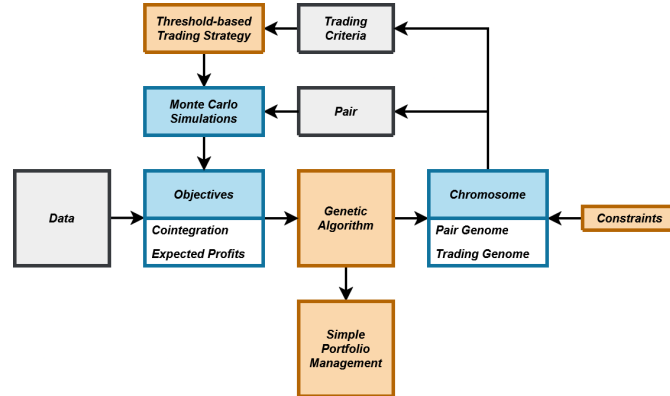


Figure 3.15: Monte Carlo Unified Pairs Trading System Architecture.

The integration of *EP* also drives structural improvements within the system. The chromosome is expanded to include a stop-loss gene, allowing direct optimization of stop-loss thresholds. Objective functions are revised to prioritize expected future profitability over historical returns. In portfolio management, filtering mechanisms now prioritize pairs with high positive *EP*, and positions are systematically closed when *EP* declines. These modifications collectively yield a more adaptive, forward-looking trading system, reducing overfitting and enhancing resilience to changing market conditions.

The revised chromosome now comprises ten genes. The additional gene encodes the stop-loss threshold, which is now subject to optimization through simulated trading outcomes, reinforcing the robustness of the optimization process. This expansion increases the theoretical solution space to

Chromosome

The revised chromosome now comprises ten genes ($G = [G^1, G^2, \dots, G^{10}]$), as illustrated in Table 3.5. The additional gene encodes the stop-loss threshold, which is now subject to optimization through simulated trading outcomes, reinforcing the robustness of the optimization process. The updated trading triggers follow equation 3.5, with $l_+ = G_9$ and $l_- = G_{10}$. This expansion increases the theoretical solution space to $\prod_{i=1}^1 |G_i| \approx 2.2 \times 10^{16}$ (1.8×10^8 pairs and 1.2×10^8 for the trading characteristics).

Table 3.5: Gene configuration for the Monte Carlo Unified Pairs Trading System.

	Pair	Open	Close	Exit Step	Window	Stop Loss
<i>Genes</i>	G_1, G_2	$G_3 - G_4$	$G_5 - G_6$	G_7	G_8	$G_9 - G_{10}$
<i>Ranges</i>	$[1, N]$	$[1.5, 2.5]$	$[-0.2, 0.2]$	$[0, 0.3]$	$[200, 400]$	$[2.5, 3.5]$
<i>Step</i>	1	0.05	0.05	0.05	25	0.1
<i>Space</i>	$\approx 500^2$	20^2	8^2	6	8	10^2

Trading Strategy

The updated trading strategy is fully governed by the optimization, with trading rules recalibrated to incorporate the stop loss trigger, governed by the genetic encoding of an optimization solution. Furthermore, $EP < 0$ also becomes an exit trigger, and an entry constrain.

The inclusion of EP in the adaptable system enables truly adaptive, day-by-day optimization of trading thresholds—entry, exit, and stop-loss—by evaluating current market conditions alongside simulated future paths. Rather than relying on static levels, the genetic algorithm uses EP to adjust thresholds dynamically: it favors immediate trades when simulations predict high returns and adopts more conservative levels when caution is warranted. This approach not only overcomes the rigidity of traditional static-threshold strategies but also stress-tests key parameters across diverse scenarios, guiding the algorithm to converge on pairs with consistently strong expected profitability.

Objective Functions

The key modification in the objective functions is the replacement of ROI with EP . Thus, we have the fitness objective $O = (EP, Coint)$. This change directs the genetic algorithm to prioritize pairs exhibiting high expected future profitability while maintaining sound cointegration characteristics. Optimizing for EP naturally favors pairs with sufficiently wide initial spreads, thereby filtering out candidates unlikely to generate tradable opportunities. As a consequence, the previously proposed profitability metric penalizations for low Mag pairs are removed, as it stops being useful.

Its relevant to note that, by incorporating a discount rate into the EP metric, the optimization inherently penalizes pairs with prolonged half-lives, without directly using it as an objective. Spreads with longer HL experience greater discounting, resulting in lower effective profits. Thus, the genetic optimization, naturally favors configurations where the half-life of the spread is minimized.

Portfolio Management

This implementation further underscores the necessity of re-optimizing the strategy every 5 days. Evolving spread dynamics require updated Monte Carlo simulations to reflect current market conditions, ensuring the derivation of optimal thresholds that adapt to the pair's latest behavior. The portfolio management process is adapted to integrate the EP metric.

In this configuration, once a position is initiated, the EP is continuously updated, and trades are closed immediately when the metric becomes negative, ensuring risk control. Furthermore, the estimator parameters are recalculated every day, on the last 60 days to capture recent market dynamics, as suggested by [42]. Figure 3.16 and Table 3.21 provides an overview of the revised portfolio management strategy.

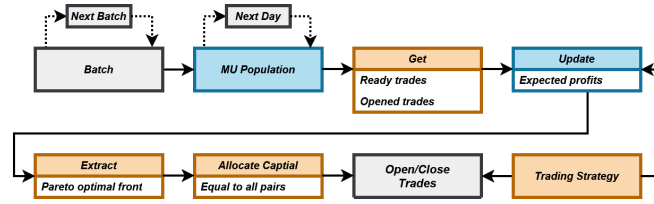


Figure 3.16: Monte Carlo Simple Portfolio Management Architecture.

Table 3.6: Monte Carlo Simple Portfolio Management Summary.

Batches	Training Windows	Portfolio Management
5 day frequency	<i>GA</i> : 900 days	<i>get</i> : ready (current batch) and opened trades
<i>MU</i> (200 runs)	<i>MC</i> : 60 days	<i>update</i> : <i>EP</i> values
		<i>extract</i> : <i>POF</i>
		<i>allocate</i> : <i>Total Capital</i> / <i>L</i> per pair
		<i>monitor</i> : exit triggers for opened trades

3.2.3 Discussion

The Monte Carlo implementation comprehensively resolves the three core limitations persistent in both traditional and SU systems. First, it eliminates historical performance dependency by replacing backtest tuning with real-time Expected Profitability (EP) metrics from Monte Carlo simulations, enabling forward-looking threshold optimization for imminent trades rather than past periods. Second, it overcomes decision rigidity through EP-guided early closure of nonviable positions and stop-loss integration, creating truly adaptive multi-exit criteria. Finally, it transcends SU's suboptimal solution aggregation by establishing EP as a universal ranking standard, enabling dynamic portfolio rebalancing based on forward-looking viability rather than historical fitness.

3.3 Sentiment-based Trading via Extreme Gradient Boosting

Traditional threshold-based pairs trading strategies, while simple and interpretable, suffer from rigidity. These systems open positions purely based on spread thresholds, without accounting for market context or short-term dynamics. As highlighted previously, this can lead to suboptimal trades—particularly in volatile or non-stationary environments.

To address these limitations, we introduce an enhanced machine learning-based approach: the Extreme Gradient Boosting based Trading Strategy (*XTS*). This method incorporates a predictive layer into the trading decision process, leveraging *XGBoost* to forecast short-term spread trends. The *XGB* model acts as a gatekeeper—predicting whether the spread is likely to expand, contract, or remain stable over the coming days, thereby refining trade timing and filtering out unpromising setups [20]. The python library used is *xgboost*. Figure 3.17 illustrates the architecture for modeling spread trends.

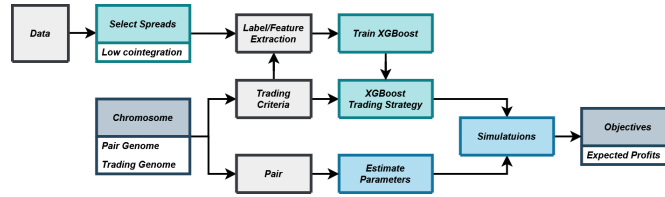


Figure 3.17: XGBoost Architecture.

A dedicated feature engineering pipeline captures relevant statistical and market features (Section 3.3.1), which are used to train multiple XGBoost classifiers. These models inform the updated trading rules by indicating the most probable spread trajectory, thus the predictive layer is seamlessly integrated into the unified optimization framework (Section 3.3.2), allowing the genetic algorithm to assess not only pair selection and threshold settings, but also the inclusion and performance of a trading strategy based on different XGB models. Through empirical evaluation (Section 4.5), we explore whether this intelligent gating system improves decision-making and trading performance relative to the previous implementations.

3.3.1 Methodology

Recurring failure modes of traditional pairs trading strategies involves either slow or no mean reversion. When the spread deviates significantly from the mean but reverts gradually, the moving average adjusts upward, preserving a high threshold and causing the delayed exit decisions to be unprofitable. Conversely, a spread that indefinitely reverts ends up triggering the stop loss threshold, resulting in a loss. These behaviors are illustrated in Figure 3.18 and Figure 3.19, highlighting the need for a more adaptive approach. As observed, the spread only becomes okay to start trading, as the trading triggers aren't stable. To address this, we propose a predictive model designed to anticipate short-term threshold behavior, enabling the identification of conditions where the threshold is likely to stabilize or reverse. By forecasting these movements, the model enhances trade timing and risk management through more informed entry and exit decisions. A strong upward trend may signal poor entry timing when shorting, while a reversal implies a very favorable entry. Identifying the emerging trend early allows for preemptive exit decisions, potentially minimizing losses.

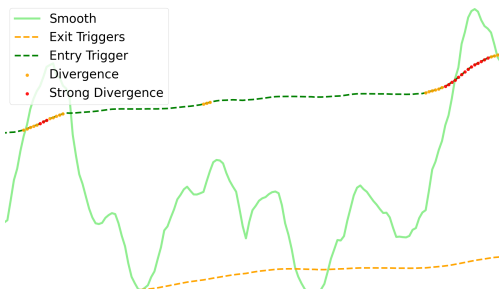


Figure 3.18: Labeled spread illustrating stability, divergence and and divergence.

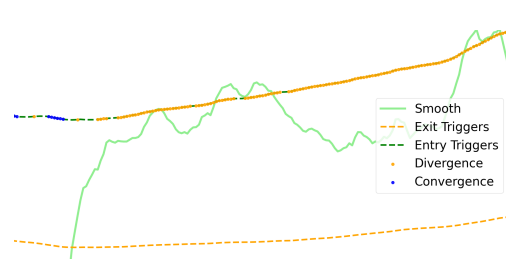


Figure 3.19: Labeled spread illustrating stability, divergence and convergence.

This approach improves upon traditional mean-reversion assumptions by explicitly modeling thresh-

old dynamics. We focus on predicting the threshold itself—rather than other metrics—because abrupt shifts in this trigger often signal significant movements in the spread. In contrast, the open threshold is a well-behaved signal, and accurately anticipating its movements, even slightly in advance, provides a substantial edge for pairs trading strategies. Furthermore, by modeling the threshold dynamics, the genetic algorithm design implicitly allows for the formation of various models, each specifically trained on the specific threshold dynamics.

Since the threshold dynamics vary for different spreads and their respective trading triggers, splitting the sentiment evaluation into various models achieves a better generalization of specific spread-threshold dynamics. The predictive machine learning algorithm used will be XGBoost, chosen for its highly efficient, parallelizable implementation of gradient boosting with built-in regularization to curb overfitting, native handling of missing values and outliers, ability to capture complex non-linear feature interactions, and straightforward extraction of feature-importance metrics—all of which have proven invaluable in accurately modeling the nuanced dynamics of financial time-series thresholds [33, 5].

Another goal of this implementation is to integrate all three methodologies—*GA*, *MC*, and *XGB*—enabling the genetic optimization process to accurately select the XGB model that performs best across the MC-generated spread trajectories. This integration facilitates rigorous stress-testing of models under various potential market scenarios, allowing for precise fine-tuning of the system's parameters.

Data

To maintain temporal consistency and mitigate overfitting, XGBoost architectures are trained on rolling two-year data windows. This approach ensures model inputs reflect the most recent market dynamics while retaining sufficient historical context for robust generalization. The training process further aligns with the genetic algorithm's optimization framework: all pairs incorporate the statistically optimal beta coefficients for their respective spreads. Consequently, the XGBoost models are explicitly trained on these optimized beta-adjusted spreads, ensuring congruence between the evolutionary selection mechanism and the predictive model's input structure. Additionally, the *GA*'s rapid convergence to populations dominated by low cointegration residuals ($coint < 0.2$) informs the XGBoost training dataset. By restricting training to pairs exhibiting stable mean-reverting properties, the models prioritize generalizable patterns over transient noise, enhancing predictive robustness across evolving market regimes.

Labels

To quantify the short-term trend of the open threshold, d_t , we define a multi-class labeling scheme based on the derivative of the open threshold's following day, split into five states, defined by $k = (k_{++}, k_+, k_-, k_{--})$:

$$y_t = \begin{cases} 0 & \rightarrow d'_{t+1} > k_{++} & \text{(Strong Divergence)} \\ 1 & \rightarrow d'_{t+1} \in]k_+, k_{++}] & \text{(Mild Divergence)} \\ 2 & \rightarrow d'_{t+1} \in [k_-, k_+] & \text{(Stable)} \\ 3 & \rightarrow d'_{t+1} \in [k_{--}, k_-[& \text{(Mild Convergence)} \\ 4 & \rightarrow d'_{t+1} < k_{--} & \text{(Strong Convergence)} \end{cases} \quad (3.15)$$

This directional classification overcomes limitations of absolute-value thresholding, especially with asymmetrical spread behavior. It enables the XGBoost model to better distinguish favorable from risky trading conditions at varying confidence levels.

Selecting the appropriate prediction horizon involves a crucial trade-off between decision-making utility and model accuracy. Although longer prediction windows can support more strategic, forward-looking decisions, model performance typically deteriorates over extended horizons due to increased uncertainty and reduced generalization capability. To mitigate variance-related errors associated with long-term forecasts, predictions are therefore focused on the next-day derivative, balancing responsiveness with reliability.

Features

XGBoost's flexibility allows the incorporation of a diverse set of features capturing both local and global dynamics. The full list is presented in Table 3.7.

Table 3.7: Extreme Gradient Boosting Input Features.

Technical Indicators	Price Dynamics	Spread Context	Market Context	Temporal Memory
$SMA_{5, 10, 25, 50}$	$\mu_{G_8} + G_3\sigma_{G_8}$	i	$S\&P500$	$labels_{lag1, lag3}$
$EMA_{10,20}$	$\mu_{G_8} - G_4\sigma_{G_8}$	j	$SMA_{50}(S\&P500)$	$\mu_{lag5, lag10}$
$MACD$	μ'_{G_8}	$i\%_{S\&P500}$		$\sigma_{lag5, lag10}$
RSI	σ'_{G_8}	$j\%_{S\&P500}$		

Here, the spread value is governed by gene configuration G : μ_{G_8} and σ_{G_8} are the rolling mean and standard deviation, computed using a window governed by gene G_8 . $i\%_{S\&P500}$ and $j\%_{S\&P500}$ denote the respective weights of each stock in the $S\&P500$ index, and i and j are the respective stock identifiers.

To train an XGBoost model on multiple stock spreads, it's essential to include identifier features that distinguish between stock pairs. When stacking the data by placing one stock's data after another, discontinuities arise between stocks, making it unclear to the model where one stock ends and the next begins. Using categorical indicators ensures the model can learn spread-specific patterns and avoid treating all inputs as a continuous, homogeneous process, which would obscure meaningful dynamics and substantially degrade performance. Furthermore, including macro-level features like index membership and index values allows the model to capture broader market movements and their influence on the spread. Feature importance analysis revealed index correlations as key predictors. Avellaneda et

al. [38] highlights the importance of sector-adjusted index weights. For example, a stock's percentage in the S&P 500 acts as a proxy for systemic risk and liquidity, which are critical for pairs trading strategies. Thus, index membership weights help contextualize market-wide dynamics. It's important to note that, due to the use of simulated price paths, we exclude fundamental indicators (e.g. Volume), which are not reliably reproducible in a simulation context. Moreover, incorporating the raw spread as an input—while beneficial for short-term predictive accuracy—was found to degrade overall system performance by introducing instability into the simulation-optimization loop. This is explained by a possible short-term low reliability moment of the simulations, that would trick the model—thus obtaining an unreliable *EP* metric. Keeping the features as rolling features increases the model selection potential of the unified pairs trading framework, while lagged features provide temporal structure.

Parameters

The model's parameters are optimized through walk-forward validation to prevent look-ahead bias, and class weights were adjusted to address imbalance. Training was performed with 3-fold temporal cross-validation. To prevent overfitting, an early stopping mechanism was implemented, halting training if the validation loss failed to improve over five consecutive epochs. Hyperparameter tuning is performed on each model, using a grid search approach.

3.3.2 Adapting the Unified System

The unified pairs trading architecture seamlessly integrates machine learning via XGBoost into its evolutionary framework. Unlike traditional approaches that optimize a single model (such as [X]), this system emphasizes label rule diversity—selecting from a set of XGBoost models, each trained to recognize different threshold dynamics. This variation aims to account for the non-stationary nature of financial time series and pair-specific spread behaviors. Incorporating multiple models introduces significant computational overhead; however, it enables the genetic algorithm to identify the most reliable XGBoost variant for each trading pair, allowing for better problem generalization. This is particularly valuable under Monte Carlo-generated price paths, which simulate a wide range of market conditions. By embedding the model selection within the optimization process, the system achieves robust generalization across both seen and unseen data. Figure 3.20 illustrated the updated system.

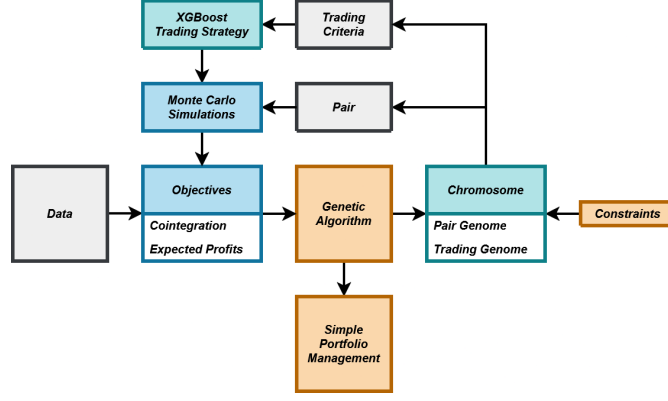


Figure 3.20: XGBoost-Monte Carlo Unified Pairs Trading System Architecture.

The selected XGBoost-enhanced strategy operates within the unified optimization loop, which combines expected profit (EP) evaluation obtained through the modified trading strategy and *coint* minimization. The expected profit metric incorporates model sentiment, thus anticipating spread widening leads to a rapid decline in projected EP , prompting premature trade closure to minimize loss. The addition of the predictive module prompts modifications in the proposed methodology, specifically in the trading rules and portfolio management.

Chromosome

The revised chromosome now comprises fourteen genes ($G = [G^1, G^2, \dots, G^{14}]$), as illustrated in Table 3.8. The additional genes encode the XGBoost sentiment prediction label thresholds (k), which are subject to optimization through simulated trading outcomes, reinforcing the robustness of the optimization process. The updated trading triggers follow equation 3.15, with $G_{11} = k_{++} - k_+$, $G_{12} = k_+$, $G_{13} = -k_-$ and $G_{14} = k_- - k_{--}$. This expansion increases the theoretical solution space to $\prod_{i=1}^1 1|G_i| \approx 2.9 \times 10^{19}$ (1.8×10^8 pairs and 1.6×10^{11} for the trading characteristics).

Table 3.8: Gene configuration for the XGBoost-Monte Carlo Unified Pairs Trading System

	Pair	Open	Close	Step	Window	Stop	Prediction
<i>Genes</i>	$G_1 - G_2$	$G_3 - G_4$	$G_5 - G_6$	G_7	G_8	$G_9 - G_{10}$	$G_{11} - G_{14}$
<i>Ranges</i>	$[1, \approx 500]$	$[1.5, 2.5]$	$[-0.2, 0.2]$	$[0, 0.3]$	$[200, 400]$	$[2.5, 3.5]$	$[0.05, 0.3]$
<i>Step</i>	1	0.05	0.05	0.05	25	0.1	0.05
<i>Space</i>	$\approx 500^2$	20^2	8^2	6	8	10^2	6^4

Trading Strategy

The proposed XGBoost-enhanced trading strategy incorporates model-driven sentiment to augment traditional threshold-based decision-making. Rather than relying solely on static conditions, the trading logic is now dynamically guided by the predicted behavior of the open threshold, allowing for more adaptive and informed trade management. The XGBoost model influences the strategy via three key rules, summarized in Table 3.9, additional to the trading trigger thresholds:

1. **Entry Confirmation:** A long position is initiated only if the spread crosses the entry threshold and the model forecasts a favorable regime—defined as stable or reverting threshold behavior (labels 0, 1, 2 for long positions; or 2, 3, 4 for short positions).
2. **Exit on Risk:** If an open position coincides with a prediction of strong divergence, the position is closed immediately to mitigate risk or lock in gains (label 4 for long positions; or 0 for short positions).
3. **Hold for Profit:** If the spread crosses the exit threshold and the model anticipates continued divergence, the position is held longer to capitalize on the momentum (label 0 for long positions; or 4 for short positions).

Table 3.9: Actions resulting from current state and label prediction (label 2 represents usual threshold-based trading strategy actions).

State		Label 0	Label 1	Label 2	Label 3	Label 4
Entry	Long	Open	Open	Open	-	-
	Short	-	-	Open	Open	Open
Hold	Long	Hold	Hold	Hold	Hold	Close
	Short	Close	Hold	Hold	Hold	Hold
Exit	Long	Hold	Close	Close	Close	Close
	Short	Close	Close	Close	Close	Hold

The updated strategy responds in real time to shifting spread dynamics, avoiding premature entries and mitigating losses during unstable regimes.

Additionally, integrating a short-term sentiment prediction (next day) enhances the system's ability to identify optimal entry and exit points, complementing the long-term (60-day) forecasts generated through Monte Carlo simulations. It is acceptable—and even expected—for short-term sentiment and long-term profitability signals to diverge, as they capture different market dynamics. By combining both perspectives within a unified framework, the system enables more informed decision-making, improving the timing of entries, exits, or trade delays. By having the *EP* metric directly linked to the models performance of possible spreads, the model that works best for the pair is selected, inherently selecting a model that fits the pair well, achieving a greater performance than applying the same fixed model.

Objectives

It's relevant to note that the integration of sentiment-based predictions into the trading strategy directly influences the optimization dynamics of the expected profitability (*EP*) metric. This occurs as the *EP* metric inherently prioritizes trading pairs that exhibit predictability under the XGBoost model's forecasts. Specifically, the fittest individuals in the genetic algorithm's population correspond to trades where the XGBoost model demonstrates consistent predictive accuracy. This relationship arises from the *EP*'s

value function: when the model performs poorly in simulations—yielding unstable or divergent predictions across trials—the simulated returns lower and the variance increases, thereby penalizing the *EP* objective. Consequently, the optimization process naturally selects pairs that have well-behaved simulations, that are easily predictable by the XGBoost framework. These pairs exhibit lower variance in their simulated outcomes, reflecting both the model’s ability to generalize and the underlying stability of the sentiment-driven price relationships. By favoring such pairs, the strategy not only reduces portfolio variance but also enhances decision-making robustness. The latter is achieved because the selected solutions remain viable not only under current sentiment conditions but also under plausible future sentiment regimes, as the XGBoost model’s predictive consistency implicitly accounts for multi-horizon dependencies.

Portfolio Management

This integration showcases how predictive modeling can be used not only for trade execution but also for strategy validation and adaptive portfolio allocation under uncertainty. Model retraining is performed every batch, using the last two years of data. As a result, the models hyperparameters are determined as to maximize the f1-score, allowing for meaningful sentiment-predictions over the last year’s data. Furthermore, the predictive model is used for determining spread divergence on opened positions. The modifications are illustrated and summarized in Figure 3.21 and Table 3.10

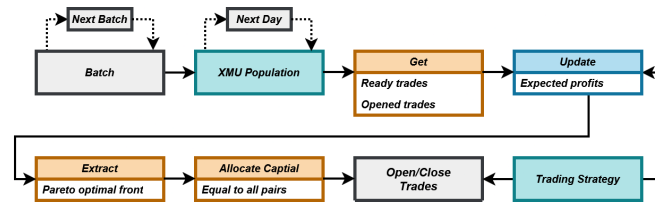


Figure 3.21: XGBoost-Monte Carlo Simple Portfolio Management Architecture.

Table 3.10: XGBoost-Monte Carlo Simple Portfolio Management Summary.

Batches	Training Windows	Portfolio Management
5 day frequency	<i>GA</i> : 900 days	<i>get</i> : ready (current batch) and opened trades
<i>XMU</i> (200 runs)	<i>MC</i> : 45 days	<i>update</i> : <i>EP</i> values
	<i>XGB</i> : 600 days	<i>extract</i> : <i>POF</i>
	(<i>coint</i> < 0.2 pairs)	<i>allocate</i> : <i>Total Capital/L</i> per pair
		<i>monitor</i> : exit triggers for opened trades

3.3.3 Discussion

The *XGB* integration fundamentally overcomes the core rigidity flaw persistent in traditional, SU, and MC systems by replacing static thresholds with context-aware forecasting. Unlike prior approaches that mandated trading on correlated pairs, the five-regime classification system (strong/mild divergence/convergence,

stability) enables anticipatory threshold adjustments. This transforms trading from reactive breakpoint detection to strategic forecasting, eliminating false breakouts that plagued threshold-dependent systems.

Crucially, the genetic optimization intrinsically selects models prioritizing stability/divergence prediction over reversion signals—validating our profit-driven design. The system dynamically updates optimal XGBoost configurations for the same pair across batches as market conditions evolve, ensuring continuous adaptation. This creates a self-tuning mechanism where Monte Carlo simulations stress-test models against forward-looking scenarios, converging only on combinations that simultaneously satisfy: Near-term regime alignment (XGBoost gating) and Simulated path profitability (EP scoring), thereby future-proofing against correlation decay.

The dual nature of trade blocking—occasionally denying profits but systematically avoiding larger losses—proves net beneficial. By permitting zero-trade outcomes during unfavorable regimes, the system transcends the “always-on” execution compulsion of prior frameworks. This intentional inaction reduces turnover while boosting returns, converting market uncertainty into selective advantage. The framework thus turns rigidity into adaptive discretion, where non-trading is as strategically sound as execution.

3.4 Graph-based Portfolio Management

Effective asset and portfolio management is fundamental to a robust trading strategy. Beyond optimizing individual trades, coordinating them within a broader portfolio framework enhances both profitability and diversification. This section presents a novel method for extracting an optimal portfolio from a pairs trading optimization problem, relying on a graph-based analysis, based on the expected profitability metric. It takes inspiration from Qureshi et al. [45], which introduced a graph-based portfolio construction methodology.

The proposed graph-based portfolio management approach for the unified pairs trading system introduces a structured, scalable solution for trade selection and capital allocation within a statistical arbitrage framework. By modeling candidate trades as a graph with weights based on EP , the system efficiently filters and prioritizes positions under various real-world constraints, such as sector diversification, exposure limits, and available capital. In the graph, edge weights represent either the expected profits or realized profits from the trade containing both securities. Figure 3.22 illustrates the overall system architecture.

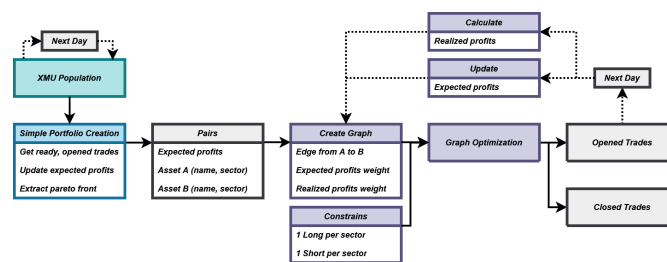


Figure 3.22: Graph-based Portfolio Management Architecture.

Optimization solutions are modeled as a directed weighted graph, each edge representing a potential or already opened trade between two securities (Section 3.4.1). A maximization process is applied to this graph to select the optimal subset of trades taking into account the current positions, expected and realized profits, sector constraints, and available capital (Section 3.4.2). The proposed method first relies on the simple portfolio management (*SPM*), as done in prior versions of the unified pairs trading system, prior to opening the trades the optimized portfolio. From there, the graph optimization is applied, and the trades will be managed. The portfolio management is integrated into the unified framework (Section 3.4.3), and subsequently evaluated as to test its overall reliability (Section 4.6).

3.4.1 Graph Representation

To formalize the selection and management of trading pairs, we introduce a directed weighted graph $G = (V, E)$, where each edge encodes a trading strategy between two stocks:

- **Vertices (V):** Each vertex represents a stock.
- **Edges (E):** Each edge ($i \rightarrow j$) with β factor, represents a trading pair where:
 - i and j are the respective assets.
 - S_i and S_j are the sectors of stocks i and j , respectively.
 - F_i and F_j represent whether each stock is being bought ($F = +1$) or sold ($F = -1$).
 - $EP_{i,j}$ denotes the expected profitability of the pair.
 - $RP_{i,j}$ captures the realized profit from an open trade at evaluation time.

This representation allows the use of graph-theoretical optimization techniques to construct portfolios under a range of practical constraints—sector diversification, capital availability, and asset reuse limitations. While possible trades are represented by the expected profits of that that trade, opened trades are encoded in the graph through two types of edges per pair: one reflecting expected profits from continuing to hold the position, and another representing realized profits if the trade were to be closed. For possible positions, the realized profit edge is absent (or equivalently zero), whereas for open positions, this edge is dynamically updated to reflect live mark-to-market performance. An example of such a graph is illustrated in Figure 3.23, and formalized below:

$$\text{Node 1} \rightarrow \text{Node 4}, \quad EP_{1,4} = 0.8, \quad RP_{1,4} = 0.0$$

$$\text{Node 1} \rightarrow \text{Node 3}, \quad EP_{1,3} = 0.9, \quad RP_{1,3} = 0.0$$

$$\text{Node 3} \rightarrow \text{Node 4}, \quad EP_{3,4} = 0.7, \quad RP_{3,4} = 0.2$$

$$S_1 = S_2 \neq S_3 = S_4$$

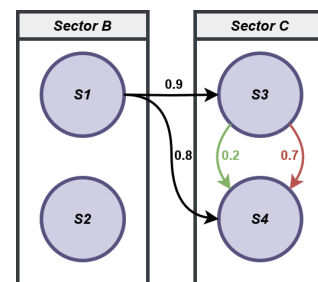


Figure 3.23: Graph example with corresponding equations.

3.4.2 Graph Optimization

It is important to note that this methodology enables the system to better leverage the expected profits metric, allowing it to dynamically choose between expected and realized profits. This adds an extra layer of decision-making for trade entries and exits, with a comprehensive view of overall portfolio capital dynamics. The graph-based implementation enhances portfolio management by incorporating multiple factors into each trading decision, significantly improving the effectiveness of the pairs trading strategy. The results of this experiment underscore the importance of robust portfolio management and highlight the potential of graph-based optimization approaches in representing and managing a pairs trading portfolio, building upon the work of Qureshi [45]. The EP metric becomes an integral component of the graph optimization, and is better than assigning a correlation metric to the edges, as the portfolio constituents were already optimized to have high correlation. Furthermore, since EP includes penalization for long or high variance trades, the portfolio maximization will inherently select the best pairs and their respective trading thresholds, while combining the solutions into a meaningful portfolio.

Having constructed a graph of candidate trading pairs P_I via genetic optimization, the next step is to extract a final, actionable portfolio P_F that maximizes profitability while satisfying structural and risk-related constraints.

Objective Function

The maximization problem can be defined as:

$$\max \sum_{i \rightarrow j} \left(EP_{i,j} \times x_{i,j} + RP_{i,j} \times (1 - x_{i,j}) \right) \quad (3.16)$$

where $x_{i,j}$ is a binary decision variable indicating whether each pair is retained in the portfolio. The model can choose to either hold an open trade and expect $EP_{i,j}$, or close it and realize $RP_{i,j}$.

Constraints

Each trading pair is either included in the portfolio ($x_{i,j} = 1$) or not included ($x_{i,j} = 0$). Each trading decision is binary and given by:

$$x_{i,j} \in \{0, 1\}, \quad \forall i, j, \quad i < j \quad (3.17)$$

As highlighted by [45], including multiple trades involving the same asset or sector often increases portfolio variance. To mitigate the risk of increased portfolio variance caused by overlapping assets or sectors, we also propose that each sector is limited to have one security bought and one sold at a time. The optimization can decide to drop a pair and cash out in order to be able to open the other trade from a common sector. To avoid over-concentration and limit portfolio variance, we impose sector-level exposure constraints. Let S be any sector out of the whole universe of security sectors, we constraint: Let S denote the set of all security sectors, and let $s \in S$ be any particular sector. We impose the following constraints to limit exposure within each sector:

$$\sum_{\substack{S_i=s \\ F_i=+1}} x_{i,j} + \sum_{\substack{S_j=s \\ F_j=+1}} x_{i,j} < 1, \quad \forall s \quad (3.18)$$

$$\sum_{\substack{S_i=s \\ F_i=-1}} x_{i,j} + \sum_{\substack{S_j=s \\ F_j=-1}} x_{i,j} < 1, \quad \forall s \quad (3.19)$$

These ensure that, at any time, only one long and one short position may involve stocks from the same sector, promoting diversification. This constraint framework allows trades to be swapped dynamically. For instance, an open position involving a sector may be closed to release exposure, enabling a new, more promising trade involving the same sector.

Portfolio Dynamics and Capital Allocation

The optimization process is fully dynamic and capital-aware: between optimization cycles, existing positions are carried forward with updated EP and RP values. Any position that no longer appears in the refreshed candidate set P_F is closed, and its realized profit $RP_{i,j}$ is returned to the capital pool, while any new candidates that resulted from the optimization have the position opened. Similar to the previous implementations, the capital is allocated according to (3.9).

Example

Figure 3.24 and Table 3.11 illustrate a concrete example of portfolio evolution under this framework. In this example, the direction of the arrow identifies the long component (origin) and short component (end).

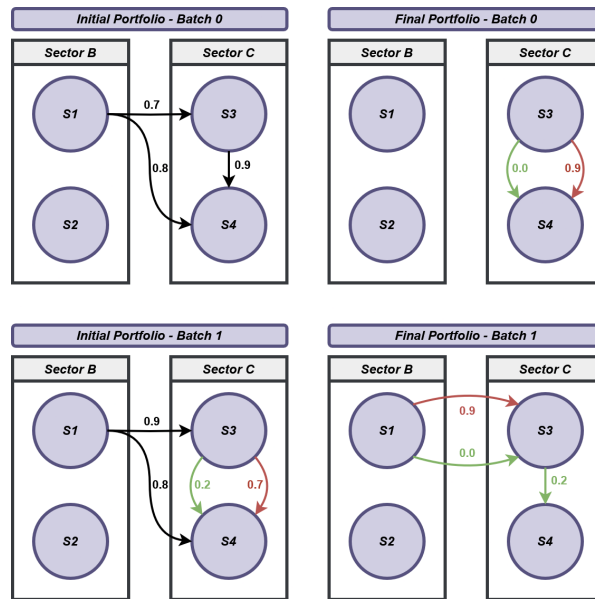


Figure 3.24: Example of evolution of a portfolio for two batches.

Table 3.11: Description for example of evolution of a portfolio for two batches.

	Initial Portfolio	Final Portfolio	Results
0	There are three edges, of which only one can be chosen, as there is no combination of pairs that follow the sector constraint.	The edge $S_3 \rightarrow S_4$ is selected by the optimization, as it has the highest $EP_{i,j}$.	$x_{3,4} = 1$ (Open long S_3 , short S_4) $EP_{Port} = EP_{3,4} = 0.9$
1	$EP_{1,3}$ and $EP_{1,4}$ get updated. Edge $S_3 \rightarrow S_4$ is kept in the graph, updating its $EP_{3,4}$ and offering the option of cashing out the current profits, $RP_{3,4}$.	The optimization cashes out $RP_{3,4}$ to unconstrain Sector C and allow trade $S_1 \rightarrow S_3$ to be opened.	$x_{3,4} = 0$ (Close long S_3 , short S_4) $x_{1,3} = 1$ (Open long S_1 , short S_3) $EP_{Port} = EP_{1,3} + RP_{3,4} = 1.1$

3.4.3 Adapting the Unified System

To evaluate the impact of the proposed Graph-based Portfolio Management, we integrate it within the broader system pipeline. The updated system, referred to as *GXMU*, is illustrated in Figure 3.25, where only adaptation to the system implementation from the previous *XMU* implementation is the introduction of the Graph-based portfolio management, *GP*.

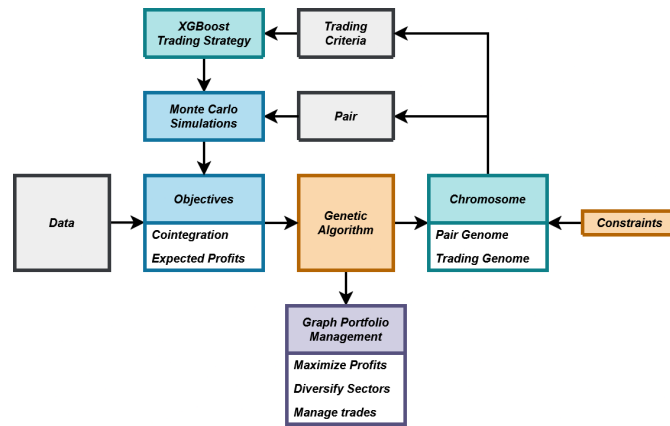


Figure 3.25: Graph-based-XGBoost-Monte Carlo Unified Pairs Trading System Architecture.

Portfolio Management

Within the *GXMU* system, portfolio management becomes more streamlined and effective. The procedure begins by extracting the *XMU* portfolio, then applying the simple portfolio extraction proposed

in previous Sections, with some updates: There is no longer a need to manually close $EP < 0$ trades, since the GP optimization inherently maximizes expected profit while respecting constraints. This results in automatic selection or exclusion of suboptimal trades. The modified portfolio generation is detailed in Figure 3.26, and summarized in Table 3.12.

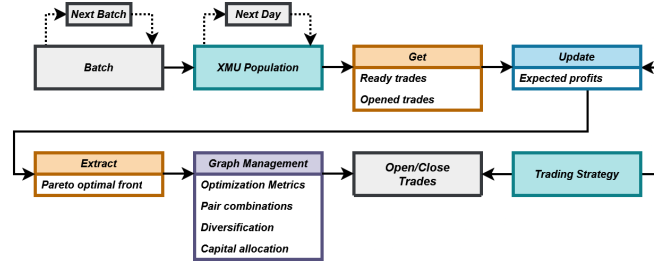


Figure 3.26: Graph-based Portfolio Management Architecture.

Table 3.12: Graph-based Portfolio Management Summary.

Batches	Training Windows	Portfolio Management
5 day frequency	GA : 900 days	get : ready and opened trades
XMU (200 runs)	MC : 60 days	$update$: EP values
	XGB : 600 days	$extract$: POF
	($coint < 0.2$ pairs)	$allocate$: GP
		$monitor$: exit triggers for opened trades

3.4.4 Discussion

The graph-based portfolio optimization fundamentally resolves the core limitation of simplistic solution aggregation that persisted through previous implementations of the framework. By reformulating portfolio construction as an edge maximization problem, the system replaces equal-weight allocation with adaptive capital distribution. This enables true strategic inaction: batches may yield zero new trades when no edge improves total portfolio EP , or dynamically liquidate existing positions to free capital for higher-conviction opportunities. Crucially, the graph structure enforces hard constraints like single-asset utilization and sector balance, transforming diversification from an afterthought into an optimization constraint.

The portfolio layer seamlessly unifies all preceding advancements: XGB regime forecasts inform edge viability, Monte Carlo's EP projections quantify edge weights, and genetic optimization continuously stress-tests different solutions across simulated regimes to use for the portfolio creation methodology. This creates a self-regulating ecosystem where capital automatically flows toward high- EP , constraint-compliant edges while avoiding overconcentration—effectively overcoming the "mandatory trading" flaw of earlier systems. The real-time rebalancing capability ensures optimal exposure management without human intervention, making portfolio adaptation as responsive as individual trade execution.

Chapter 4

Results

This research advances pairs trading and market prediction by introducing a novel framework that integrates machine learning techniques—multi-objective genetic algorithms, Monte Carlo simulations, and XGBoost models—to address critical limitations in traditional strategies. Building on the methodologies outlined previously, this chapter evaluates the performance of the proposed framework across four interconnected case studies, each focusing on an independent component of the full system. These components are designed to be modular, enabling further optimization of individual elements while collectively contributing to a robust and adaptive trading strategy. The framework’s architecture emphasizes a balance between modularity and integration. Each case study targets a distinct aspect of the trading process—pair selection, profit optimization, trade execution, and portfolio management—allowing for independent refinement of methodologies. Notably, each subsequent case study builds upon and significantly improves the results of its predecessors, demonstrating the value of decomposing a complex strategy into specialized, optimized subsystems, while combining methodologies where possible. This Chapter presents the empirical outcomes of the four case studies:

- **4.3 Simple Unified Pairs Trading System (*SU*)** The unification of pair formation (*PF*) and trading strategy (*TS*) within a single optimization process demonstrates significant reliability improvements over separate methodologies. Augmenting traditional threshold-based strategy performance is significantly improved by the proposed reversion test, indicating the potential of rigorous validation of spread dynamics in aiding trading decisions. Incorporating the cointegration-optimized β coefficient into spread construction elevates pair quality during evolutionary selection. This econometrically grounded approach prioritizes equilibrium-driven relationships, enhancing the fidelity of mean-reversion signals. Imposing search space constraints are able to reduce portfolio variance compared to unconstrained baselines, mitigating tail risk without sacrificing return potential. Performing objective penalization reveals slight performance improvements and co-optimizing returns and cointegration stability reliably optimizes the pair formation and trading strategy. Furthermore, magnitude penalization directs evolutionary pressure toward solutions with economically meaningful spread deviations. Systematic re-calibration of the system improves the overall solution quality over time, confirming the necessity of adaptive position management.

- **4.4 Expected Profitability via Monte Carlo Simulation (*MU*)** Evaluates the accuracy of simulated price paths and quantifies the role of expected profit metrics in guiding optimization. Notes the importance of an accurate simulation methodology, as to correctly evolve the solutions towards real profitable trades. The system prioritizes strategies resilient to diverse market scenarios, resulting in a significant improvements in performance. The access to fully dynamic entry triggers significantly improves the returns and decreases the risk of the strategy.
- **4.5 Sentiment-based Trading via Extreme Gradient Boosting (*XMU*)** Demonstrates the *XGB* model's effectiveness in enhancing trade timing by predicting mean-reverting behavior. The models accurately capture threshold dynamics, validating the approach of basing trading rules on alternative predictive signals rather than solely on raw spread values. Moreover, the genetic optimization framework successfully identifies a single *XGB* model from the ensemble that consistently performs well on out-of-sample spread outcomes, by stress-testing various models in the available simulations. This selection process significantly improves risk-adjusted returns and effectively integrates with the broader unified trading system.
- **4.6 Graph-based Portfolio Management (*GXMU*)** Validates the graph-driven portfolio construction methodology, showcasing how diversification constraints applied to a capital-aware optimization can successfully maximize the portfolio returns.

The results underscore the synergistic benefits of combining modular, independently optimizable components with strategic integration. For instance, while separating *PF* and *TS* could theoretically allow deeper specialization, their unification ensures cohesive strategy evolution without computational trade-offs. Similarly, the XGBoost model and Monte Carlo simulations, though developed as standalone modules, amplify the system's robustness when combined. Key demonstrate that the framework outperforms conventional pairs trading strategies and benchmark indexes in consistency, risk mitigation, and profitability. Ultimately, the findings advocate for a hybrid approach: decomposing complex trading systems into optimizable modules while selectively integrating critical phases to preserve flexibility and efficiency. This methodology not only addresses the challenges of pairs trading but also offers a scalable blueprint for adaptive pairs trading systems in dynamic markets.

4.1 Transaction Costs

The costs and fees associated with each market operation gradually significantly erode potential profits and therefore must be accounted for. An otherwise profitable strategy might become unprofitable due to the costly nature of the market: The back-testing simulation incorporates short-selling fees, market slippage, and operational costs. Table 4.1, based on estimations from [4] (and similarly adopted in [18], [6], and [5]), presents these cost components. Explicit costs encompass direct trading expenses, including broker commissions, transaction taxes, stamp duties, and exchange fees. Implicit costs, by contrast, include indirect expenses such as slippage—the discrepancy between the requested trade price and the actual execution price.

Additionally, transaction costs are a function of the β factor. The resulting rules are presented in Table 4.2, following the methodology of [5].

Table 4.1: Transaction costs considered.

Cost	Explicit	Implicit	Short Rental
Description	Commissions, exchange fees	Slippage, market impact, delays	Short position fee
Rate	0.08%	0.2%	1% (p.a.)

Table 4.2: Transaction costs calculation.

Position		Explicit	Implicit	Rental
$\beta > 1$	Long	$(1 + 1/\beta) \times 0.08\%$	$(1 + 1/\beta) \times 0.2\%$	$(1 + 1/\beta) \times (1\%/252) \times tradingDays$
	Short	$(1 + 1/\beta) \times 0.08\%$	$(1 + 1/\beta) \times 0.2\%$	$(1 + 1/\beta) \times (1\%/252) \times tradingDays$
$\beta \leq 1$	Long	$(1 + \beta) \times 0.08\%$	$(1 + \beta) \times 0.2\%$	$(1 + \beta) \times (1\%/252) \times tradingDays$
	Short	$(1 + \beta) \times 0.08\%$	$(1 + \beta) \times 0.2\%$	$(1 + \beta) \times (1\%/252) \times tradingDays$

4.2 Implementation Results

Table 4.5 shows the performance of SU , MU , XMU , $GXMU$, and Figure 4.14 compares the portfolio performance to the $S\&P500$ index returns from 2000 – 2024. The system is executed starting each period (2000 – 2008, 2008 – 2016, 2016 – 2024), and its performance is analyzed for the eight year period. The performance metrics are averaged for each analysis: The total ROI for the testing period (with and without transaction costs); the annual ROI ; the Sharpe Ratio and MDD for the whole period; the system's win ratio; and the trade total. These metrics are averaged for the three periods. For subsequent tests in this Chapter, the exact same conditions are applied.

Table 4.3: Strategy performance metrics averaged for the testing periods: 2000 – 2008, 2008 – 2016, 2016 – 2024.

Strategy	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
SU	56%	101%	5.0%	0.99	21%	56%	462
MU	81%	158%	7.5%	1.66	10%	59%	488
XMU	90%	169%	8.4%	3.61	4%	65%	276
$GXMU$	109%	186%	8.9%	3.71	5%	65%	186
$S\&P500$	73%	89%	5.4%	0.87	39%	67%	1

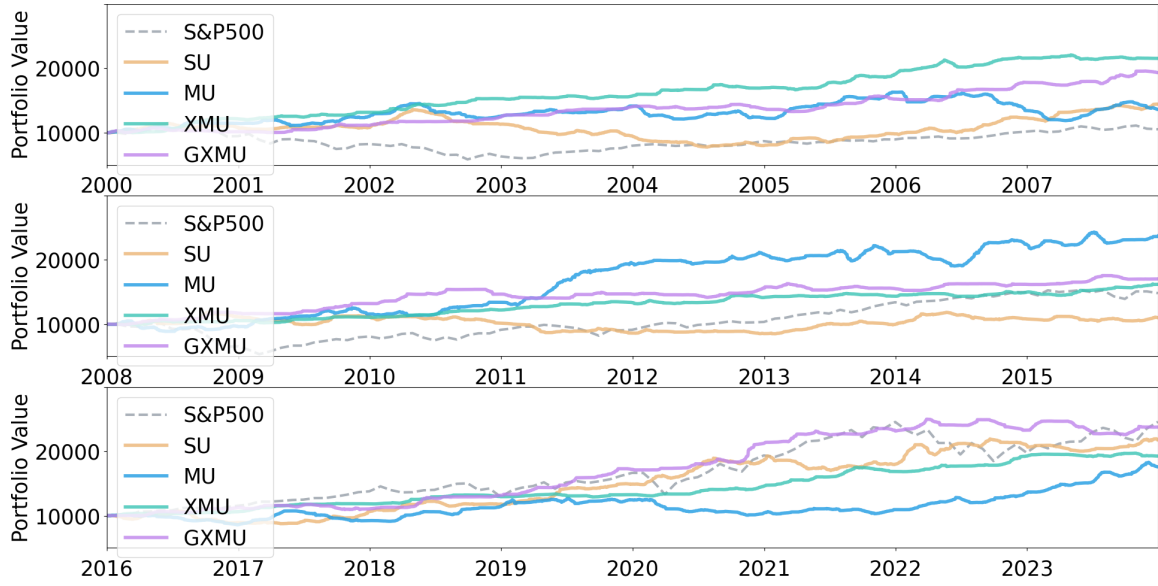


Figure 4.1: Strategy performance for the testing periods: 2000 – 2008, 2008 – 2016, 2016 – 2024.

All applications of the core framework delivered positive returns, with each version outperforming its predecessor across key metrics. However, the relative performance rankings do not remain consistent across all time periods. This underscores the importance of evaluating results across multiple time frames, as relying on a single period can lead to misleading conclusions about the robustness of a system.

Embedding a Monte Carlo–based profitability estimator into the genetic algorithm’s fitness function enabled *MU* to achieve a substantially higher win rate and risk-adjusted return than *SU*. The Monte Carlo layer filtered out low-probability candidates, resulting in a portfolio skewed toward higher-confidence trades. Consequently, the win rate rose notably, net returns increased, and maximum drawdowns halved, demonstrating that downside risk did not grow despite selecting higher-return opportunities. *XMU* incorporates an *XGB*–based sentiment classifier. Adding sentiment filtering led to a marked reduction in trade count and increased the win rate, indicating higher-confidence signals being executed. Maximum drawdown noticeably shrank compared to *MU*, the Sharpe Ratio edged higher, and net returns climbed further. During the sample’s most volatile periods, equity drawdown troughs were noticeably shallower than previous iterations, confirming the sentiment filter’s ability to preemptively reduce allocation when downside risk spiked. The final *GXMU* iteration merges the previous implementations with a graph-based portfolio analysis, constraining trades in common sectors. This combination drove a further reduction in total trades while delivering the highest net returns of all configurations. *GXMU* achieved the peak Sharpe Ratio observed across versions, reflecting superior risk-adjusted performance. Maximum drawdown was slightly higher than in *XMU*—because overlapping signals sometimes amplified intra-period volatility—but this uptick was offset by improved overall returns and broader diversification. The win rate remained robust, and average profit per trade rose materially due to better timing and dynamic position sizing. In the equity curve, *GXMU* and *XMU* paths are the smoothest, with only shallow dips even during its most severe drawdown, whereas *SU* and *MU* experienced much deeper troughs.

Across all out-of-sample subperiods—from the early years of the dataset to more recent market environments, *MU*, *XMU* and *GXMU* consistently outperformed the *S&P500* Total Return Index in terms of cumulative return, Sharpe Ratio, and maximum drawdown. During bull markets, the strategies either matched or outperformed the index’s upside potential, while during volatile, sideways, or bearish market phases, they demonstrated significantly better downside protection.

These findings highlight the robustness and adaptability of the proposed methodologies, particularly the integration of Monte Carlo-based pair selection with sentiment-based filtering, which together deliver added economic value beyond traditional market-cap-weighted strategies. Notably, the implementations maintained minimal correlation with the *S&P500* throughout all periods, reinforcing their market neutrality and validating their role as viable components in a diversified investment portfolio. However, a key limitation observed is the gradual decay in performance over time. For example, while all implementations remained profitable, their returns in the 2016–2024 period were markedly lower than those achieved between 2000–2008, reflecting a diminishing alpha generation capability as markets evolve and arbitrage opportunities become increasingly compressed. In contrast, the *S&P500* has shown stronger performance in recent years, possibly due to structural shifts in market dynamics, monetary policy, or increased investor participation. These observations underscore the importance of ongoing model recalibration and innovation to sustain alpha in quantitative strategies over extended horizons. In conclusion, the proposed system implementations offer moderate annual returns with relatively low risk, making them a compelling market-neutral addition to a trader’s diversified portfolio.

4.3 Simple Unified Pairs Trading System

The *SU* system implementation achieved positive annual returns, with relatively low drawdowns, indicating the low-risk potential of pairs trading. These results are indicative of the potential of this integrated system as a baseline for a more complex system. This methodology can be improved many ways. Either by improving the objective configuration, or to improve the trading strategy. A smart approach would be to limit the trades, as to not blindly enter all trades equally when the open condition is met. Additionally, from the final portfolio, a wiser population can be selected.

4.3.1 Hyperparameters

The genetic algorithm’s hyperparameter optimization identified to be 80 generations, with a population size of 250 as the configuration yielding peak performance, with minimum computational complexity. Sensitivity testing of crossover and mutation operators revealed negligible performance variation across probability thresholds and methodological implementations. This invariance suggests the algorithm’s robustness to operator parameterization, enabling retention of standard genetic programming conventions without loss of optimization efficacy.

4.3.2 Cointegration Coefficient

To evaluate the benefit of using cointegration-derived weights, the performance of the unified system is compared, over the 24-year out-of-sample period, for: (i) equal-weight spreads and (ii) optimal spreads constructed via the estimated cointegration coefficient. Table 4.4 summarizes key performance metrics. The β -adjusted configurations exhibit a statistically significant increase in the system performance, indicating both superior risk-adjusted returns and enhanced consistency. By estimating the equilibrium coefficient β that minimizes the cointegration residual, the strategy concentrates on pairs whose price relationship is most persistent and mean-reverting. This leads to more consistent risk-adjusted returns, reduced downside exposure, and a higher probability of successful trades. In turn, evolutionary optimization guided by a cointegration objective produces a more robust and reliable trading universe than simple equal-weight pairings.

Table 4.4: Performance analysis for the cointegration factor.

Cointegration Factor	<i>ROI</i>	<i>ROI</i> (no costs)	<i>ROI</i> (p.a.)	<i>SR</i>	<i>MDD</i>	Win Rate	Trades
	29%	51%	1.38%	0.58	29%	54%	455
✓	56%	101%	4.98%	0.99	21%	56%	462

4.3.3 Momentum Constraint

The impact of the momentum-based constraint is evaluated by comparing system performance with and without the constraint over the 24-year out-of-sample period, as presented in Table 4.5. Introducing the momentum-based constraint slightly raises the annualized return, while reducing the maximum draw-down. By prohibiting short positions when the spread exceeds its 30-day moving average, and long positions when it falls below, the constraint prevents trades against prevailing momentum. This refinement narrows the search space to pairs whose spread dynamics align more consistently with market trends, thereby enhancing risk-adjusted returns, lowering downside exposure and increasing trade success probability. Overall, incorporating a simple momentum-based filter into the evolutionary search yields a more robust and economically sensible trading universe than an unconstrained approach, highlighting the importance of search space constraints, as opposed to the typical sector constraints.

Table 4.5: Performance analysis for the momentum constraint.

Momentum Constraint	<i>ROI</i>	<i>ROI</i> (no costs)	<i>ROI</i> (p.a.)	<i>SR</i>	<i>MDD</i>	Win Rate	Trades
	51%	93%	2.5%	0.84	24%	56%	477
✓	56%	101%	4.98%	0.99	21%	56%	462

4.3.4 Simple Reversion Test

The impact of the simple reversion confirmation test is evaluated by comparing the baseline threshold-based strategy against the same strategy augmented with the delayed entry and exit rules, over the 24-year out-of-sample period, presented in Table 4.6.

The addition of the simple reversion test increases annualized return. By requiring a second threshold crossing before opening a position, and adaptively adjusting the exit threshold on subsequent crossings, the enhanced strategy filters out false signals generated by transient volatility spikes. This confirmation mechanism allows the spread to exhibit genuine mean-reversion before entry, and to capture more of the full reversal before exit. As a result, risk-adjusted returns improve, downside exposure is curtailed, and trade success probability is higher compared to the unmodified threshold-based approach. Notably, delaying the exit also proves to be very effective, indicating the potential of delaying exits in pairs trading. Furthermore, adding the entry delay proves to be very effective, even on its own. It indicates that not every pair should be bought, and performing an additional test can significantly improve a pairs trading strategy. Finally, the implementation of signal smoothing significantly enhances system performance, particularly improving the exit delay timing.

Table 4.6: Performance analysis for the simple reversion test.

<i>Entry</i>	<i>Exit</i>	<i>Smooth</i>	<i>ROI</i>	<i>ROI (no costs)</i>	<i>ROI (p.a.)</i>	<i>SR</i>	<i>MDD</i>	<i>Win Rate</i>	<i>Trades</i>
			−14%	22%	−0.9%	0.44	38%	53%	499
✓			18%	31%	0.3%	0.51	23%	54%	454
✓		✓	29%	49%	1.7%	0.33	30%	54%	501
	✓		6%	10%	0.1%	0.61	28%	54%	462
	✓	✓	38%	61%	2.1%	0.81	23%	54%	444
✓	✓	✓	56%	101%	4.98%	0.99	21%	56%	462

In summary, incorporating simple mean-reversion test substantially strengthens the effectiveness of the pairs trading strategy compared to traditional threshold-based approaches. Notably, the strategy fails to generate positive net returns after transaction costs when these enhancements are excluded.

4.3.5 Chromosome

The optimal gene ranges are determined by iteratively comparing the performance of different gene configuration during the genetic optimization. Figure 4.2, Figure 4.3, Figure 4.4 and Figure 4.5 compare the Sharpe Ratios of pair-strategy combinations, selected in the first year of the 2000 – 2008 execution. These show that, for different pairs, the optimal threshold configurations vary, and present a clear performance peak, thus confirming the potential of assigning a specific trading strategy to each specific pair. Furthermore, by averaging the curves, we can detect lower and upper thresholds for the gene configurations. The parameter selected by the optimization is also shown, revealing an acceptable selection of these parameters.

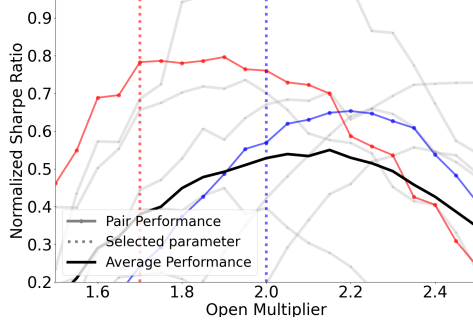


Figure 4.2: Pair-Strategy Sharpe Ratio by varying the open threshold multiplier.

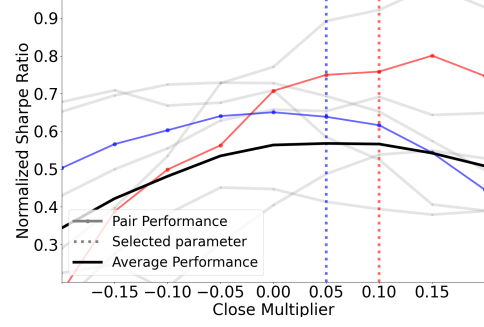


Figure 4.3: Pair-Strategy Sharpe Ratio by varying the close threshold multiplier.

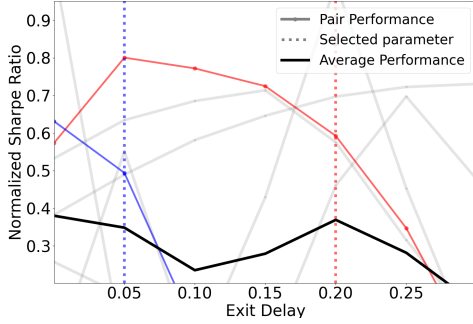


Figure 4.4: Pair-Strategy Sharpe Ratio by varying the exit delay.

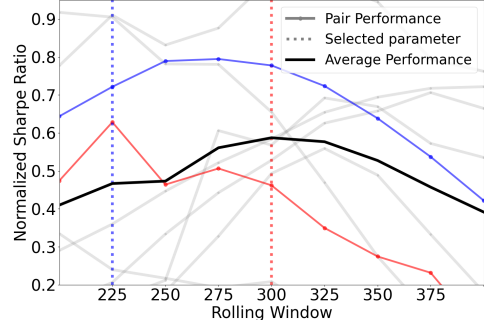


Figure 4.5: Pair-Strategy Sharpe Ratio by varying the rolling window.

This analysis supports the ranges set in Table 3.1, as the plotted ranges are able to define each curve accurately. It is important to note that for parameters such as the entry and exit triggers, as well as the rolling window length, the performance strategy performance remains relatively flat around the optimal values. This indicates that near-optimal parameter choices yield profitability comparable to the true maximum. In contrast, the exit delay parameter exerts a significantly greater influence on strategy performance, and incorrect calibration of can lead to a substantial decline in profitability, underscoring the critical importance of precisely tuning this parameter.

4.3.6 Objectives

The implementation is locked into the *ROI* and *Coint* objectives due to the previous implementation choices. Thus, magnitude enters the optimization only as a penalization on *ROI*. Penalizing the fitness of individuals whose initial spread magnitude falls outside the desired (tradable) range not only expands the final portfolio of tradable pairs but also slightly improves overall strategy robustness. Table 4.7 compares key performance metrics for different penalization levels applied to the *ROI* score of low and high *Mag* candidates. Even when the *ROI* contribution of all low and high magnitude individuals is set to zero (100% penalization), the system still outperforms the unconstrained baseline, highlighting the benefit of excluding pairs unlikely to trade at the start of testing. The best risk-adjusted returns and lowest drawdown occur with penalization levels around 50%, which effectively prune poorly conditioned spreads while retaining sufficient evolutionary diversity. Consequently, applying a moderate magnitude-based penalty strikes an optimal balance between portfolio breadth and trading performance.

Table 4.7: Performance analysis for different Mag penalization ratios.

Q	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
0	48%	89%	3.59%	0.79	26%	55%	456
0.25	54%	96%	4.51%	0.87	24%	56%	462
0.5	56%	101%	4.98%	0.99	21%	56%	462
0.75	52%	95%	4.58%	0.84	19%	55%	481
1	51%	92%	3.99%	0.80	25%	55%	470

4.3.7 Portfolio Management

To assess the benefit of the proposed Simple Portfolio management (SP) procedure, we compare the performance of the unified GA-optimized system with and without the portfolio re-optimization step, over the 24-year out-of-sample period. Furthermore, it is determined the optimal value for L , responsible for defining the allocation portion of capital that gets allocated per trade. Table 4.8 and Table 4.9 report key performance metrics for these tests. The results demonstrate that continuous re-calibration of the portfolio concentrates capital into higher-quality, diversified pairs, thereby enhancing both return and risk metrics compared to a naive allocation of all GA candidates. This happens as the various parameters in the system are always up to date. The optimal value for allocating capital is determined to be $L = 75$. A low value for L presents low returns and high drawdowns, since the capital is spent at an higher rate.

Table 4.8: Performance analysis for the system recalibration.

Recalibrate	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
	29%	48%	2.5%	0.44	25%	53%	499
✓	56%	101%	4.98%	0.99	21%	56%	462

Table 4.9: Performance analysis for different capital allocation ratios.

L	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
25	29%	48%	2.5%	0.44	25%	56%	462
50	33%	69%	3.1%	0.51	23%	56%	462
75	56%	101%	4.98%	0.99	21%	56%	462
100	45%	76%	2.95%	0.51	23%	56%	462

Figure 4.6 shows that less than five days yields no significant improvement in results, when re-calibrating the system. However, it's noted that not re-calibrating the system results in a loss of performance. Figure 4.7 illustrates the new pareto solution count, every 25 new runs. It reveals that the new solution count does not increase substantially after 200 runs, making it sufficient to thoroughly explore the entire search space.

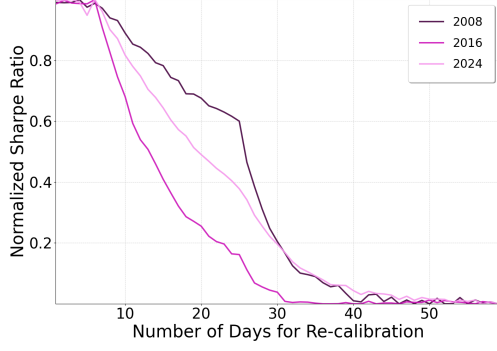


Figure 4.6: Sharpe Ratio for the system by varying the number of days for recalibration.

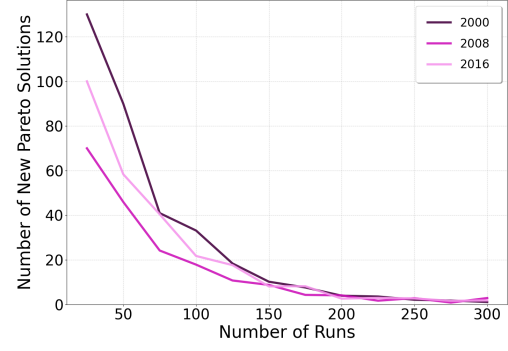


Figure 4.7: Number of new solutions in the pareto front in intervals of 25 runs.

4.3.8 Benchmark

Table 4.10 reports the out-of-sample performance of the unified system (SU) against the modular benchmarks (P_1 , P_2 , T_1 , T_2) over the 2000–2024 period. P_1 and P_2 illustrate the poor performance of a static threshold-based trading strategy. Although applying an adaptable trading strategy to T_1 and T_2 pairs increases the profitability, the unified system outperforms the separate systems in all metrics, even without explicit optimization of an additional objective (HL , Vol , MDD). The improved performance of the unified system over the separately optimized counterparts validates the integration of PF and TS within a single optimization process. Note that the T_2 and P_2 configuration (minimize Half-life) achieves greater performance overall than T_1 and P_2 (maximizing volatility), respectively, thus underscoring the viability of HL integration in a pairs trading system. This is relevant as the MU implementation inherently minimizes the half life from the EP maximization, due to the discounting nature of the value function estimation, also validating the inclusion of the half-life metric in such a system.

Table 4.10: Performance analysis for the benchmark systems.

Implementation	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
SU	56%	101%	4.98%	0.99	21%	56%	462
P_1	−13%	17%	−1.6%	−0.32	29%	54%	431
P_2	−2%	22%	−0.4%	−0.15	24%	54%	512
T_1	−5%	30%	−0.9%	−0.07	31%	54%	412
T_2	13%	39%	1.05%	0.11	27%	55%	491

4.3.9 Full System

Figure 4.12 and Figure 4.13 illustrate a profitable and an unprofitable trade from 2016. In the first case, the optimized triggers fit the testing period, and the entry delay allows for greater gains. In fact, weren't it for the simple reversal test, the position would have been opened the first time it crosses the threshold, and would have been closed as it crossed the static stop loss trigger, thus resulting in a loss. In the second example, the spread slowly diverges after entry, and the static stop loss trigger isn't able to close the position.

A trade being opened for a long time isn't optimal, as the capital could be used elsewhere. This analysis underscores the importance of closing a trade earlier, relying on other factors other than the typical pairs trading thresholds. This demonstrates the rigidity of the threshold-based trading strategy, even if armed with a reversion test.

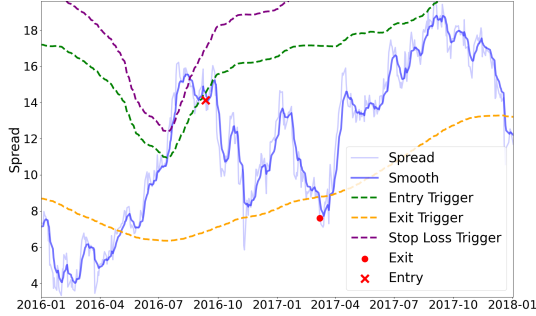


Figure 4.8: Pair *PLD* – *ACN* (2016 – 2018)

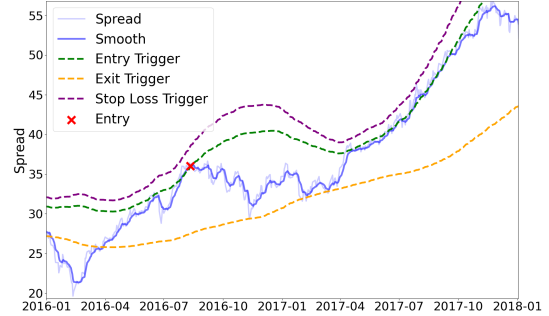


Figure 4.9: Pair *PLD* – *SCG* (2016 – 2018)

4.4 Expected Profitability via Monte Carlo Simulation

The *MU* system replaces past performance maximization with a value function representing the expected profitability of a pair. By optimizing this value function, the final portfolio naturally favors the most promising trades while filtering out suboptimal ones. In addition, trading triggers are optimized for the next trade rather than the entire training period, enabling more adaptive decision-making. The results highlight the importance of using an expected profitability metric as an optimization guide, rather than solely relying on past data, which is often insufficient and unrepresentative of future performance. In contrast to the previous system, the *EP*-based system yields more actionable, trade-ready signals, as those present an higher expected profitability.

4.4.1 Calibration

The simulation parameters m , Δt , b , r and γ are calibrated through backtesting. Regarding the number of simulations, m , and time step, Δt , Figure 4.10 and Figure 4.11 reveal that the *EP* metric converges towards a stable value at roughly $m = 1000$ and $\Delta t = 0.05$.

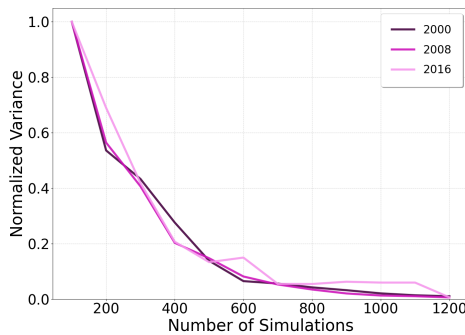


Figure 4.10: Normalized simulation solution variance according to number of simulations for the first batch of 2000, 2008 and 2016.

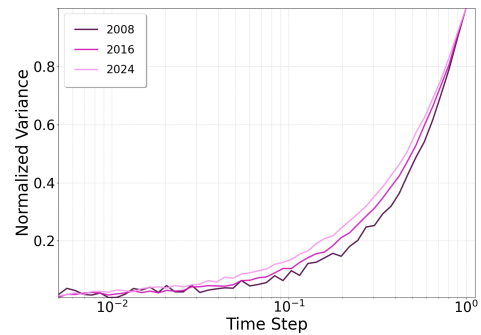


Figure 4.11: Normalized simulation solution variance according to time step for the first batch of 2000, 2008 and 2016.

Furthermore, model performance is tested by varying parameters b , r and γ . The b value that maximizes system profitability is determined to be $b = 25$ (Table 4.11), representing occasional moderate simulated jumps. At both very low and very high discount rates, the forward-looking EP deviates downward relative to realized historical profits, confirming $r = 0.05$ (Table 4.12) as the optimal balance. Similarly, γ maximizes profits at $\gamma = 0.1$ (Table 4.13), being the best trade-off point between mean expected profit and variance penalization. Lower or higher risk-aversion weights produce smaller gains. It's important to note that these parameters aren't thoroughly optimized, as obtaining the value that exactly maximizes the system performance would result in an overfit model. However, since the optimal parameter values remain stable over the 24 year testing period, they are roughly obtained through backtesting.

Table 4.11: Performance analysis for different b values.

b	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
5	29%	48%	2.5%	0.34	25%	57%	499
25	81%	158%	7.5%	1.66	10%	59%	488
50	67%	88%	4.9%	1.31	15%	57%	463

Table 4.12: Performance analysis for different r values.

r	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
0.005	72%	139%	6.4%	1.16	15%	58%	441
0.01	81%	158%	7.5%	1.66	10%	59%	488
0.05	63%	109%	4.9%	1.31	23%	59%	462

Table 4.13: Performance analysis for different γ values.

γ	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
0.05	77%	131%	4.5%	1.24	25%	55%	499
0.1	81%	158%	7.5%	1.66	10%	59%	488
0.2	79%	144%	5.9%	1.41	23%	56%	485

The dynamic exit trigger conditions successfully improve the overall system performance. Incorporating stop-loss optimization increases the median Sharpe ratio and reduces variance, supporting the effectiveness of adaptable the stop-loss mechanism within the optimization. The additional sell trigger, that closes positions when the expected profit turns negative, reveals a slight reduction in drawdowns, underscoring the value of continuous EP monitoring for risk management. Table 4.14 and Table 4.15 show the performance improvements with each of the new exit triggers.

Table 4.14: Performance analysis for stop loss trigger.

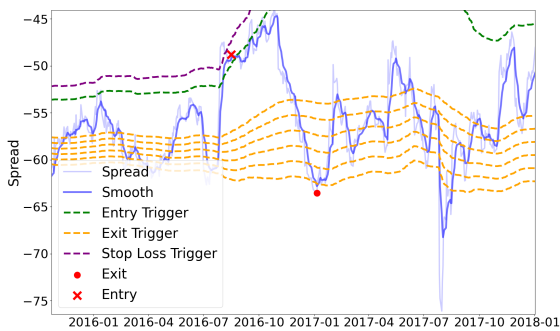
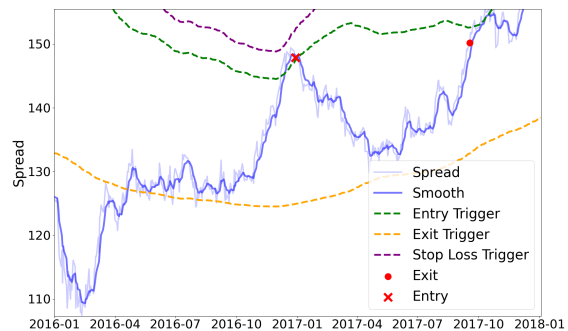
Stop Loss	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
	77%	141%	7.1%	1.51	15%	59%	484
✓	81%	158%	7.5%	1.66	10%	59%	488

Table 4.15: Performance analysis for early close trigger.

Early Close	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
	71%	133%	6.8%	1.54	14%	59%	459
✓	81%	158%	7.5%	1.66	10%	59%	488

4.4.2 Full System

Figure 4.12 and Figure 4.13 illustrate two positions from a tradable batch of 2016 – 2024. In the first case, the simulated outcomes are accurately able to predict the real path of the spread, thus correctly optimizing the thresholds for maximizing trade profitability. In the second case, the spread doesn't reach the exit threshold before it starts reverting. However, negative *EP* sell condition is able to close the position before more losses were accumulated. Even if unprofitable, the current iteration of the system is capable of determining if a trade would become unprofitable, before it reaches the typical stop loss triggers. The *MU* system outperforms the previous implementation, proving the robustness of optimizing for an expected profitability metric, and stress-testing the various trading parameters on simulated paths. However, trade timings could still be further improved, as this iteration of the system relies on long-term predictions (60 days). Adding a short-term prediction to the trading strategy can massively improve the overall profitability of the strategy, as to fine-tune trade entries, exits and delays.

Figure 4.12: Pair *TXN* – *ADP* (2016 – 2018)Figure 4.13: Pair *CVX* – *ANDV* (2016 – 2018)

4.5 Sentiment-based Trading via Extreme Gradient Boosting

The system is adapted to incorporate an ensemble of XGBoost models within the trading strategy, allowing early detection of spread reversals. The adaptation allows the system to select the optimal model that maximizes the expected profitability of each solution.

Evaluation metrics confirmed that the *XMU* system outperformed the previous implementations, and demonstrated robust performance under varied input conditions. The solution is capable of providing real-time feedback with minimal latency, affirming the feasibility of using an XGBoost prediction module in practical, high-stakes environments, in addition to the typical threshold-based trading triggers.

4.5.1 Calibration

To optimize the predictive power of the XGBoost classifier, a calibration phase is dynamically conducted (every batch) using the best-performing subset of features from the training data. Grid search is employed as to obtain tuned key hyperparameters, including the number of estimators, learning rate, maximum depth, subsample ratio, and column subsample ratio. The optimal configurations achieve high precision and recall metrics, which are verified using cross-validation. This tuning ensured the model was neither underfitting nor overfitting, providing a strong foundation for downstream integration. Figure 4.14 shows the average feature importance from all models of the first batch of 2000. According to the results, each feature strikes an important role in predicting the next day derivative.

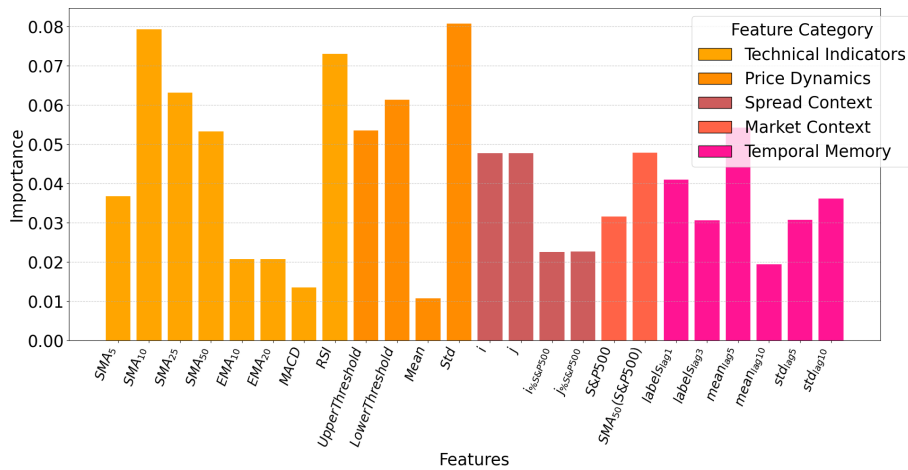


Figure 4.14: Feature importance for the first batch of 2000.

4.5.2 Full System

Figure 4.12 and Figure 4.13 illustrate a favorable and unfavorable trade from the *XMU* system, from the testing period of 2016 – 2024. In the first case, the prediction module accurately predicts the optimal entry and exit timings. In the second case, the same pair gets chosen twice, nine months later. In both occasions, the *XGB* model blocks the entries. In the first time, it blocked an unprofitable trade, while in the second, it denied a profitable one. This shows the duality of such an entry block rule. The *XMU* implementation effectively integrates multiple trading triggers into a cohesive system that analyzes the spread using diverse factors, enabling more informed and data-driven trading decisions.

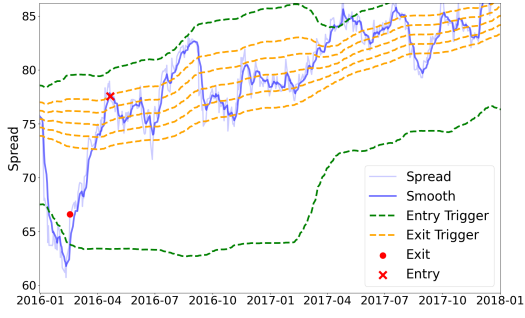


Figure 4.15: Pair *FBHS* – *FTI* (2016 – 2018)

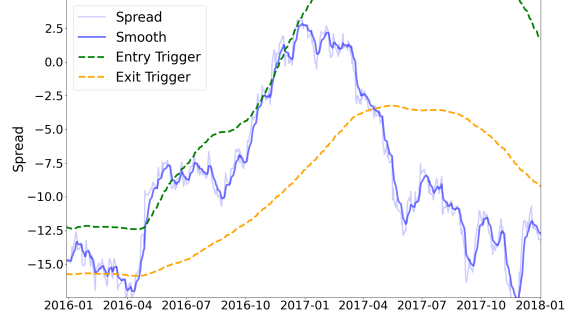


Figure 4.16: Pair *STI* – *AME* (2016 – 2018)

Its relevant to note that the selected model often presents classification such as the one presented in Table 4.16. Interestingly, the chosen configurations often disregard the prediction performance for the reversion (0, 1) labels, in order to more accurately predict the stability (2) and divergence (3, 4) labels. This validates the implementation decision of allowing the genetic optimization to select specific model configurations that maximize potential profits, and allowing the optimization to evaluate each model in the simulated outcomes, as it optimizes for the model that boosts the profitability. Moreover, along the batches, the optimal *XGBoost* model configuration will change for the same pair, if it happens to be selected over various batches, as the system selects the parametrization that outperforms in the simulated environment.

Table 4.16: Classification for Pair *FBHS* – *FTI* (2016, $k = 0.3, 0.1, 0.05, 0.2$)

	precision	recall	f1-score	support
0	0.96	0.75	0.84	2529
1	0.26	0.67	0.38	17758
2	0.81	0.64	0.72	126882
3	0.73	0.71	0.72	106336
4	0.79	0.84	0.81	53360
accuracy			0.70	306865
macro avg	0.71	0.72	0.69	306865
weighted avg	0.75	0.70	0.72	306865

4.6 Graph-based Portfolio Management

The full GXMU system integrates all enhancements—expected profitability metrics (EP), XGBoost trading signals, and graph-based portfolio optimization—into a unified and adaptive trading framework.

The graph-based portfolio layer selects trades by solving an edge maximization problem that considers the *EP* score, cointegration quality, trade compatibility, and diversification. The result is a portfolio that not only selects high-quality trades but also ensures that capital is optimally distributed across non-overlapping and sector-balanced positions.

4.6.1 Calibration

The graph-based implementation is calibrated by redefining the optimal L allocation ratio. According to Table 4.17, the new optimal value is $L = 50$, balancing profitability with risk. The lower value for applying capital shows the need for higher capital allocations, possibly due to the smaller trade counts, compared to the previous iterations of the system. Furthermore, the sector constraint is validated by allowing more than one asset acquisition per sector, and comparing the performance, as shown in Table 4.18. Results indicate that a single trade per sector outperforms the less constrained configurations.

Table 4.17: Performance analysis for different capital allocation ratios.

L	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
25	77%	131%	5.5%	2.94	24%	65%	186
50	109%	186%	8.9%	3.71	5%	65%	186
75	99%	169%	8.1%	3.41	15%	65%	186

Table 4.18: Performance analysis for maximum pairs per sector.

Max Pairs per Sector	ROI	ROI (no costs)	ROI (p.a.)	SR	MDD	Win Rate	Trades
1	109%	186%	8.9%	3.71	5%	65%	186
2	98%	177%	8.3%	3.21	8%	65%	201
5	90%	169%	8.0%	3.03	13%	65%	246

4.6.2 Full System

Figure 4.12 and Figure 4.13 showcase the two solutions from the $GXMU$ optimization. In the first example, the system accurately predicts the optimal entry and exit timings. In the second example, although the trade would have been profitable, the system can't open the trade as it shares a common asset with the previously mentioned trade. Although this case resulted in missed returns, the strategy performs better than the previous iteration. The capital allocation graph adapts dynamically as new optimization solutions are introduced, enabling real-time decision-making and effective trade rebalancing. In conclusion, $GXMU$ demonstrates that intelligent capital distribution—guided by predictive metrics and graph connectivity—enhances portfolio robustness without sacrificing profitability.

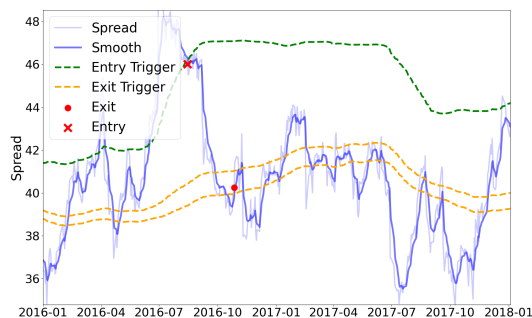


Figure 4.17: Pair $GIS - AYI$ (2016 – 2018)

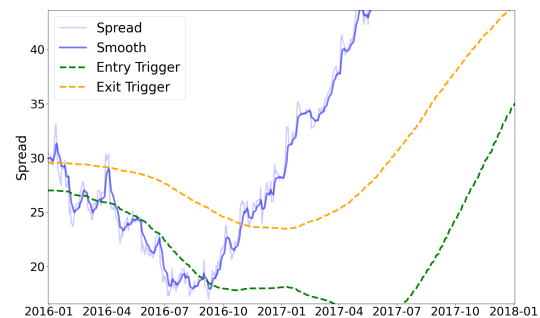


Figure 4.18: Pair $CCL - AYI$ (2016 – 2018)

Chapter 5

Conclusions

This thesis has presented a novel, modular framework for pairs trading that integrates multi-objective *GA*, *MC* simulations, *XGB*-based execution models, and graph-driven portfolio construction. The results demonstrate meaningful improvements over traditional strategies in risk-adjusted performance, adaptability, and robustness across dynamic market regimes. Each component was carefully designed and empirically validated, contributing to a cohesive and forward-looking trading system.

5.1 Achievements

- **Unified Pairs Trading System:** The integration of pair formation (*PF*) and trading strategy (*TS*) into a single evolutionary optimization process yielded notable performance enhancements over decoupled approaches. The introduction of dynamic mean-reversion validation and adaptive threshold tuning led to more stable trade entries and exits, ultimately resulting in improved profitability and lower drawdowns.
- **Monte Carlo Profit Optimization:** By replacing historical return-based objectives with expected profit (*EP*) derived from forward-looking simulations, the framework mitigated overfitting and improved generalization. The use of stochastic price path simulations within the optimization allowed the strategy to adapt to uncertainty and model real-world market behaviors more effectively.
- **XGBoost-Based Execution:** Incorporating machine learning through *XGB* introduced predictive capabilities into trade execution. This component served as an intelligent gatekeeper, filtering trade signals based on forecasted mean-reversion behavior. Furthermore, the inclusion of various predictive models enabled the optimization procedure to select the most favorable candidate through *EP* maximization on simulated spreads.
- **Portfolio Creation via Graph Analysis:** A graph-based optimization layer enabled dynamic capital allocation across the portfolio, minimizing exposure to systemic risks arising from correlated trades. This approach fostered diversification and enhanced stability, showing that intelligent portfolio structuring is essential for sustaining profitability across multiple simultaneous positions.

5.2 Future Work

The success of this framework highlights the potential of combining simulation, optimization, and machine learning into a modular system. Several directions can be pursued to expand upon this foundation:

- **System Design:** The implementations proposed can each be improved by adapting the various components of the unified framework. The objective configuration could be explored, and although the *EP-Coint* combination works well, perhaps a multi-objective configuration could be explored. The trading strategy component could also be improved, although the various trading triggers proposed work well in unison.
- **Simulation Accuracy:** While the Monte Carlo-based EP objective offers robustness, simulation quality can be further improved using adaptive stochastic volatility models or incorporating external market regimes into path generation.
- **Portfolio Intelligence:** The graph portfolio optimizer can be made more aware of trade state (profit/loss, correlation clusters, etc.), potentially improving capital allocation and trade selection under market stress.
- **Incorporating Fundamentals:** Currently, the framework relies purely on technical analysis. Integrating fundamental analysis (e.g., earnings, valuation metrics) could significantly improve the economic soundness of selected trades and reduce exposure to structurally mispriced securities.

5.3 Final Remarks

This thesis presents a robust, scalable, and modular framework for market-neutral investing, leveraging advanced computational tools. By integrating traditional statistical techniques with machine learning and graph theory, it outlines a practical path for the development of adaptive quantitative strategies. The proposed framework bridges the gap between theoretical models and real-world implementation, offering a solid foundation for further innovation in algorithmic pairs trading.

Through a series of complementary methodologies, this work demonstrates consistent improvements over baseline models. These enhancements not only validate the flexibility of the framework but also highlight key insights for practitioners and researchers alike. The core takeaways from this study are: (i) the diminishing returns of over-reliance on historical performance as a predictive tool; (ii) the limitations of rigid, threshold-based trading strategies; (iii) the importance of continual recalibration and dynamic portfolio management; and (iv) the advantages of incorporating multiple, diverse factors into trade decision-making processes. Together, these findings underscore the necessity of a flexible, data-driven approach to quantitative finance, and set the stage for future research into more adaptive and resilient trading systems.

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